Blood-based Biomarkers for the Prediction of Hypertrophic Cardiomyopathy Prognosis

A Systematic Review & Meta-analysis

M. Jansen et al. 2021

Supplemental Material

Table of Contents

Supplemental Methods 2
Study eligibility 2
Definitions 2
Statistical analysis 2
Supplemental references 3
Supplemental Tables 6
Supplemental Table 1: Search strategy 6
Supplemental Table 2: Reported in-/exclusion criteria and biomarker platforms 10
Supplemental Table 3: Excluded studies & reason for exclusion 14
Supplemental Table 4: Biomarkers for heart failure 25
Supplemental Table 5: Biomarkers for malignant ventricular arrhythmia 28
Supplemental Table 6: Biomarkers for outflow tract obstruction 30
Supplemental Table 7: Event rates of composite endpoints (including surrogate endpoints) 30
Supplemental Table 8: Biomarkers for composite endpoints 31
Supplemental Table 9: Biomarkers for composite endpoints including surrogate endpoints 32
Supplemental Figures 46
Supplemental Figure 1: Quantitative analysis endpoints 46
References 53

Supplemental Methods

Study eligibility

Plasma or serum biomarkers obtained from central (arterial or venous) sources or the coronary sinus were excluded from analysis, as the invasiveness of these procedures was deemed incompatible with risk stratification in the general HCM population.

Studies consisting exclusively of subjects with phenocopies (e.g. Amyloidosis, Fabry's disease, Friedrich's ataxia) or subjects aged <18 years were excluded.

Definitions

Hypertrophic cardiomyopathy (HCM) was defined in accordance with the 2014 European Society of Cardiology criteria, summarised as a maximum wall thickness ≥15 mm or ≥13 mm in first degree family members of patients with confirmed HCM, not explained by abnormal loading conditions.[1] However, to avoid selection bias, studies defining HCM in accordance with other criteria or classifications, such as the 1980 and 1995 World Health Organization/International Society and Federation of Cardiology Classification of cardiomyopathies,[2, 3] were also included.

Left ventricular outflow tract obstruction was defined as left ventricular outflow tract gradient at rest and/or after provocation with cut-off values left to the discretion of the studies or septal reduction therapy, i.e. septal myectomy or alcohol septum ablation.

Heart failure (HF) included congestive HF (New York Heart association [NYHA] class, progression to a higher NYHA class and/or signs of congestion, or HF-related hospitalisation), systolic dysfunction (left ventricular ejection fraction <50% or left ventricular fractional shortening <25%) or end-stage HF (left ventricular assist device implantation, heart transplantation, HF-related mortality). Combined congestive HF and end-stage HF outcomes were grouped under congestive HF.

Malignant ventricular arrhythmia (MVA) included sustained ventricular tachycardia (>100 beats per minute and ≥30 seconds in duration or requiring earlier termination due to haemodynamic instability), ventricular fibrillation, appropriate implantable cardioverter-defibrillator therapy, resuscitated cardiac arrest or sudden cardiac death.

Surrogate endpoints for HCM progression included atrial fibrillation, unexplained syncope, non-sustained ventricular tachycardia, ICD implantation and stroke/thromboembolism. Cardiovascular mortality and all-cause mortality were included as surrogate endpoints comprising end-stage HF, MVA and stroke- or other thromboembolism-related mortality; cardiovascular mortality outcomes including intervention- and myocardial infarction-related mortality were likewise included but were specified in the Supplemental Table 9.

Statistical analysis

Missing summary data were calculated in line with the Cochrane handbook.[4] Means were calculated from medians in accordance with Luo et al. (2018) and Wan et al. (2014),[5, 6] using an online calculator (available on

http://www.math.hkbu.edu.hk/~tongt/papers/median2mean.html, last accessed 19-10-2021). Outcomes of left ventricular outflow tract (LVOT) obstruction, left ventricular ejection fraction (LVEF) and left ventricular fractional shortening using cut-off values greater/smaller than a certain value were combined with outcomes using cut-off values greater/smaller than or equal to the same cut-off value.

Analyses performed on mathematically transformed biomarker values (e.g. log transformed) or categorised according to quantiles or hypothesis- or data-driven cut-off values were pooled separately. As event rates in HCM are low and inclusion of the surrogate endpoints may heavily contribute to the total number of observed events, outcomes encompassing surrogate endpoints were analysed separately from outcomes without surrogate endpoints. Results from multivariable analyses reporting aHR were only pooled for models adjusting for the same variables, with particular attention given to assessments of MVA adjusting to current or previous established risk factors, including the 2011 ACCF/AHA guideline SCD risk factors,[7] the 2014 ESC HCM-risk SCD calculator[8] and the 2019 Enhanced ACC/AHA strategy.[9]

Supplemental references

S1. Aizawa Y, Tanimoto Y, Hirata Y, Fujisawa T, Fukuoka R, Nakajima K, et al. Incidence, Clinical Characteristics, and Long-term Outcome of the Dilated Phase of Hypertrophic Cardiomyopathy. The Keio journal of medicine. 2019;68(4):87-94.

S2. Ayça B, Sahin I, Kucuk SH, Akin F, Kafadar D, Avşar M, et al. Increased Transforming Growth Factor-β Levels Associated With Cardiac Adverse Events in Hypertrophic Cardiomyopathy. Clinical cardiology. 2015;38(6):371-7.

S3. Begue C, Morner S, Brito D, Hengstenberg C, Cleland JGF, Arbustini E, et al. Mid-regional proatrial natriuretic peptide for predicting prognosis in hypertrophic cardiomyopathy. Heart rhythm. 2020;106(3):196-202.

S4. Bi X, Yang C, Song Y, et al. Matrix Metalloproteinases Increase Because of Hypoperfusion in Obstructive Hypertrophic Cardiomyopathy. Ann Thorac Surg. 2021;111(3):915-22.

S5. Coats CJ, Gallagher MJ, Foley M, O'Mahony C, Critoph C, Gimeno J, et al. Relation between serum N-terminal pro-brain natriuretic peptide and prognosis in patients with hypertrophic cardiomyopathy. European heart journal. 2013;34(32):2529-37.

S6. D'Amato R, Tomberli B, Castelli G, Spoladore R, Girolami F, Fornaro A, et al. Prognostic value of N-terminal pro-brain natriuretic Peptide in outpatients with hypertrophic cardiomyopathy. The American journal of cardiology. 2013;112(8):1190-6.

S7. Ekizler FA, Cay S, Acar B, Tak BT, Kafes H, Ozeke O, et al. Monocyte to high-density lipoprotein cholesterol ratio predicts adverse cardiac events in patients with hypertrophic cardiomyopathy. Biomarkers in medicine. 2019;13(14):1175-86.

S8. Gastl M, Gruner C, Labucay K, Gotschy A, Von Spiczak J, Polacin M, et al. Cardiovascular magnetic resonance T2* mapping for the assessment of cardiovascular events in hypertrophic cardiomyopathy. Open heart. 2020;7(1):e001152.

S9. Geske JB, McKie PM, Ommen SR, Sorajja P. B-type natriuretic peptide and survival in hypertrophic cardiomyopathy. Journal of the American College of Cardiology. 2013;61(24):2456-60.

S10. Gommans DHF, Cramer GE, Fouraux MA, et al. Usefulness of High-Sensitivity Cardiac Troponin T to Predict Long-Term Outcome in Patients with Hypertrophic Cardiomyopathy. Am J Cardiol. 2021;152:120-4.

S11. Hamada M, Shigematsu Y, Ohtani T, Ikeda S. Elevated Cardiac Enzymes in Hypertrophic Cardiomyopathy Patients With Heart Failure - A 20-Year Prospective Follow-up Study. Circulation journal : official journal of the Japanese Circulation Society. 2016;80(1):218-26.

S12. Hasler S, Manka R, Greutmann M, Gämperli O, Schmied C, Tanner FC, et al. Elevated high-sensitivity troponin T levels are associated with adverse cardiac remodelling and myocardial fibrosis in hypertrophic cardiomyopathy. Swiss medical weekly. 2016;146:w14285.

S13. Hu DJ, Xu J, Du W, Zhang JX, Zhong M, Zhou YN. Cardiac magnetic resonance and galectin-3 level as predictors of prognostic outcomes for non-ischemic cardiomyopathy patients. The international journal of cardiovascular imaging. 2016;32(12):1725-33.

S14. Imazu M, Fukuda H, Kanzaki H, et al. Plasma indoxyl sulfate levels predict cardiovascular events in patients with mild chronic heart failure. Sci Rep. 2020;10(1):16528.

S15. Kitaoka H, Hitomi N, Yabe T, Furuno T, Doi YL. Cardiovascular events and plasma atrial natriuretic peptide level in patients with hypertrophic cardiomyopathy. The American journal of cardiology. 2001;87(11):1318-20.

S16. Kitaoka H, Kubo T, Okawa M, Hayato K, Yamasaki N, Matsumura Y, et al. Impact of metalloproteinases on left ventricular remodeling and heart failure events in patients with hypertrophic cardiomyopathy. Circulation journal : official journal of the Japanese Circulation Society. 2010;74(6):1191-6.

S17. Kitaoka H, Kubo T, Okawa M, Takenaka N, Sakamoto C, Baba Y, et al. Tissue doppler imaging and plasma BNP levels to assess the prognosis in patients with hypertrophic cardiomyopathy. Journal of the American Society of Echocardiography : official publication of the American Society of Echocardiography. 2011;24(9):1020-5.

S18. Kitaoka H, Kubo T, Baba Y, Yamasaki N, Matsumura Y, Furuno T, et al. Serum tenascin-C levels as a prognostic biomarker of heart failure events in patients with hypertrophic cardiomyopathy. Journal of cardiology. 2012;59(2):209-14.

S19. Kubo T, Kitaoka H, Okawa M, Yamanaka S, Hirota T, Baba Y, et al. Combined measurements of cardiac troponin I and brain natriuretic peptide are useful for predicting adverse outcomes in hypertrophic cardiomyopathy. Circulation journal : official journal of the Japanese Circulation Society. 2011;75(4):919-26.

S20. Kubo T, Kitaoka H, Yamanaka S, Hirota T, Baba Y, Hayashi K, et al. Significance of high-sensitivity cardiac troponin T in hypertrophic cardiomyopathy. Journal of the American College of Cardiology. 2013;62(14):1252-9.

S21. Kubo T, Ochi Y, Baba Y, et al. Elevation of high-sensitivity cardiac troponin T and left ventricular remodelling in hypertrophic cardiomyopathy. ESC Heart Fail. 2020;7(6):3593-600.

S22. Maczynska-Mazuruk R, Malek LA, Klopotowski M, Kukula K, Dabrowski M, Baranowski R, et al. Serum Nterminal pro-brain natriuretic peptide as a prognostic marker in patients with hypertrophic cardiomyopathy. Kardiologia polska. 2019;77(5):571-3.

S23. Minami Y, Haruki S, Kanbayashi K, Maeda R, Itani R, Hagiwara N. B-type natriuretic peptide and risk of sudden death in patients with hypertrophic cardiomyopathy. Heart rhythm. 2018;15(10):1484-90.

S24. Miyaji Y, Iwanaga Y, Nakamura T, Yasuda M, Kawamura T, Miyazaki S. Interrelationship between the Myocardial Mass, Fibrosis, BNP, and Clinical Outcomes in Hypertrophic Cardiomyopathy. Internal medicine (Tokyo, Japan). 2016;55(10):1261-8.

S25. Murakami K, Shigematsu Y, Hamada M, Higaki J. Insulin resistance in patients with hypertrophic cardiomyopathy. Circulation journal : official journal of the Japanese Circulation Society. 2004;68(7):650-5.

S26. Mutlu B, Bayrak F, Kahveci G, Degertekin M, Eroglu E, Basaran Y. Usefulness of N-terminal pro-B-type natriuretic peptide to predict clinical course in patients with hypertrophic cardiomyopathy. The American journal of cardiology. 2006;98(11):1504-6.

S27. Özyılmaz S, Satılmışoğlu MH, Gül M, Uyarel H, Serdar OA. Evaluation of the association between serum uric acid level and the predicted risk score of sudden cardiac death in five years in patients with hypertrophic cardiomyopathy. Turk Kardiyoloji Dernegi arsivi : Turk Kardiyoloji Derneginin yayin organidir. 2018;46(2):111-20.

S28. Pieroni M, Bellocci F, Sanna T, Verardo R, Ierardi C, Maseri A, et al. Increased brain natriuretic peptide secretion is a marker of disease progression in nonobstructive hypertrophic cardiomyopathy. Journal of cardiac failure. 2007;13(5):380-8.

S29. Sahin I, Gungor B, Ozkaynak B, Uzun F, Kucuk SH, Avci, II, et al. Higher copeptin levels are associated with worse outcome in patients with hypertrophic cardiomyopathy. Clinical cardiology. 2017;40(1):32-7.

S30. Scott MKD, Quinn K, Li Q, Carroll R, Warsinske H, Vallania F, et al. Increased monocyte count as a cellular biomarker for poor outcomes in fibrotic diseases: a retrospective, multicentre cohort study. The Lancet Respiratory medicine. 2019;7(6):497-508.

S31. Shirotani S, Minami Y, Saito C, et al. B-type natriuretic peptide and outcome in patients with apical hypertrophic cardiomyopathy. J Cardiol. 2020;76(4):357-63.

S32. Siriwardena M, Bagai A, Delgado D, Ross HJ, Rozenblyum E, Rakowski H, et al. Prognostic Implications of Point-of-Care and Serial B-type Natriuretic Peptide Levels in Patients With Hypertrophic Cardiomyopathy. The American journal of cardiology. 2018;122(8):1421-8.

S33. Song C, Wang S, Guo Y, Zheng X, Lu J, Fang X, et al. Preoperative NT-proBNP Predicts Midterm Outcome After Septal Myectomy. Journal of the American Heart Association. 2019;8(4):e011075.

S34. Song B, Yao B, Dang H, et al. Soluble ST2, Galectin-3 and clinical prognosis of patients with hypertrophic cardiomyopathy undergoing ventricular septal myectomy: a correlation analysis. Cardiovasc Diagn Ther. 2020;10(2):145-52.

S35. Sugiura Y, Morimoto R, Aoki S, Yamaguchi S, Haga T, Kuwayama T, et al. Prognostic impact of mitral L-wave in patients with hypertrophic cardiomyopathy without risk factors for sudden cardiac death. 2019;34(12):2002-10.

S36. Wang Y, Tang Y, Zou Y, Wang D, Zhu L, Tian T, et al. Plasma level of big endothelin-1 predicts the prognosis in patients with hypertrophic cardiomyopathy. International journal of cardiology. 2017;243:283-9.

S37. Wang Z, Xu Y, Liao H, Chen X, He S. U-shaped association between serum uric acid concentration and mortality in hypertrophic cardiomyopathy patients. Upsala journal of medical sciences. 2020;125(1):44-51.

S38. Wang S, Cui H, Ji K, et al. Impact of type 2 diabetes mellitus on mid-term mortality for hypertrophic cardiomyopathy patients who underwent septal myectomy. Cardiovasc Diabetol. 2020;19(1):64.

S39. Wang S, Song C, Cui H, et al. Red blood cell distribution width is associated with adverse cardiovascular events after septal myectomy. J Thorac Dis. 2021;13(2):582-91.

S40. Wang Z, Zhao L, He S. Prognostic nutritional index and the risk of mortality in patients with hypertrophic cardiomyopathy. Int J Cardiol. 2021;331:152-7.

S41. Yang HJ, Liu X, Qu C, et al. Usefulness of Red Blood Cell Distribution Width to Predict Heart Failure Hospitalization in Patients with Hypertrophic Cardiomyopathy. Int Heart J. 2018;59(4):779-85.

S42. Yildiz SS, Sahin I, Cetinkal G, Aksan G, Kucuk SH, Keskin K, et al. Usefulness of Serum Omentin-1 Levels for the Prediction of Adverse Cardiac Events in Patients with Hypertrophic Cardiomyopathy. Medical principles and practice : international journal of the Kuwait University, Health Science Centre. 2018;27(2):107-14.

S43. Yoshihisa A, Yokokawa T, Ichijo Y, et al. Soluble Neprilysin - Cardiac Function and Outcome in Hypertrophic Cardiomyopathy. Circ Rep. 2019;1(6):261-7.

S44. Zen K, Irie H, Doue T, Takamiya M, Yamano T, Sawada T, et al. Analysis of circulating apoptosis mediators and proinflammatory cytokines in patients with idiopathic hypertrophic cardiomyopathy: comparison between nonobstructive and dilated-phase hypertrophic cardiomyopathy. International heart journal. 2005;46(2):231-44.

S45. Zhang K, Meng X, Wang W, Zheng J, An S, Wang S, et al. Prognostic Value of Free Triiodothyronine Level in Patients With Hypertrophic Obstructive Cardiomyopathy. The Journal of clinical endocrinology and metabolism. 2018;103(3):1198-205.

S46. Zhu L, Wang J, Wang Y, Jia L, Sun K, Wang H, et al. Plasma Uric Acid as a Prognostic Marker in Patients With Hypertrophic Cardiomyopathy. The Canadian journal of cardiology. 2015;31(10):1252-8.

S47. Zhu L, Zou Y, Wang Y, Luo X, Sun K, Wang H, et al. Prognostic Significance of Plasma High-Sensitivity C-Reactive Protein in Patients With Hypertrophic Cardiomyopathy. Journal of the American Heart Association. 2017;6(2).

Supplemental Tables

Supplemental Table 1: Search strategy

	Search #1 (broad)		
	Pubmed	**Embase**	**Cochrane library**
Domain	("Cardiomyopathy, Hypertrophic"[mesh] OR “hypertrophic cardiomyopathy”[tiab] OR “hypertrophic cardiomyopathies”[tiab] OR “hypertrophic nonobstructive cardiomyopathy”[tiab] OR “HCM”[tiab] OR "CMH"[tiab] OR “hypertrophic nonobstructive cardiomyopathies”[tiab] OR "HNOCM"[tiab] OR "NOHCM"[tiab] OR "HNCM"[tiab] OR "NOCM"[tiab] OR “hypertrophic obstructive cardiomyopathy”[tiab] OR “hypertrophic obstructive cardiomyopathies”[tiab] OR “HOCM”[tiab] OR “asymmetric septal hypertrophy”[tiab] OR "ASH"[tiab] OR “hypertrophic subaortic stenosis”[tiab] OR “IHSS”[tiab])	(‘Cardiomyopathy, Hypertrophic’/exp OR ‘hypertrophic cardiomyopathy’:ti,ab OR ‘hypertrophic cardiomyopathies’:ti,ab OR ‘hypertrophic nonobstructive cardiomyopathy’:ti,ab OR ‘HCM’:ti,ab OR ‘CMH’:ti,ab OR ‘hypertrophic nonobstructive cardiomyopathies’:ti,ab OR ‘HNOCM’:ti,ab OR ‘NOHCM’:ti,ab OR ‘HNCM’:ti,ab OR ‘NOCM’:ti,ab OR ‘hypertrophic obstructive cardiomyopathy’:ti,ab OR ‘hypertrophic obstructive cardiomyopathies’:ti,ab OR ‘HOCM’:ti,ab OR ‘asymmetric septal hypertrophy’:ti,ab OR ‘ASH’:ti,ab OR ‘hypertrophic subaortic stenosis’:ti,ab OR ‘IHSS’:ti,ab)	(“hypertrophic cardiomyopathy”:ti,ab OR “hypertrophic cardiomyopathies”:ti,ab OR “hypertrophic nonobstructive cardiomyopathy”:ti,ab OR “HCM”:ti,ab OR “CMH”:ti,ab OR “hypertrophic nonobstructive cardiomyopathies”:ti,ab OR “HNOCM”:ti,ab OR “NOHCM”:ti,ab OR “HNCM”:ti,ab OR “NOCM”:ti,ab OR “hypertrophic obstructive cardiomyopathy”:ti,ab OR “hypertrophic obstructive cardiomyopathies”:ti,ab OR “HOCM”:ti,ab OR “asymmetric septal hypertrophy”:ti,ab OR “ASH”:ti,ab OR “hypertrophic subaortic stenosis”:ti,ab OR “IHSS”:ti,ab)
	AND	AND	AND
Determinant	("biomarkers"[mesh] OR "biomarker" OR "biomarkers" OR "biological marker" OR “biological markers” OR "molecular marker" OR "molecular markers" OR "genetic marker" OR "genetic markers" OR "DNA marker" OR "DNA markers" OR "cytogenetic marker" OR "cytogenetic markers" OR "proteome"[mesh] OR "proteomics"[mesh] OR "proteome" OR "proteomes" OR "proteomics" OR "metabolome"[mesh] OR "metabolomics"[mesh] OR "metabolome" OR "metabolomes" OR "metabolic profile" OR "metabolic profiles" OR "metabolomics" OR "metabonomic" OR "metabonomics" OR "biochemical marker" OR "biochemical markers" OR "circulating marker" OR "circulating markers" OR "serum marker" OR "serum markers" OR "plasma marker" OR "plasma markers" OR "laboratory marker" OR "laboratory markers" OR ("plasma" AND "marker") OR ("serum" AND "marker") OR ("blood" AND "marker") OR ("circulating" AND "marker") OR ("laboratory" AND "marker") OR ("plasma level") OR ("serum level") OR ("blood level") OR ("circulating level") OR ("laboratory level") OR ("plasma levels") OR ("serum levels") OR ("blood levels") OR ("circulating levels") OR ("laboratory levels") OR	(‘biomarkers’/exp OR ‘biomarker’ OR ‘biomarkers’ OR ‘biological marker’ OR ‘biological markers’ OR ‘molecular marker’ OR ‘molecular markers’ OR ‘genetic marker’ OR ‘genetic markers’ OR ‘DNA marker’ OR ‘DNA markers’ OR ‘cytogenetic marker’ OR ‘cytogenetic markers’ OR ‘proteome’/exp OR ‘proteomics’/exp OR ‘proteome’ OR ‘proteomes’ OR ‘proteomics’ OR ‘metabolome’/exp OR ‘metabolomics’/exp OR ‘metabolome’ OR ‘metabolomes’ OR ‘metabolic profile’ OR ‘metabolic profiles’ OR ‘metabolomics’ OR ‘metabonomic’ OR ‘metabonomics’ OR ‘biochemical marker’ OR ‘biochemical markers’ OR ‘circulating marker’ OR ‘circulating markers’ OR ‘serum marker’ OR ‘serum markers’ OR ‘plasma marker’ OR ‘plasma markers’ OR ‘laboratory marker’ OR ‘laboratory markers’ OR (‘plasma’ AND ‘marker’) OR (‘serum’ AND ‘marker’) OR (‘blood’ AND ‘marker’) OR (‘circulating’ AND ‘marker’) OR (‘laboratory’ AND ‘marker’) OR ('plasma level') OR ('serum level') OR ('blood level') OR ('circulating level') OR ('laboratory level') OR ('plasma levels') OR ('serum levels') OR ('blood levels') OR ('circulating levels') OR ('laboratory levels') OR ('plasma level') OR ('serum level') OR ('blood	(“biomarker” OR “biomarkers” OR “biological marker” OR “biological markers” OR “molecular marker” OR “molecular markers” OR “genetic marker” OR “genetic markers” OR “DNA marker” OR “DNA markers” OR “cytogenetic marker” OR “cytogenetic markers” OR “proteome” OR “proteomes” OR “proteomics” OR “metabolome” OR “metabolomes” OR “metabolic profile” OR “metabolic profiles” OR “metabolomics” OR “metabonomic” OR “metabonomics” OR “biochemical marker” OR “biochemical markers” OR “circulating marker” OR “circulating markers” OR “serum marker” OR “serum markers” OR “plasma marker” OR “plasma markers” OR “laboratory marker” OR “laboratory markers” OR (“plasma” AND “marker”) OR (“serum” AND “marker”) OR (“blood” AND “marker”) OR (“circulating” AND “marker”) OR (“laboratory” AND “marker”) OR OR ("plasma level") OR ("serum level") OR ("blood level") OR ("circulating level") OR ("laboratory level") OR ("plasma levels") OR ("serum levels") OR ("blood levels") OR ("circulating levels") OR ("laboratory levels") OR ("plasma level") OR ("serum level") OR ("blood level") OR ("circulating level") OR ("laboratory level") OR ("plasma levels") OR ("serum levels") OR

	("plasma level") OR ("serum level") OR ("blood level") OR ("circulating level") OR ("laboratory level") OR ("plasma levels") OR ("serum levels") OR ("blood levels") OR ("circulating levels") OR ("laboratory levels") OR ("plasma concentration") OR ("serum concentration") OR ("blood concentration") OR ("plasma concentrations") OR ("serum concentrations") OR ("blood concentrations") OR ("circulating plasma") OR ("circulating serum"))	level') OR ('circulating level') OR ('laboratory level') OR ('plasma levels') OR ('serum levels') OR ('blood levels') OR ('circulating levels') OR ('laboratory levels') OR ('plasma concentration') OR ('serum concentration') OR ('blood concentration') OR ('plasma concentrations') OR ('serum concentrations') OR ('blood concentrations') OR ('circulating plasma') OR ('circulating serum'))	("blood levels") OR ("circulating levels") OR ("laboratory levels") OR ("plasma concentration") OR ("serum concentration") OR ("blood concentration") OR ("plasma concentrations") OR ("serum concentrations") OR ("blood concentrations") OR ("circulating plasma") OR ("circulating serum"))
Filters	-	Sources: Embase Publication types: Article, Review, Letter, Article in Press	-
Records identified (as of 11-10-2021)	1548	587	235
		Search #2 (specific biomarkers)	
	Pubmed	**Embase**	**Cochrane library**
Domain	("Cardiomyopathy, Hypertrophic"[mesh] OR “hypertrophic cardiomyopathy”[tiab] OR “hypertrophic cardiomyopathies”[tiab] OR “hypertrophic nonobstructive cardiomyopathy”[tiab] OR “HCM”[tiab] OR "CMH"[tiab] OR “hypertrophic nonobstructive cardiomyopathies”[tiab] OR "HNOCM"[tiab] OR "NOHCM"[tiab] OR "HNCM"[tiab] OR "NOCM"[tiab] OR “hypertrophic obstructive cardiomyopathy”[tiab] OR “hypertrophic obstructive cardiomyopathies”[tiab] OR “HOCM”[tiab] OR “asymmetric septal hypertrophy”[tiab] OR “hypertrophic subaortic stenosis”[tiab] OR “IHSS”[tiab]) AND	(‘Cardiomyopathy, Hypertrophic’/exp OR ‘hypertrophic cardiomyopathy’:ti,ab OR ‘hypertrophic cardiomyopathies’:ti,ab OR ‘hypertrophic nonobstructive cardiomyopathy’:ti,ab OR ‘HCM’:ti,ab OR ‘CMH’:ti,ab OR ‘hypertrophic nonobstructive cardiomyopathies’:ti,ab OR ‘HNOCM’:ti,ab OR ‘NOHCM’:ti,ab OR ‘HNCM’:ti,ab OR ‘NOCM’:ti,ab OR ‘hypertrophic obstructive cardiomyopathy’:ti,ab OR ‘hypertrophic obstructive cardiomyopathies’:ti,ab OR ‘HOCM’:ti,ab OR ‘asymmetric septal hypertrophy’:ti,ab OR ‘hypertrophic subaortic stenosis’:ti,ab OR ‘IHSS’:ti,ab) AND	(“hypertrophic cardiomyopathy”:ti,ab OR “hypertrophic cardiomyopathies”:ti,ab OR “hypertrophic nonobstructive cardiomyopathy”:ti,ab OR “HCM”:ti,ab OR “CMH”:ti,ab OR “hypertrophic nonobstructive cardiomyopathies”:ti,ab OR “HNOCM”:ti,ab OR “NOHCM”:ti,ab OR “HNCM”:ti,ab OR “NOCM”:ti,ab OR “hypertrophic obstructive cardiomyopathy”:ti,ab OR “hypertrophic obstructive cardiomyopathies”:ti,ab OR “HOCM”:ti,ab OR “asymmetric septal hypertrophy”:ti,ab OR “hypertrophic subaortic stenosis”:ti,ab OR “IHSS”:ti,ab) AND
Determinant	("BNP" OR "proBNP" OR "NTproBNP" OR "BNPT" OR "brain natriuretic peptide" OR "ventricular natriuretic peptide" OR "B-type natriuretic peptide" OR "TnI" OR "cTnI" OR "hscTnI" OR "TnT" OR "cTnT" OR "hscTnT" OR "Tn" OR "cTn" OR "hscTn" OR "troponin I" OR "troponin T" OR "ANP" OR "ANF" OR "proANP" OR "MRproANP" OR "atrial natriuretic peptide" OR "atrial natriuretic factor" OR "CA-125" OR "CA125" OR "cancer antigen 125" OR "carcinoma antigen 125" OR "carbohydrate antigen 125" OR "mucin16" OR "mucin 16" OR "MUC16" OR "CK" OR "CPK" OR "CKM" OR "CKMM" OR "CKMM1" OR "CKMM2" OR "CKMM3" OR "CKMB" OR "Creatine kinase" OR "Creatinekinase" OR "creatine	('BNP' OR 'proBNP' OR 'NTproBNP' OR 'BNPT' OR 'brain natriuretic peptide' OR 'ventricular natriuretic peptide' OR 'B-type natriuretic peptide' OR 'TnI' OR 'cTnI' OR 'hscTnI' OR 'TnT' OR 'cTnT' OR 'hscTnT' OR 'Tn' OR 'cTn' OR 'hscTn' OR 'troponin I' OR 'troponin T' OR 'ANP' OR 'ANF' OR 'proANP' OR 'MRproANP' OR 'atrial natriuretic peptide' OR 'atrial natriuretic factor' OR 'CA-125' OR 'CA125' OR 'cancer antigen 125' OR 'carcinoma antigen 125' OR 'carbohydrate antigen 125' OR 'mucin16' OR 'mucin 16' OR 'MUC16' OR 'CK' OR 'CPK' OR 'CKM' OR 'CKMM' OR 'CKMM1' OR 'CKMM2' OR 'CKMM3' OR 'CKMB' OR 'Creatine kinase' OR 'Creatinekinase' OR 'creatine phosphokinase' OR	("BNP" OR "proBNP" OR "NTproBNP" OR "BNPT" OR "brain natriuretic peptide" OR "ventricular natriuretic peptide" OR "B-type natriuretic peptide" OR "TnI" OR "cTnI" OR "hscTnI" OR "TnT" OR "cTnT" OR "hscTnT" OR "Tn" OR "cTn" OR "hscTn" OR "troponin I" OR "troponin T" OR "ANP" OR "ANF" OR "proANP" OR "MRproANP" OR "atrial natriuretic peptide" OR "atrial natriuretic factor" OR "CA-125" OR "CA125" OR "cancer antigen 125" OR "carcinoma antigen 125" OR "carbohydrate antigen 125" OR "mucin16" OR "mucin 16" OR "MUC16" OR "CK" OR "CPK" OR "CKM" OR "CKMM" OR "CKMM1" OR "CKMM2" OR "CKMM3" OR "CKMB" OR "Creatine kinase" OR "Creatinekinase" OR "creatine

phosphokinase" OR "creatinephosphokinase" OR "phosphocreatine kinase" OR "phosphocreatinekinase" OR "gal-3" OR "galectin" OR "galectin3" OR "LDH" OR "LDH1" OR "Lactate dehydrogenase" OR "Lactatedehydrogenase" OR "myoglobin" OR "sIL" OR "soluble IL" OR "soluble interleukin" OR "ST2" OR "sST2" OR "IL1RL1" OR "suppression of tumorigenicity 2" OR "Interleukin 1 receptor-like 1" OR "TN-C" OR "TNC" OR "Tenascin-C" OR "UA" OR "SUA" OR "Uric acid" OR "BAP" OR "BSAP" OR "BALP" OR "Bone ALP" OR "Bone-specific alkaline phosphatase" OR "Bone alkaline phosphatase" OR "Alkaline Phosphatase Bone" OR "skeletal alkaline phosphatase" OR "CITP" OR "CTX" OR "C-terminal telopeptide" OR "carboxy-terminal collagen crosslinks" OR "MMP" OR "MMP1" OR "MMP2" OR "MMP3" OR "MMP9" OR "MMPs" OR "Matrix metallopeptidase" OR "Matrix metallopeptidases" OR "Matrix metalloproteinase" OR "Matrix metalloproteinases" OR "matrixin" OR "matrixins" OR "OPN" OR "BSP-1" OR "BSP1" OR "BNSP" OR "ETA-1" OR "ETA1" OR "SPP-1" OR "SPP1" OR "Osteopontin" OR "bone sialoprotein I" OR "early T-lymphocyte activation" OR "secreted phosphoprotein 1" OR "POSTN" OR "OSF-2" OR "OSF2" OR "periostin" OR "osteoblast-specific factor" OR "PICP" OR "P1CP" OR "CICP" OR "C1CP" OR "Procollagen type I carboxy-terminal propeptide" OR "Procollagen type I C-terminal propeptide" OR "C-terminal Propeptide of Type-I Procollagen" OR "type I procollagen C-terminal propeptide" OR "collagen I carboxy terminal propeptide" OR "TIMP" OR "TIMP1" OR "TIMP2" OR "TIMP4" OR "TIMP metallopeptidase inhibitor" OR "tissue inhibitor of metalloproteinases" OR "ALCAR" OR "ALC" OR "Acetylcarnitine" OR "Acetyl-L-carnitine" OR "GBP-28" OR "apM1" OR "AdipoQ" OR "Acrp30" OR "Adiponectin" OR "Carnitine" OR "Glucose" OR "HDL" OR "High-density lipoprotein" OR "hFABP" OR "FABP" OR "Heart-type fatty acid binding protein" OR "LDL" OR "Low-density lipoprotein" OR "NEFA" OR "non-esterified fatty acids" OR "non-esterified fatty acid" OR "free fatty acid" OR "Omentin-1" OR "Triglyceride" OR "Triglycerides" OR

'creatinephosphokinase' OR 'phosphocreatine kinase' OR 'phosphocreatinekinase' OR 'gal-3' OR 'galectin' OR 'galectin3' OR 'LDH' OR 'LDH1' OR 'Lactate dehydrogenase' OR 'Lactatedehydrogenase' OR 'myoglobin' OR 'sIL' OR 'soluble IL' OR 'soluble interleukin' OR 'ST2' OR 'sST2' OR 'IL1RL1' OR 'suppression of tumorigenicity 2' OR 'Interleukin 1 receptor-like 1' OR 'TN-C' OR 'TNC' OR 'Tenascin-C' OR 'UA' OR 'SUA' OR 'Uric acid' OR 'BAP' OR 'BSAP' OR 'BALP' OR 'Bone ALP' OR 'Bone-specific alkaline phosphatase' OR 'Bone alkaline phosphatase' OR 'Alkaline Phosphatase Bone' OR 'skeletal alkaline phosphatase' OR 'CITP' OR 'CTX' OR 'C-terminal telopeptide' OR 'carboxy-terminal collagen crosslinks' OR 'MMP' OR 'MMP1' OR 'MMP2' OR 'MMP3' OR 'MMP9' OR 'MMPs' OR 'Matrix metallopeptidase' OR 'Matrix metallopeptidases' OR 'Matrix metalloproteinase' OR 'Matrix metalloproteinases' OR 'matrixin' OR 'matrixins' OR 'OPN' OR 'BSP-1' OR 'BSP1' OR 'BNSP' OR 'ETA-1' OR 'ETA1' OR 'SPP-1' OR 'SPP1' OR 'Osteopontin' OR 'bone sialoprotein I' OR 'early T-lymphocyte activation' OR 'secreted phosphoprotein 1' OR 'POSTN' OR 'OSF-2' OR 'OSF2' OR 'periostin' OR 'osteoblast-specific factor' OR 'PICP' OR 'P1CP' OR 'CICP' OR 'C1CP' OR 'Procollagen type I carboxy-terminal propeptide' OR 'Procollagen type I C-terminal propeptide' OR 'C-terminal Propeptide of Type-I Procollagen' OR 'type I procollagen C-terminal propeptide' OR 'collagen I carboxy terminal propeptide' OR 'TIMP' OR 'TIMP1' OR 'TIMP2' OR 'TIMP4' OR 'TIMP metallopeptidase inhibitor' OR 'tissue inhibitor of metalloproteinases' OR 'ALCAR' OR 'ALC' OR 'Acetylcarnitine' OR 'Acetyl-L-carnitine' OR 'GBP-28' OR 'apM1' OR 'AdipoQ' OR 'Acrp30' OR 'Adiponectin' OR 'Carnitine' OR 'Glucose' OR 'HDL' OR 'High-density lipoprotein' OR 'hFABP' OR 'FABP' OR 'Heart-type fatty acid binding protein' OR 'LDL' OR 'Low-density lipoprotein' OR 'NEFA' OR 'non-esterified fatty acids' OR 'non-esterified fatty acid' OR 'free fatty acid' OR 'Omentin-1' OR 'Triglyceride' OR 'Triglycerides' OR 'triacylglycerol' OR 'triacylglyceride' OR 'triacylglycerides' OR 'TG' OR

phosphokinase" OR "creatinephosphokinase" OR "phosphocreatine kinase" OR "phosphocreatinekinase" OR "gal-3" OR "galectin" OR "galectin3" OR "LDH" OR "LDH1" OR "Lactate dehydrogenase" OR "Lactatedehydrogenase" OR "myoglobin" OR "sIL" OR "soluble IL" OR "soluble interleukin" OR "ST2" OR "sST2" OR "IL1RL1" OR "suppression of tumorigenicity 2" OR "Interleukin 1 receptor-like 1" OR "TN-C" OR "TNC" OR "Tenascin-C" OR "UA" OR "SUA" OR "Uric acid" OR "BAP" OR "BSAP" OR "BALP" OR "Bone ALP" OR "Bone-specific alkaline phosphatase" OR "Bone alkaline phosphatase" OR "Alkaline Phosphatase Bone" OR "skeletal alkaline phosphatase" OR "CITP" OR "CTX" OR "C-terminal telopeptide" OR "carboxy-terminal collagen crosslinks" OR "MMP" OR "MMP1" OR "MMP2" OR "MMP3" OR "MMP9" OR "MMPs" OR "Matrix metallopeptidase" OR "Matrix metallopeptidases" OR "Matrix metalloproteinase" OR "Matrix metalloproteinases" OR "matrixin" OR "matrixins" OR "OPN" OR "BSP-1" OR "BSP1" OR "BNSP" OR "ETA-1" OR "ETA1" OR "SPP-1" OR "SPP1" OR "Osteopontin" OR "bone sialoprotein I" OR "early T-lymphocyte activation" OR "secreted phosphoprotein 1" OR "POSTN" OR "OSF-2" OR "OSF2" OR "periostin" OR "osteoblast-specific factor" OR "PICP" OR "P1CP" OR "CICP" OR "C1CP" OR "Procollagen type I carboxy-terminal propeptide" OR "Procollagen type I C-terminal propeptide" OR "C-terminal Propeptide of Type-I Procollagen" OR "type I procollagen C-terminal propeptide" OR "collagen I carboxy terminal propeptide" OR "TIMP" OR "TIMP1" OR "TIMP2" OR "TIMP4" OR "TIMP metallopeptidase inhibitor" OR "tissue inhibitor of metalloproteinases" OR "ALCAR" OR "ALC" OR "Acetylcarnitine" OR "Acetyl-L-carnitine" OR "GBP-28" OR "apM1" OR "AdipoQ" OR "Acrp30" OR "Adiponectin" OR "Carnitine" OR "Glucose" OR "HDL" OR "High-density lipoprotein" OR "hFABP" OR "FABP" OR "Heart-type fatty acid binding protein" OR "LDL" OR "Low-density lipoprotein" OR "NEFA" OR "non-esterified fatty acids" OR "non-esterified fatty acid" OR "free fatty acid" OR "Omentin-1" OR "Triglyceride" OR "Triglycerides" OR

	"triacylglycerol" OR "triacylglyceride" OR "triacylglycerides" OR "TG" OR "thromboglobulin" OR "thromboglobulin" OR "d-dimer" OR "platelet" OR "platelets" OR "GMP-140" OR "CD62P" OR "PADGEM" OR "P-selectin" OR "Prothrombin fragment" OR "sCD40L" OR "soluble CD40 ligand" OR "TAT" OR "thrombin-antithrombin complex" OR "vWF" OR "von Willebrand factor" OR "cRNA" OR "telomere" OR "TSH" OR "Thyroid-stimulating hormone" OR "thyrotropin" OR "thyrotropic hormone" OR "T4" OR "FT4" OR "TT4" OR "thyroxine" OR "T3" OR "FT3" OR "TT3" OR "Triiodothyronine" OR "PGF" OR "prostaglandin" OR "ADM" OR "adrenomedullin" OR "ET-1" OR "endothelin" OR "L-arginine" OR "dimethylarginine" OR "urotensin" OR "CT-1" OR "cardiotrophin" OR "CRP" OR "hsCRP" OR "C-reactive protein" OR "IL-6" OR "IL6" OR "Interleukin" OR "sFas" OR "soluble Fas" OR "sFasL" OR "FasL" OR "Fas ligand" OR "TNF" OR "TNFA" OR "Tumor necrosis factor" OR "cachexin" OR "cachectin" OR "autoantibody" OR "autoantibodies" OR "Adrenaline" OR "adrenalin" OR "Epinephrine" OR "CT-proAVP" OR "Copeptin" OR "DLIS" OR "Digoxin-like immunoreactive substance" OR "Digitalis-like immunoreactive substances" OR "noradrenaline" OR "noradrenalin" OR "Norepinephrine" OR "Creatinine" OR "eGFR" OR "GFR" OR "Glomerular filtration rate" OR "erythrocyte creatine" OR "Hb" OR "Haemoglobin" OR "Hemoglobin" OR "HbA1C" OR "A1C" OR "GDF15" OR "GDF-15" OR "Growth differentiation factor-15" OR "MIC-1" OR "MIC1" OR "Macrophage inhibitory cytokine-1" OR "IGF" OR "IGF1" OR "IGF1A" OR "IGFI" OR "MGF" OR "Insulin-like growth factor 1" OR "somatomedin C" OR "IGFBP" OR "IGFBP1" OR "IGFBP3" OR "IBP" OR "IBP1" OR "IBP3" OR "Insulin-like growth factor-binding protein" OR "placental protein 12" OR "VEGF" OR "VPF" OR "vascular endothelial growth factor" OR "vascular permeability factor") AND ("plasma" OR "serum" OR "blood" OR "circulating")	'thromboglobulin' OR 'thromboglobulin' OR 'd-dimer' OR 'platelet' OR 'platelets' OR 'GMP-140' OR 'CD62P' OR 'PADGEM' OR 'P-selectin' OR 'Prothrombin fragment' OR 'sCD40L' OR 'soluble CD40 ligand' OR 'TAT' OR 'thrombin-antithrombin complex' OR 'vWF' OR 'von Willebrand factor' OR 'cRNA' OR 'telomere' OR 'TSH' OR 'Thyroid-stimulating hormone' OR 'thyrotropin' OR 'thyrotropic hormone' OR 'T4' OR 'FT4' OR 'TT4' OR 'thyroxine' OR 'T3' OR 'FT3' OR 'TT3' OR 'Triiodothyronine' OR 'PGF' OR 'prostaglandin' OR 'ADM' OR 'adrenomedullin' OR 'ET-1' OR 'endothelin' OR 'L-arginine' OR 'dimethylarginine' OR 'urotensin' OR 'CT-1' OR 'cardiotrophin' OR 'CRP' OR 'hsCRP' OR 'C-reactive protein' OR 'IL-6' OR 'IL6' OR 'Interleukin' OR 'sFas' OR 'soluble Fas' OR 'sFasL' OR 'FasL' OR 'Fas ligand' OR 'TNF' OR 'TNFA' OR 'Tumor necrosis factor' OR 'cachexin' OR 'cachectin' OR 'autoantibody' OR 'autoantibodies' OR 'Adrenaline' OR 'adrenalin' OR 'Epinephrine' OR 'CT-proAVP' OR 'Copeptin' OR 'DLIS' OR 'Digoxin-like immunoreactive substance' OR 'Digitalis-like immunoreactive substances' OR 'noradrenaline' OR 'noradrenalin' OR 'Norepinephrine' OR 'Creatinine' OR 'eGFR' OR 'GFR' OR 'Glomerular filtration rate' OR 'erythrocyte creatine' OR 'Hb' OR 'Haemoglobin' OR 'Hemoglobin' OR 'HbA1C' OR 'A1C' OR 'GDF15' OR 'GDF-15' OR 'Growth differentiation factor-15' OR 'MIC-1' OR 'MIC1' OR 'Macrophage inhibitory cytokine-1' OR 'IGF' OR 'IGF1' OR 'IGF1A' OR 'IGFI' OR 'MGF' OR 'Insulin-like growth factor 1' OR 'somatomedin C' OR 'IGFBP' OR 'IGFBP1' OR 'IGFBP3' OR 'IBP' OR 'IBP1' OR 'IBP3' OR 'Insulin-like growth factor-binding protein' OR 'placental protein 12' OR 'VEGF' OR 'VPF' OR 'vascular endothelial growth factor' OR 'vascular permeability factor') AND ('plasma' OR 'serum' OR 'blood' OR 'circulating')	"triacylglycerol" OR "triacylglyceride" OR "triacylglycerides" OR "TG" OR "thromboglobulin" OR "thromboglobulin" OR "d-dimer" OR "platelet" OR "platelets" OR "GMP-140" OR "CD62P" OR "PADGEM" OR "P-selectin" OR "Prothrombin fragment" OR "sCD40L" OR "soluble CD40 ligand" OR "TAT" OR "thrombin-antithrombin complex" OR "vWF" OR "von Willebrand factor" OR "cRNA" OR "telomere" OR "TSH" OR "Thyroid-stimulating hormone" OR "thyrotropin" OR "thyrotropic hormone" OR "T4" OR "FT4" OR "TT4" OR "thyroxine" OR "T3" OR "FT3" OR "TT3" OR "Triiodothyronine" OR "PGF" OR "prostaglandin" OR "ADM" OR "adrenomedullin" OR "ET-1" OR "endothelin" OR "L-arginine" OR "dimethylarginine" OR "urotensin" OR "CT-1" OR "cardiotrophin" OR "CRP" OR "hsCRP" OR "C-reactive protein" OR "IL-6" OR "IL6" OR "Interleukin" OR "sFas" OR "soluble Fas" OR "sFasL" OR "FasL" OR "Fas ligand" OR "TNF" OR "TNFA" OR "Tumor necrosis factor" OR "cachexin" OR "cachectin" OR "autoantibody" OR "autoantibodies" OR "Adrenaline" OR "adrenalin" OR "Epinephrine" OR "CT-proAVP" OR "Copeptin" OR "DLIS" OR "Digoxin-like immunoreactive substance" OR "Digitalis-like immunoreactive substances" OR "noradrenaline" OR "noradrenalin" OR "Norepinephrine" OR "Creatinine" OR "eGFR" OR "GFR" OR "Glomerular filtration rate" OR "erythrocyte creatine" OR "Hb" OR "Haemoglobin" OR "Hemoglobin" OR "HbA1C" OR "A1C" OR "GDF15" OR "GDF-15" OR "Growth differentiation factor-15" OR "MIC-1" OR "MIC1" OR "Macrophage inhibitory cytokine-1" OR "IGF" OR "IGF1" OR "IGF1A" OR "IGFI" OR "MGF" OR "Insulin-like growth factor 1" OR "somatomedin C" OR "IGFBP" OR "IGFBP1" OR "IGFBP3" OR "IBP" OR "IBP1" OR "IBP3" OR "Insulin-like growth factor-binding protein" OR "placental protein 12" OR "VEGF" OR "VPF" OR "vascular endothelial growth factor" OR "vascular permeability factor") AND ("plasma" OR "serum" OR "blood" OR "circulating")
Filters	-	Sources: Embase Publication types: Article, Review, Letter, Article in Press	-
Records identified (as of 11-10-2021)	1106	415	123

Supplemental Table 2: Reported in-/exclusion criteria and biomarker platforms

Study (reference)	Domain	Biomarker(s)	Platform
Aizawa 2019 (S1)	HCM (LVH in absence of other cardiac or noncardiac diseases) Exclusion criteria: -	BNP Creatinine eGFR	NA NA NA
Ayca 2015 (S2)	HCM (MWT ≥15 mm not explained by causes of secondary LVH) Exclusion criteria: hypertension grade ≥2, with end-organ damage or treated with multiple drugs, acute inflammation, chronic liver or kidney failure, hematological disease, current use of angiotensin-converting enzyme inhibitors, and use of angiotensin-converting enzyme inhibitors, angiotensin II type 1 receptor blockers or aldosterone inhibitors prior to HCM diagnosis	BNP TGF-β1	[Cobas 8000 Modular ECLIA (Roche Diagnostics, NA, UK)] (S33) Human TGF-β Cytoset ELISA (BioSource, Camarillo, CA, USA)
Begue 2020 (S3)	HCM (MWT >15 mm, >13 mm or Romhilt-Estes score ≥4/pathological Q waves/significant ST-T changes for relatives, in absence of causes of secondary LVH) Exclusion criteria: -	MR-proANP NT-proBNP	KRYPTOR (BRAHMS, Hennigsdorf, Germany) E170 ECLIA (Roche Diagnostics, NA)
Bi 2021 (S4)	HOCM (MWT ≥15 mm or ≥13 mm in relatives unexplained by other conditions, LVOT gradient ≥40 mmHg at rest or ≥50 mmHg during provocation, undergoing myectomy) Exclusion criteria: coronary artery disease, severe valvular disease, stage 3-5 chronic kidney disease, connective tissue disease and osteoarthropathy	PICP ICTP MMP-2 TIMP1	E-EL-H0196c ELISA (Elabscience, Wuhan, China) CSB-E10363h ELISA (Cusabio, Wuhan, China) MMP200 ELISA (R&D Systems, Minneapolis, Minnesota, USA) DTL100 ELISA (R&D systems, Minneapolis, Minnesota, USA)
Coats 2013 (S5)	HCM (MWT >15 mm unexplained by abnormal loading conditions, McKenna 1997 criteria in first degree relatives) Exclusion criteria: Friedrich's ataxia, Noonan syndrome and metabolic disorders	NT-proBNP	E170 ECLIA (Roche, NA)
D'Amato 2013 (S6)	HCM (LVH without dilatation, unexplained by causes of secondary LVH) Exclusion criteria: LVEF <50% at baseline and known CAD	NT-proBNP	Cobas E601 Elecsys proBNP II (Roche Diagnostics, Indianapolis, Indiana, USA)
Ekizler 2019 (S7)	HCM (MWT ≥15 mm not explained by causes of secondary LVH) Exclusion criteria: metabolic diseases, syndromic phenocopies	Creatinine Glucose Haemoglobin HDL-cholesterol hs-CRP Lymphocyte count Monocyte count Neutrophil count Platelet count TSH Uric acid White blood cell count	NA NA XE-1200 (Sysmex, Kobe, Japan) Selective solubilisation method (Determiner L HDL, Kyowa Medex, Tokyo, Japan) NA XE-1200 (Sysmex) XE-1200 (Sysmex) XE-1200 (Sysmex) XE-1200 (Sysmex) NA NA XE-1200 (Sysmex)
Gastl 2020 (S8)	HCM (MWT ≥15 mm, unexplained by other causes of LVH) Exclusion criteria: CAD, exclusion criteria for CMR	[hs-]cTnT (S10)	[Cobas 8000 ECLIA (Roche Diagnostics, Rotkreuz, Switzerland)] (S10)
Geske 2013 (S9)	HCM (myocardial hypertrophy in absence of cardiac or systemic causes)	BNP	DXI 800 Fluorescence immunoassay (Biosite Diagnostics, San Diego, CA, USA/Beckman Coulter, NA)
Gommans 2021 (S10)	HCM (MWT ≥15 mm or ≥13 mm in pathogenic variant carriers) Exclusion criteria: cardiac hypertrophy with known cause other than sarcomeric gene mutation, coronary artery disease, stroke, septal reduction therapy, contraindication for exercise testing	hs-cTnT	Elecsys 2010 (Roche Diagnostics, Almere, the Netherlands)
Hamada 2016 (S11)	HCM (1995 WHO/ISFC classification) Exclusion criteria: HF, AF, LV end diastolic diameter ≥50 mm, LVFS <30%, notched R wave on ECG, creatinine ≥1.2 mg/dL	CK-MB	Cellulose acetate electrophoresis (Cellogel) (Chemetron, Milan, Italy) and fluorescence (NA)
Hasler 2016 (S12)	HCM (MWT ≥15 mm unexplained by other cardiac or systemic causes of LVH) Exclusion criteria: CAD, creatinine >260 nmol/mL	hs-cTnT	Cobas 8000 ECLIA (Roche Diagnostics, Rotkreuz, Switzerland)

Study (reference)	Domain	Biomarker(s)	Platform
Hu 2016 (S13)	HCM (MWT ≥15 mm) Exclusion criteria: ischemic cardiomyopathy, myocarditis, hypertensive heart disease, valvular heart disease, secondary cardiomyopathy, chronic or acute inflammatory disease, hematological malignancies, severe chronic kidney disease	Galectin-3	ELISA (Waltham, NA, USA)
Imazu 2020 (S14)	HCM after HF-hopsitalisation (criteria NA) Exclusion criteria: unstable congestive HF, eGFR <40 ml/min/1.73m^2, NYHA class IV	BNP eGFR Indoxyl sulfate	NA NA High-performance liquid chromatography (Shimadzu, Kyoto, Japan)
Kitaoka 2001 (S15)	HCM (LVH, non-dilated, in absence of other cardiac disease or hypertension) Exclusion criteria: systolic dysfunction, severe mitral regurgitation due to chordal rupture and <6 months follow-up duration	ANP	Shionoria ANP (Shionogi, Osaka, Japan)
Kitaoka 2010 (S16)	HCM (MWT ≥15 mm, unexplained by other cardiac or systemic causes of LVH) Exclusion criteria: myocardial infarction due to CAD, inflammatory disease, malignancy and creatinine ≥2.0 mg/dL	BNP MMP-2 MMP-9 TIMP1	TOSOH II (TOSOH, Tokyo, Japan) One-step sandwich ELISA (clone 42-5D11) (Daiichi Fine Chemical, Toyama, Japan) One-step sandwich ELISA (clone 56-2A4) (Daiichi Fine Chemical) One-step sandwich ELISA (clone 147-6D11) (Daiichi Fine Chemical)
Kitaoka 2011 (S17)	HCM (MWT ≥15 mm, non-dilated, unexplained by other cardiac or systemic causes of LVH) Exclusion criteria: LVFS <25%, time interval between BNP measurement and echocardiography >28 days, creatinine ≥2.0 mg/dL, myocardial infarction	BNP	TOSOH II (TOSOH, Tokyo, Japan)
Kitaoka 2012 (S18)	HCM (MWT ≥15 mm, non-dilated, unexplained by other cardiac or systemic causes of LVH) Exclusion criteria: CAD, lung, liver and collagen disease, malignancy and creatinine ≥2.0 mg/dL	BNP eGFR Tenascin-C	TOSOH II (TOSOH, Tokyo, Japan) NA One-step sandwich ELISA (Immuno-Biological Laboratories, Takasaki, Japan)
Kubo 2011 (S19)	HCM (unexplained MWT ≥15 mm) Exclusion criteria: CAD and creatinine ≥3.0 mg/dL	BNP cTnI	RIA (Shionogi, Osaka, Japan) or [TOSOH II] (TOSOH, Tokyo, Japan) (S14) Unicel DxI 800 Access Immunoassay (Beckman Coultier, NA)
Kubo 2013 (S20)	HCM (unexplained MWT ≥15 mm) Exclusion criteria: CAD and creatinine ≥3.0 mg/dL	hs-cTnT	Elecsys immunoassay (Roche Diagnostics, Rotkreuz, Switzerland)
Kubo 2020 (S21)	HCM (unexplained MWT ≥15 mm) Exclusion criteria: LVEF <50%, CAD and creatinine ≥3.0 mg/dL		
Maczynska-Mazuruk 2019 (S22)	HCM (MWT ≥15 mm in the absence of abnormal loading conditions or in accordance with the criteria for first-degree relatives, criteria not defined) Exclusion criteria: -	NT-proBNP	Cobas e601 sandwich ECLIA (Roche Diagnostics, Indianapolis, IN, USA)
Minami 2018 (S23)	HCM (LVH, non-dilated, unexplained by other cardiac or systemic causes of LVH) Exclusion criteria: missing BNP measurement	BNP	AIA-1800 FEIA (TOSOH, Tokyo, Japan)
Miyaji 2016 (S24)	HCM (Assymmetric/focal LVH ≥15 mm unexplained by other cardiac or systemic causes of LVH) Exclusion criteria: -	BNP	FIA (LSI Medience, NA, Japan)
Murakami 2004 (S25)	HCM (LVH, non-dilated, in the absence of cardiac or systemic causes of LVH) Exclusion criteria: diabetes	Glucose Immunoreactive insulin (together used to calculate insulin resistance)	Glucose oxidase method RIA (Abbott, NA, Japan)
Mutlu 2006 (S26)	HCM (MWT ≥15 mm, non-dilated, unexplained by other cardiac or systemic causes of LVH) Exclusion criteria: concommitant cardiovascular, pulmonary or renal conditions	NT-proBNP	NA (Roche Diagnostics, Mannheim, Germany)
Ozyilmaz 2018 (S27)	HCM (MWT ≥15 mm or ≥13 mm combined with family history, positive gene mutation or electrocardiogam abnormalities, not explained by abnormal loading conditions) Exclusion criteria: prior septal reduction therapy, CAD, prior myocardial infarction, aortic valve stenosis, uncontrolled hypertension, renal failure, malignancy, hypoxanthine-guanine phosphoribosyl transferase enzyme deficiency, gout, psoriasis, drugs affecting uric acid levels (salicates, thiazide diuretics, allopurinol, cytotoxic drugs, ethambutol, pyrazinamide), alcohol use, missing UA measurement, lost to follow-up	Uric acid	Cobas 6000 Enzymatic colorimetric method (Roche Diagnostics, Basel, Switzerland)

Study (reference)	Domain	Biomarker(s)	Platform
Pieroni 2007 (S28)	HCM (LVH, non-dilated, in absence of other cardiac or systemic causes of LVH) Exclusion criteria: HOCM (LVOT gradient >30 mmHg), Fabry disease	BNP	Shionoria BNP (Shionogi, NA, Japan)
Sahin 2017 (S29)	HCM (MWT ≥15 mm in absence of any other cause of LVH) Exclusion criteria: acute congestive HF, LVEF <50%, CAD, eGFR <30 mL/min/m2 and blood pressure >140/90 mmHg	Copeptin NT-proBNP	ELISA (Wuhan ElAab Science, Wuhan China) Elecsys 2010 ECLIA (Roche Diagnostics, São Paulo, Brazil)
Scott 2019 (S30)	HCM (criteria not provided) Exclusion criteria: -	Monocyte count	NA
Shirotani 2020 (S31)	Apical HCM (MWT ≥15 mm unexplained by other cardiac or systemic causes of LVH, predominantly involving the LV apex [basal wall thickness <15mm]) Exclusion criteria: no BNP measurement at baseline	BNP	AIA-1800 (TOSOH Inc., Tokyo, Japan)
Siriwardena 2018 (S32)	HCM (MWT ≥15 mm, non-dilated, unexplained by other cardiac or systemic causes of LVH) Exclusion criteria: -	BNP	POC Triage BNP (Biosite Diagnostics/Alere, San Diego, Ca, USA)
Song 2019 (S33)	HOCM (MWT ≥15 mm in absence of other cardiac or systemic causes of LVH, myectomy) Exclusion criteria: -	NT-proBNP	ECLIA (Roche, Mannheim, Germany)
Song 2020 (S34)	HOCM (MWT ≥15 mm in absence of other cardiac or systemic causes of LVH, symptomatic LVOT gradient >50 mmHg at rest undergoing myectomy) Exclusion criteria: prior cardiac surgery, alcohol septal ablation, severe liver and renal insufficiency	Galectin-3 Soluble ST2	ELISA x28408/x29505 (NA, Xihui, China) ELISA x28408/x29505 (NA, Xihui, China)
Sugiura 2019 (S35)	HCM (MWT ≥15 mm in the absence of other cardiac or systemic causes of LVH) with >1 echocardiography at >2 week interval Exclusion criteria: 2011 ACCF/AHA SCD risk factors, syndromic or metabolic cause of LVH, CAD requiring therapy, primary valvular heart disease, pacemaker and ICD	BNP	NA
Wang 2017 (S36)	HCM (MWT ≥15 mm, or ≥13 mm in relatives, unexplained by other cardiac or systemic causes of LVH) Exclusion criteria: CAD	Big endothelin-1	BI-20082H (Biomedica Medizinprodukte, NA, Austria)
Wang 2020a (S37)	HCM (MWT ≥15 mm unexplained by abnormal loading conditions) Exclusion criteria: metabolic disease, syndromic causes of LVH, loss to follow-up, incomplete biochemistry data	eGFR Glucose LDL-cholesterol Triglycerides Uric acid	NA NA NA NA Uricase method (NA)
Wang 2020b (S38)	HOCM (MWT ≥15 mm in absence of other cardiac or systemic causes of LVH, symptomatic LVOT gradient >50 mmHg at rest or with provocation undergoing myectomy, with diabetes according to medical chart) Exclusion criteria: -	eGFR NT-proBNP	NA NA
Wang 2021a (S39)	HOCM (MWT ≥15 mm unexplained by abnormal loading conditions, symptomatic LVOT gradient >50 mmHg at rest or with provocation undergoing myectomy)	Haemoglobin LDL NT-proBNP Red blood cell distribution width	XT1800i hematology analyser (Sysmex, NA, Japan) NA NA XT1800i hematology analyser (Sysmex, NA, Japan)
Wang 2021b (S40)	HCM (MWT ≥15 mm, unexplained by loading conditions) Exclusion criteria: lost to follow-up, <18 years, no albumin level or lymphocyte count available, known malignancy, active inflammation, end stage renal or liver failure	Prognostic nutritional index (10 × albumin [g/dL] + 5 × lymphocyte count [nL])	XN-9000 (Sysmex, Kobe, Japan) & Cobas 8000 (Roche, NA)
Yang 2018 (S41)	HCM (according to 2011 ACCF/AHA guidelines) Exclusion criteria: prior HF, loss to follow-up	BNP Haemoglobin Mean corpuscular volume Red blood cell distribution width	Dimension-RXL (Dupont, NA, USA) XN-9000 (Sysmex, Kobe, Japan) XN-9000 (Sysmex, Kobe, Japan) XN-9000 (Sysmex, Kobe, Japan)
Yildiz 2018 (S42)	HCM (MWT ≥15 mm, unexplained by loading conditions) Exclusion criteria: NYHA class III-IV, CAD, valvular heart disease, blood pressure >140/90	BNP Creatinine Intelectin-1	Cobas 8000 Modular ECLIA (Roche Diagnostics, NA, UK) Hitachi 7600 (Hitachi, NA, Japan) ELISA (Sunred Biotechnology, Shanghai, China)

Study (reference)	Domain	Biomarker(s)	Platform
	mmHg, active or chronic inflammatory disease, aortic valve stenosis, eGFR <30 ml/min/[1.73]m^2)		
Yoshihisa 2019 (S43)	HCM (according to 2014 ESC and 2011 ACCF/AHA guidelines)	Soluble neprilysin	ELH-Neprilysin-1 RIA (RayBiotech, Norcross, GA, USA)
Zen 2005 (S44)	HCM (no definition provided) in dilated stage (LVEF ≤50% & LVEDD ≥55 mm) Exclusion criteria: HOCM (not defined), significant concomitant disease (including pulmonary disease, malignancy, autoimmune disorders, neurodegenerative disorders, diabetes mellitus, thyroid disease or viral infection)	Soluble Fas	Sandwich ELISA (Medical & Biological Laboratories, Nagoya, Japan)
Zhang 2018 (S45)	HOCM (MWT ≥15 mm or ≥13 mm in relatives or combined with non-cardiac symptoms and signs, electrogram abnormalities, laboratory tests and multimodality cardiac imaging in absence of any other cause of LVH; LVOT gradient ≥30 mmHg in rest/after provocation) Exclusion criteria: age <16 years, incomplete information medical history/clinical data/thyroid function, FT3 >4.09 pg/mL, history of thyroid disorder, hypothalamic-pituitary disorder and autoimmune disease, drugs affecting thyroid function (including corticosteroids and amiodarone)	Creatinine Free T3 Free T4 NT-proBNP TSH	NA Immulite 2000 RIA (Siemens, NA, Germany) Immulite 2000 RIA (Siemens) [ECLIA (Roche, Mannheim, Germany)] (S29) Immulite 2000 RIA (Siemens)
Zhu 2015 (S46)	HCM (MWT ≥15 mm in absence of other cardiac or systemic causes of LVH) Exclusion criteria: loss to follow-up, chronic kidney disease	Uric acid	Technicon SMA 12-60 (Technicon, NA, NA)
Zhu 2017 (S47)	HCM (MWT ≥15 mm unexplained by abnormal loading conditions) Exclusion criteria: autoimmune or inflammatory disease, active infection and malignancy	hs-CRP	Immunoturbidimetric Assay (Orion Diagnostica, NA, Finland)

Table specifying the reported in- and exclusion criteria and platforms used for biomarker analyses in individual studies. Diagnostic criteria for HCM used as inclusion criteria are provided as reported by the included studies. Cited diagnostic guidelines are provided if no diagnostic criteria were specified (1995 WHO/ISFC classification [3], McKenna 1997 criteria for first-degree relatives [10], 2011 ACCF/AHA guidelines [7], 2014 ESC guidelines [1]). Biomarkers and platforms within square brackets were not specified in the study but were reported in another study by the same group (references provided within brackets).

AF, atrial fibrillation; ANP, atrial natriuretic peptide; BNP, ventricular (brain or B-type) natriuretic peptide; CAD, coronary artery disease; CK-MB, creatine (phospho)kinase MB isoform; CLIA, chemiluminescent immunoassay; CRP, C-reactive protein; cTnI/cTnT, cardiac troponin I/T; ECLIA, electro-chemiluminescence immunoassay; eGFR, estimated glomerular filtration rate; EIA, enzyme immunoassay; ELISA, enzyme-linked immunosorbent assay; ICTP, C-terminal telopeptide of type I collagen; FEIA, fluorimetric enzyme-linked immunoassay; FIA, fluorescence immunoassay; HCM, hypertrophic cardiomyopathy; HDL, high-density lipoprotein; HOCM, hypertrophic obstructive cardiomyopathy; HF, heart failure; hs, high-sensitivity; ICD, implantable cardioverter-defibrillator; LDL, low-density lipoprotein; LVEF, left ventricular ejection fraction; LVFS, left ventricular fractional shortening; LVH, left ventricular hypertrophy; LVOT, left ventricular outflow tract; MMP, matrix metallopeptidase (metalloproteinases); MR-proANP, midregional pro-atrial natriuretic peptide; MWT, maximum wall thickness; NA, not available; NT-proBNP, N-terminal prohormone of brain natriuretic peptide; NYHA, New York Heart Association; PICP, Propeptide of procollagen type I; RIA, radioimmunoassay; TIMP, TIMP (tissue inhibitor of metalloproteinases) metallopeptidase inhibitor 1; TGF, transforming growth factor; TSH, thyroid-stimulating hormone; T3, triiodothyronine; T4, thyroxine.

Supplemental Table 3: Excluded studies & reason for exclusion

First author & year	Reference	Reason
Adalsteinsdottir 2020	Adalsteinsdottir B, Burke M, Maron BJ, et al. Hypertrophic cardiomyopathy in myosin-binding protein C (MYBPC3) Icelandic founder mutation carriers. Open Heart. 2020;7(1):e001220.	Wrong design
Akita 2018	Akita K, Tsuruta H, Yuasa S, et al. Prognostic significance of repeated brain natriuretic peptide measurements after percutaneous transluminal septal myocardial ablation in patients with drug-refractory hypertrophic obstructive cardiomyopathy. Open Heart. 2018;5(1):e000786.	Wrong design
Arteaga 2005	Arteaga E, Araujo AQ, Buck P, et al. Plasma amino-terminal pro-B-type natriuretic peptide quantification in hypertrophic cardiomyopathy. Am Heart J. 2005;150(6):1228-32.	Wrong design
Autore 1988	Autore C, Fiorito S, Pelliccia A, et al. Antimitochondrial autoantibodies in myocardial hypertrophy: comparison between hypertrophic cardiomyopathy, hypertensive heart disease, and athlete's heart. Am Heart J. 1988;116(2 Pt 1):496-500.	Wrong outcome
Bayrak 2005	Bayrak F, Kahveci G, Fotbolcu H, et al. The role of plasma N-terminal pro B-type natriuretic peptide to predict the clinical course of patients with hypertrophic cardiomyopathy. Turk Kardiyol Dernegi Ars. 2005;33(2):77-83.	Wrong language
Biagini 2005	Biagini E, Coccolo F, Ferlito M, et al. Dilated-hypokinetic evolution of hypertrophic cardiomyopathy: prevalence, incidence, risk factors, and prognostic implications in pediatric and adult patients. J Am Coll Cardiol. 2005;46(8):1543-50.	Wrong determinant
Binder 2007	Binder J, Ommen SR, Chen HH, et al. Usefulness of brain natriuretic peptide levels in the clinical evaluation of patients with hypertrophic cardiomyopathy. Am J Cardiol. 2007;100(4):712-4.	Wrong design
Blackshear 2016	Blackshear JL, Kusumoto H, Safford RE, et al. Usefulness of Von Willebrand Factor Activity Indexes to Predict Therapeutic Response in Hypertrophic Cardiomyopathy. Am J Cardiol. 2016;117(3):436-42.	Wrong design
Blackshear 2018	Blackshear JL, Safford RE, Thomas CS, et al. Platelet Function Analyzer 100 and Brain Natriuretic Peptide as Biomarkers in Obstructive Hypertrophic Cardiomyopathy. Am J Cardiol. 2018;121(6):768-74.	Wrong design
Boudonas 1994	Boudonas G, Boura P, Lefkos N, et al. A possible role for autoantibodies in left ventricular hypertrophy. Cardiology. 1994;84(4-5):278-83.	Wrong outcome
Briguori 2001	Briguori C, Betocchi S, Manganelli F, et al. Determinants and clinical significance of natriuretic peptides and hypertrophic cardiomyopathy. Eur Heart J. 2001;22(15):1328-36.	Wrong design
Brito 2004	Brito D, Matias JS, Sargento L, et al. Plasma N-terminal pro-brain natriuretic peptide: a marker of left ventricular hypertrophy in hypertrophic cardiomyopathy. Rev Port Cardiol. 2004;23(12):1557-82.	Full text not available
Buzas 2004	Buzás K, Megyeri K, Hõgye M, et al. Comparative study of the roles of cytokines and apoptosis in dilated and hypertrophic cardiomyopathies. Eur Cytokine Netw. 2004;15(1):53-9.	Full text not available
Cambronero 2010	Cambronero F, Vilchez JA, Garcia-Honrubia A, et al. Plasma levels of von Willebrand factor are increased in patients with hypertrophic cardiomyopathy. Thromb Res. 2010;126(1):e46-50.	Wrong design
Captur 2020	Captur G, Heywood WE, Coats C, et al. Identification of a Multiplex Biomarker Panel for Hypertrophic Cardiomyopathy Using Quantitative Proteomics and Machine Learning. Mol Cell Proteomics. 2020;19(1):114-27.	Wrong design
Chatterjee 2018	Chatterjee S, de Gonzalo-Calvo D, Derda AA, et al. Leukocyte telomere length correlates with hypertrophic cardiomyopathy severity. Sci Rep. 2018;8(1):11227.	Wrong design
Chen 2013	Chen S, Yuan J, Qiao S, et al. A high-normal thyrotropin level is associated with the severity of left ventricular diastolic dysfunction in patients with hypertrophic cardiomyopathy. Kardiol Pol. 2013;71(2):143-51.	Wrong design

First author & year	Reference	Reason
Cramer 2014	Cramer G, Bakker J, Gommans F, et al. Relation of highly sensitive cardiac troponin T in hypertrophic cardiomyopathy to left ventricular mass and cardiovascular risk. Am J Cardiol. 2014;113(7):1240-5.	Wrong design
Cramer 2020	Cramer GE, Gommans DHF, Dieker HJ, et al. Exercise and myocardial injury in hypertrophic cardiomyopathy. Heart. 2020.	Wrong design
Cui 2017	Cui H, Wu X, Wang S, et al. Time and age dependent decrease of NT-proBNP after septal myectomy in hypertrophic obstructive cardiomyopathy. Clin Chem Lab Med. 2017;55(10):1614-20.	Wrong outcome
Derchi 1992	Derchi G, Bellone P, Chiarella F, et al. Plasma levels of atrial natriuretic peptide in hypertrophic cardiomyopathy. Am J Cardiol. 1992;70(18):1502-4.	Wrong design
Derda 2015	Derda AA, Thum S, Lorenzen JM, et al. Blood-based microRNA signatures differentiate various forms of cardiac hypertrophy. Int J Cardiol. 2015;196:115-22.	Wrong outcome
Dimitrow 2007	Dimitrow PP, Undas A, Bober M, et al. Plasma biomarkers of endothelial dysfunction in patients with hypertrophic cardiomyopathy. Pharmacol Rep. 2007;59(6):715-20.	Wrong design
Dimitrow 2008	Dimitrow PP, Undas A, Bober M, et al. Obstructive hypertrophic cardiomyopathy is associated with enhanced thrombin generation and platelet activation. Heart. 2008;94(6):e21.	Wrong design
Dimitrow 2009a	Dimitrow PP, Undas A. Letter by Dimitrow and Undas regarding article, "Functional impairment of von Willebrand factor in hypertrophic cardiomyopathy: relation to rest and exercise obstruction". Circulation. 2009;119(23):e590; author reply e1.	Wrong publication type
Dimitrow 2009b	Dimitrow PP, Undas A, Wolkow P, et al. Enhanced oxidative stress in hypertrophic cardiomyopathy. Pharmacol Rep. 2009;61(3):491-5.	Wrong design
Dimitrow 2011	Dimitrow PP, Undas A, Cheng TO. Exercise modulates circulating adipokine levels in hypertrophic cardiomyopathy. Pol Arch Med Wewn. 2011;121(11):384-90.	Wrong outcome
Dimitrow 2014	Dimitrow PP, Cheng TO. Hypertrophic cardiomyopathy: genotype-positive, phenotype-"almost" negative. Diagnostic role of exercise in provocation of left ventricular outflow tract gradient. Int J Cardiol. 2014;177(3):736-7.	Wrong publication type
Duan 2020	Duan X, Liu R, Luo XL, et al. The relationship between β(1) -adrenergic and M(2) -muscarinic receptor autoantibodies and hypertrophic cardiomyopathy. Exp Physiol. 2020;105(3):522-30.	Wrong design
Efthimiadis 2007	Efthimiadis GK, Hitoglou-Makedou A, Giannakoulas G, et al. Clinical significance of N-terminal-probrain natriuretic peptide in hypertrophic cardiomyopathy. Heart Vessels. 2007;22(5):322-7.	Wrong design
Elliott 2008	Elliott P. Relevance of platelet activation in obstructive hypertrophic cardiomyopathy. Heart. 2008;94(6):688-9.	Wrong publication type
Elmas 2011	Elmas E, Doesch C, Fluechter S, et al. Midregional pro-atrial natriuretic peptide: a novel marker of myocardial fibrosis in patients with hypertrophic cardiomyopathy. Int J Cardiovasc Imaging. 2011;27(4):547-56.	Wrong outcome
Ettinger 1971	Ettinger PO, Oldewurtel HA, Dzindzio B, et al. Glucose intolerance in nonischemic cardiac disease. Role of cardiac output and adrenergic function. Circulation. 1971;43(6):809-23.	Wrong population
Fahy 1996	Fahy GJ, McCreery CJ, O'Sullivan F, et al. Plasma atrial natriuretic peptide is elevated in patients with hypertrophic cardiomyopathy. Int J Cardiol. 1996;55(2):149-55.	Wrong design
Fang 2013	Fang L, Beale A, Ellims AH, et al. Associations between fibrocytes and postcontrast myocardial T1 times in hypertrophic cardiomyopathy. J Am Heart Assoc. 2013;2(5):e000270.	Wrong outcome
Fang 2017	Fang L, Ellims AH, Beale AL, et al. Systemic inflammation is associated with myocardial fibrosis, diastolic dysfunction, and cardiac hypertrophy in patients with hypertrophic cardiomyopathy. Am J Transl Res. 2017;9(11):5063-73.	Wrong design
Fassbach 2005	Fassbach M, Schwartzkopff B. Elevated serum markers for collagen synthesis in patients with hypertrophic cardiomyopathy and diastolic dysfunction. Z Kardiol. 2005;94(5):328-35.	Wrong design

First author & year	Reference	Reason
Fernandes 2012	Fernandes F, Arteaga-Fernandez E, Antunes Mde O, et al. Plasma pro-B-type natriuretic peptide testing as a screening method for hypertrophic cardiomyopathy. J Card Fail. 2012;18(7):564-8.	Wrong population
Franz 2009	Franz M, Berndt A, Altendorf-Hofmann A, et al. Serum levels of large tenascin-C variants, matrix metalloproteinase-9, and tissue inhibitors of matrix metalloproteinases in concentric versus eccentric left ventricular hypertrophy. Eur J Heart Fail. 2009;11(11):1057-62.	Wrong population
Fu 1994	Fu ML, Hoebeke J, Matsui S, et al. Autoantibodies against cardiac G-protein-coupled receptors define different populations with cardiomyopathies but not with hypertension. Clin Immunol Immunopathol. 1994;72(1):15-20.	Wrong outcome
Fucikova 2016	Fucikova A, Lenco J, Tambor V, et al. Plasma concentration of fibronectin is decreased in patients with hypertrophic cardiomyopathy. Clin Chim Acta. 2016;463:62-6.	Wrong outcome
Gawor 2017	Gawor M, Spiewak M, Janas J, et al. The usefulness of sST2 and galectin-3 as novel biomarkers for better risk stratification in hypertrophic cardiomyopathy. Kardiol Pol. 2017;75(10):997-1004.	Wrong design
Gawor 2018	Gawor M, Śpiewak M, Kubik A, et al. Circulating biomarkers of hypertrophy and fibrosis in patients with hypertrophic cardiomyopathy assessed by cardiac magnetic resonance. Biomarkers. 2018;23(7):676-82.	Wrong outcome
Gebka 2020	Gebka A, Rajtar-Salwa R, Dziewierz A, et al. Serum N-terminal pro-B-type natriuretic peptide levels are associated with left atrial dilation, resting left ventricular outflow tract gradient, and pulmonary hypertension in patients with hypertrophic cardiomyopathy. Postepy Kardiol Interwencyjnej. 2020;16(1):97-101.	Wrong design
Gommans 2017	Gommans DF, Cramer GE, Bakker J, et al. High T2-weighted signal intensity is associated with elevated troponin T in hypertrophic cardiomyopathy. Heart. 2017;103(4):293-9.	Wrong outcome
Gommans 2018a	Gommans DHF, Cramer GE, Bakker J, et al. High T2-weighted signal intensity for risk prediction of sudden cardiac death in hypertrophic cardiomyopathy. Int J Cardiovasc Imaging. 2018;34(1):113-20.	Wrong outcome
Gommans 2018b	Gommans DHF, Cramer GE, Fouraux MA, et al. Prediction of Extensive Myocardial Fibrosis in Nonhigh Risk Patients With Hypertrophic Cardiomyopathy. Am J Cardiol. 2018;122(3):483-9.	Wrong outcome
Goto 2013	Goto D, Kinugawa S, Hamaguchi S, et al. Clinical characteristics and outcomes of dilated phase of hypertrophic cardiomyopathy: report from the registry data in Japan. J Cardiol. 2013;61(1):65-70.	Wrong determinant
Gregor 1987	Gregor P, Jira M, Raska I, et al. Autoantibodies in hypertrophic cardiomyopathy and their clinical significance. Eur Heart J. 1987;8(7):773-8.	Wrong design
Gregor 1990	Gregor P, Zichová M, Preucil P, et al. [Are there abnormalities of parathormone, calcium and phosphorus metabolism in hypertrophic cardiomyopathy?]. Vnitr Lek. 1990;36(5):446-52.	Full text not available
Guo 2017	Guo Y, Wu X, Zheng X, et al. Usefulness of Preoperative Transforming Growth Factor-Beta to Predict New Onset Atrial Fibrillation After Surgical Ventricular Septal Myectomy in Patients With Obstructive Hypertrophic Cardiomyopathy. Am J Cardiol. 2017;120(1):118-23.	Wrong outcome
Hamada 1991	Hamada M, Ohtani T, Sekiya M, et al. Serum creatine kinase MM isoforms in hypertrophic cardiomyopathy. Clin Sci (Lond). 1991;81(6):723-6.	Wrong design
Hamada 1998	Hamada M, Shigematsu Y, Kawakami H, et al. Increased plasma levels of adrenomedullin in patients with hypertrophic cardiomyopathy: its relation to endothelin-I, natriuretic peptides and noradrenaline. Clin Sci (Lond). 1998;94(1):21-8.	Wrong design
Han 2009	Han ZH, Li Y, Lin J, et al. [Serum troponin I level in patients with hypertrophic cardiomyopathy]. Zhonghua Xin Xue Guan Bing Za Zhi. 2009;37(12):1085-7.	Full text not available
Hasegawa 1993	Hasegawa K, Fujiwara H, Doyama K, et al. Ventricular expression of brain natriuretic peptide in hypertrophic cardiomyopathy. Circulation. 1993;88(2):372-80.	Wrong design

First author & year	Reference	Reason
Hasegawa 1996	Hasegawa K, Fujiwara H, Koshiji M, et al. Endothelin-1 and its receptor in hypertrophic cardiomyopathy. Hypertension. 1996;27(2):259-64.	Wrong design
Hayashi 2000	Hayashi T, Ijiri Y, Toko H, et al. Increased digitalis-like immunoreactive substances in patients with hypertrophic cardiomyopathy. Eur Heart J. 2000;21(4):296-305.	Wrong design
Hernandez-Romero 2015	Hernandez-Romero D, Jover E, Martinez CM, et al. TWEAK and NT-proBNP levels predict exercise capacity in hypertrophic cardiomyopathy. Eur J Clin Invest. 2015;45(2):179-86.	Wrong outcome
Hina 1997	Hina K, Kusachi S, Iwasaki K, et al. Use of serum creatine kinase MM isoforms for predicting the progression of left ventricular dilation in patients with hypertrophic cardiomyopathy. Jpn Circ J. 1997;61(4):315-22.	Wrong design
Hirata 2001	Hirata Y, Matsumoto A, Aoyagi T, et al. Measurement of plasma brain natriuretic peptide level as a guide for cardiac overload. Cardiovasc Res. 2001;51(3):585-91.	Wrong population
Hladij 2017a	Hładij R, Rajtar-Salwa R, Dimitrow PP. Troponin as ischemic biomarker is related with all three echocardiographic risk factors for sudden death in hypertrophic cardiomyopathy (ESC Guidelines 2014). Cardiovasc Ultrasound. 2017;15(1):24.	Wrong outcome
Hladij 2017b	Hładij R, Rajtar-Salwa R, Petkow Dimitrow P. Associaton of elevated troponin levels with increased heart rate and higher frequency of nonsustained ventricular tachycardia in hypertrophic cardiomyopathy. Pol Arch Intern Med. 2017;126(6):445-7.	Wrong outcome
Ho 2010	Ho CY, Lopez B, Coelho-Filho OR, et al. Myocardial fibrosis as an early manifestation of hypertrophic cardiomyopathy. N Engl J Med. 2010;363(6):552-63.	Wrong design
Ho 2013	Ho CY, Abbasi SA, Neilan TG, et al. T1 measurements identify extracellular volume expansion in hypertrophic cardiomyopathy sarcomere mutation carriers with and without left ventricular hypertrophy. Circ Cardiovasc Imaging. 2013;6(3):415-22.	Wrong design
Ho 2016	Ho CY, Cirino AL, Lakdawala NK, et al. Evolution of hypertrophic cardiomyopathy in sarcomere mutation carriers. Heart. 2016;102(22):1805-12.	Wrong population
Ho 2017a	Ho CY, Day SM, Colan SD, et al. The Burden of Early Phenotypes and the Influence of Wall Thickness in Hypertrophic Cardiomyopathy Mutation Carriers: Findings From the HCMNet Study. JAMA Cardiol. 2017;2(4):419-28.	Wrong design
Ho 2017b	Ho JE, Shi L, Day SM, et al. Biomarkers of cardiovascular stress and fibrosis in preclinical hypertrophic cardiomyopathy. Open Heart. 2017;4(2):e000615.	Wrong design
Hogye 2004	Högye M, Mándi Y, Csanády M, et al. Comparison of circulating levels of interleukin-6 and tumor necrosis factor-alpha in hypertrophic cardiomyopathy and in idiopathic dilated cardiomyopathy. Am J Cardiol. 2004;94(2):249-51.	Wrong outcome
Icli 2014	Icli A, Aksoy F, Dogan A, et al. Increased mean platelet volume in hypertrophic cardiomyopathy. Angiology. 2014;65(5):420-4.	Wrong design
Ishii 1997	Ishii Y, Kawashima E, Kawabe J, et al. [Secretion of atrial natriuretic peptide from the left ventricle and heart in patients with hypertrophic cardiomyopathy: relationship with hemodynamic and echocardiographic profiles]. J Cardiol. 1997;30(1):19-28.	Full text not available
Iwasaki 2009	Iwasaki J, Nakamura K, Matsubara H, et al. Relationship between circulating levels of monocyte chemoattractant protein-1 and systolic dysfunction in patients with hypertrophic cardiomyopathy. Cardiovasc Pathol. 2009;18(6):317-22.	Wrong design
Jenab 2014	Jenab Y, Pourjafari M, Darabi F, et al. Prevalence and determinants of elevated high-sensitivity cardiac troponin T in hypertrophic cardiomyopathy. J Cardiol. 2014;63(2):140-4.	Wrong design
Jorgenrud 2015	Jorgenrud B, Jalanko M, Helio T, et al. The Metabolome in Finnish Carriers of the MYBPC3-Q1061X Mutation for Hypertrophic Cardiomyopathy. PLoS One. 2015;10(8):e0134184.	Wrong design

First author & year	Reference	Reason
Jumaah 2017	Jumaah S, Çelekli A, Sucu M. Roles of human urotensin-II, creatine kinase-MB, and uric acid serum levels in hypertrophic cardiomyopathy patients. Anatol J Cardiol. 2017;18:90.	Full text not available
Jumaah 2018	Jumaah S, Celekli A, Sucu M. The role of human urotensin-II in patients with hypertrophic cardiomyopathy. J Immunoassay Immunochem. 2018;39(2):150-62.	Wrong design
Kahveci 2009	Kahveci G, Bayrak F, Mutlu B, et al. Determinants of elevated NT-proBNP levels in patients with hypertrophic cardiomyopathy: an echocardiographic study. Heart Lung Circ. 2009;18(4):266-70.	Wrong design
Kalyva 2016	Kalyva A, Marketou ME, Parthenakis FI, et al. Endothelial progenitor cells as markers of severity in hypertrophic cardiomyopathy. Eur J Heart Fail. 2016;18(2):179-84.	Wrong outcome
Katyshkina 1992	Katyshkina NI, Ovchinikov PL. [Hormonal aspects of the pathogenesis of cardiac arrhythmias in patients with hypertrophic cardiomyopathy]. Kardiologiia. 1992;32(5):8-10.	Full text not available
Kawasaki 2013	Kawasaki T, Sakai C, Harimoto K, et al. Usefulness of high-sensitivity cardiac troponin T and brain natriuretic peptide as biomarkers of myocardial fibrosis in patients with hypertrophic cardiomyopathy. Am J Cardiol. 2013;112(6):867-72.	Wrong outcome
Khan 1999	Khan K, Talwar S. Screening for familial hypertrophic cardiomyopathy using brain natriuretic peptide. Eur Heart J. 1999;20(7):550.	Wrong publication type
Kim 2006	Kim SW, Park SW, Lim SH, et al. Amount of left ventricular hypertrophy determines the plasma N-terminal pro-brain natriuretic peptide level in patients with hypertrophic cardiomyopathy and normal left ventricular ejection fraction. Clin Cardiol. 2006;29(4):155-60.	Wrong design
Kitaoka 2009	Kitaoka H, Kubo T, Okawa M, et al. Utility of tissue Doppler imaging to predict exercise capacity in hypertrophic cardiomyopathy: comparison with B-type natriuretic peptide. J Cardiol. 2009;53(3):361-7.	Wrong design
Kitaoka 2010	Kitaoka H, Kubo T, Okawa M, et al. Plasma adiponectin levels and left ventricular remodeling in hypertrophic cardiomyopathy. Int Heart J. 2010;51(1):51-5.	Wrong design
Kitow 2016	Kitow J, Derda AA, Beermann J, et al. Mitochondrial long noncoding RNAs as blood based biomarkers for cardiac remodeling in patients with hypertrophic cardiomyopathy. Am J Physiol Heart Circ Physiol. 2016;311(3):H707-12.	Wrong design
Kubo 2010	Kubo T, Kitaoka H, Okawa M, et al. Serum cardiac troponin I is related to increased left ventricular wall thickness, left ventricular dysfunction, and male gender in hypertrophic cardiomyopathy. Clin Cardiol. 2010;33(2):E1-7.	Wrong design
Kubo 2016	Kubo T, Okumiya T, Baba Y, et al. Erythrocyte creatine as a marker of intravascular hemolysis due to left ventricular outflow tract obstruction in hypertrophic cardiomyopathy. J Cardiol. 2016;67(3):274-8.	Wrong design
Kuusisto 2012	Kuusisto J, Kärjä V, Sipola P, et al. Low-grade inflammation and the phenotypic expression of myocardial fibrosis in hypertrophic cardiomyopathy. Heart. 2012;98(13):1007-13.	Wrong outcome
Lanser 2020	Lanser L, Nemati N, Seifert M, et al. Inflammation, iron and vitamin D metabolism in different cardiomyopathy aetiologies. Pteridines. 2020;31(1):28-37.	Wrong design
Lee 2020	Lee HJ, Kim HK, Kim M, et al. Clinical impact of atrial fibrillation in a nationwide cohort of hypertrophic cardiomyopathy patients. Ann Transl Med. 2020;8(21):1386.	Wrong outcome
Le Tourneau 2008	Le Tourneau T, Susen S, Caron C, et al. Functional impairment of von Willebrand factor in hypertrophic cardiomyopathy: relation to rest and exercise obstruction. Circulation. 2008;118(15):1550-7.	Wrong design
Li 2021a	Li Y, Liu J, Cao Y, et al. Predictive values of multiple non-invasive markers for myocardial fibrosis in hypertrophic cardiomyopathy patients with preserved ejection fraction. Sci Rep. 2021;11(1):4297.	Wrong outcome

First author & year	Reference	Reason
Li 2021b	Li ZL, He S, Xia CC, et al. Global longitudinal diastolic strain rate as a novel marker for predicting adverse outcomes in hypertrophic cardiomyopathy by cardiac magnetic resonance tissue tracking. Clin Radiol. 2021;76(1):78.e19-78.e25.	Wrong determinant
Liu 2019	Liu L, Wu L, Zheng L, et al. Associations Between Multiple Circulating Biomarkers and the Presence of Atrial Fibrillation in Hypertrophic Cardiomyopathy with or Without Left Ventricular Outflow Tract Obstruction. Int Heart J. 2019;60(2):327-35.	Wrong outcome
Liu 2020a	Liu LM, Shen LS, Liu SY, et al. Correlations between low thyroid function and incidence of atrial fibrillation in hypertrophic obstructive cardiomyopathy. Chronic Dis Transl Med. 2020;6(1):35-45.	Wrong design
Liu 2020b	Liu L, Liu S, Shen L, et al. Correlations between cardiac troponin I and nonsustained ventricular tachycardia in hypertrophic obstructive cardiomyopathy. Clin Cardiol. 2020;43(10):1150-9.	Wrong outcome
Liu 2020c	Liu W, Zhang Y, Liu Y, et al. Assessment of left ventricular systolic function in hypertrophic cardiomyopathy patients with myocardial injury: a study based on layer-specific speckle tracking echocardiaography. Int J Cardiovasc Imaging. 2020;36(11):2129-37.	Wrong outcome
Lombardi 2003	Lombardi R, Betocchi S, Losi MA, et al. Myocardial collagen turnover in hypertrophic cardiomyopathy. Circulation. 2003;108(12):1455-60.	Wrong outcome
Longo 1984	Longo G, Cecchi F, Grossi A, et al. Coagulation and platelet function in hypertrophic cardiomyopathy. Thromb Haemost. 1984;51(2):299.	Full text not available
Mady 2008	Mady C, Fernandes F, Arteaga E, et al. Serum NT pro-BNP: relation to systolic and diastolic function in cardiomyopathies and pericardiopathies. Arq Bras Cardiol. 2008;91(1):46-54.	Wrong design
Magga 2008	Magga J, Sipola P, Vuolteenaho O, et al. Significance of plasma levels of N-terminal Pro-B-type natriuretic peptide on left ventricular remodeling in non-obstructive hypertrophic cardiomyopathy attributable to the Asp175Asn mutation in the alpha-tropomyosin gene. Am J Cardiol. 2008;101(8):1185-90.	Wrong outcome
Magri 2020	Magrì D, Mastromarino V, Gallo G, et al. Risk Stratification in Hypertrophic Cardiomyopathy. Insights from Genetic Analysis and Cardiopulmonary Exercise Testing. J Clin Med. 2020;9(6).	Wrong determinant
Maisch 1980	Maisch B, Berg PA, Kochsiek K. Immunological parameters in patients with congestive cardiomyopathy. Basic Res Cardiol. 1980;75(1):221-2.	Wrong outcome
Maisch 1992	Maisch B. Anticardiac antibodies in hypertrophic cardiomyopathy as a marker of severity. Postgrad Med J. 1992;68 Suppl 1:S29-35.	Full text not available
Mándi 2000	Mándi Y, Hogye M, Talha EM, et al. Cytokine production and antibodies against heat shock protein 60 in cardiomyopathies of different origins. Pathobiology. 2000;68(3):150-8.	Wrong outcome
Marketou 2015	Marketou ME, Parthenakis FI, Kalyva A, et al. Circulating mesenchymal stem cells in patients with hypertrophic cardiomyopathy. Cardiovasc Pathol. 2015;24(3):149-53.	Wrong outcome
Maron 2004	Maron BJ, Tholakanahalli VN, Zenovich AG, et al. Usefulness of B-type natriuretic peptide assay in the assessment of symptomatic state in hypertrophic cardiomyopathy. Circulation. 2004;109(8):984-9.	Wrong design
Matsumori 1994	Matsumori A, Yamada T, Suzuki H, et al. Increased circulating cytokines in patients with myocarditis and cardiomyopathy. Br Heart J. 1994;72(6):561-6.	Wrong outcome
Matsuo 2002	Matsuo S, Nakamura Y, Tsutamoto T, et al. Impairments of myocardial sympathetic activity may reflect the progression of myocardial damage or dysfunction in hypertrophic cardiomyopathy. J Nucl Cardiol. 2002;9(4):407-12.	Wrong outcome
Matsuura 2008	Matsuura H, Murakami T, Hina K, et al. Association of elevated plasma B-type natriuretic peptide levels with paroxysmal atrial fibrillation in patients with nonobstructive hypertrophic cardiomyopathy. Clin Biochem. 2008;41(3):134-9.	Wrong outcome
McGorrian 2013	McGorrian CM, Lyster S, Roy A, et al. Use of a highly-sensitive cardiac troponin I assay in a screening population for hypertrophic cardiomyopathy: a case-referent study. BMC Cardiovasc Disord. 2013;13:70.	Wrong outcome

First author & year	Reference	Reason
Meng 2018	Meng XB, Wang WY, Zhang K, et al. [Value of N-terminal pro-B-type natriuretic peptide on long-term outcome of patients with hypertrophic cardiomyopathy]. Zhonghua Xin Xue Guan Bing Za Zhi. 2018;46(3):192-7.	Full text not available
Meng 2021	Meng Y, Wang S, Liu P, et al. The preoperative glomerular filtration rate predicts new-onset postoperative atrial fibrillation in patients with hypertrophic obstructive cardiomyopathy who undergo isolated septal myectomy. J Thorac Dis. 2021;13(3):1612-23.	Wrong determinant
Monserrat 2011	Monserrat L, Lopez B, Gonzalez A, et al. Cardiotrophin-1 plasma levels are associated with the severity of hypertrophy in hypertrophic cardiomyopathy. Eur Heart J. 2011;32(2):177-83.	Wrong design
Montoro-Garcia 2012	Montoro-Garcia S, Hernandez-Romero D, Jover E, et al. Growth differentiation factor-15, a novel biomarker related with disease severity in patients with hypertrophic cardiomyopathy. Eur J Intern Med. 2012;23(2):169-74.	Wrong design
Moreno 2010	Moreno V, Hernández-Romero D, Vilchez JA, et al. Serum levels of high-sensitivity troponin T: a novel marker for cardiac remodeling in hypertrophic cardiomyopathy. J Card Fail. 2010;16(12):950-6.	Wrong design
Mori 1998	Mori K, Komamura K, Ohta N, et al. [Relationship between apoptosis-related markers and cardiomyopathy]. Rinsho Byori. 1998;46(12):1235-40.	Full text not available
Morioka 2005	Morioka N, Shigematsu Y, Hamada M, et al. Circulating levels of heart-type fatty acid-binding protein and its relation to thallium-201 perfusion defects in patients with hypertrophic cardiomyopathy. Am J Cardiol. 2005;95(11):1334-7.	Wrong design
Munch 2016	Munch J, Avanesov M, Bannas P, et al. Serum Matrix Metalloproteinases as Quantitative Biomarkers for Myocardial Fibrosis and Sudden Cardiac Death Risk Stratification in Patients With Hypertrophic Cardiomyopathy. J Card Fail. 2016;22(10):845-50.	Wrong outcome
Nagata 1985	Nagata S, Park Y, Minamikawa T, et al. Thallium perfusion and cardiac enzyme abnormalities in patients with familial hypertrophic cardiomyopathy. Am Heart J. 1985;109(6):1317-22.	Wrong design
Nakamura 1999a	Nakamura T, Sugihara H, Kinoshita N, et al. Serum carnitine concentrations in patients with idiopathic hypertrophic cardiomyopathy: relationship with impaired myocardial fatty acid metabolism. Clin Sci (Lond). 1999;97(4):493-501.	Wrong design
Nakamura 1999b	Nakamura T, Sugihara H, Inaba T, et al. CD36 deficiency has little influence on the pathophysiology of hypertrophic cardiomyopathy. J Mol Cell Cardiol. 1999;31(6):1253-9.	Wrong design
Nakamura 2002	Nakamura T, Sakamoto K, Yamano T, et al. Increased plasma brain natriuretic peptide level as a guide for silent myocardial ischemia in patients with non-obstructive hypertrophic cardiomyopathy. J Am Coll Cardiol. 2002;39(10):1657-63.	Wrong outcome
Nakamura 2014	Nakamura S, Takano H, Matsuda J, et al. Prognostic values of highly sensitive cardiac troponin T and B-type natriuretic peptide for clinical features in hypertrophic obstructive cardiomyopathy: a cross-sectional study. BMJ Open. 2014;4(9):e005968.	Wrong design
Nct 2017	Nct. Nitrite in Hypertrophic Cardiomyopathy (HCM) Study. https://clinicaltrialsgov/show/NCT03251287. 2017.	Wrong publication type
Neubauer 2019	Neubauer S, Kolm P, Ho CY, et al. Distinct Subgroups in Hypertrophic Cardiomyopathy in the NHLBI HCM Registry. J Am Coll Cardiol. 2019;74(19):2333-45.	Wrong design
Nishigaki 1996	Nishigaki K, Tomita M, Kagawa K, et al. Marked expression of plasma brain natriuretic peptide is a special feature of hypertrophic obstructive cardiomyopathy. J Am Coll Cardiol. 1996;28(5):1234-42.	Wrong design
Noji 2004	Noji Y, Shimizu M, Ino H, et al. Increased circulating matrix metalloproteinase-2 in patients with hypertrophic cardiomyopathy with systolic dysfunction. Circ J. 2004;68(4):355-60.	Wrong design
Ntelios 2020	Ntelios D, Efthimiadis G, Zegkos T, et al. Correlation of miR-146a-5p plasma levels and rs2910164 polymorphism with left ventricle	Wrong design

First author & year	Reference	Reason
	outflow tract obstruction in hypertrophic cardiomyopathy. Hellenic J Cardiol. 2020.	
Ogino 2004	Ogino K, Ogura K, Kinugawa T, et al. Neurohumoral profiles in patients with hypertrophic cardiomyopathy: differences to hypertensive left ventricular hypertrophy. Circ J. 2004;68(5):444-50.	Wrong outcome
Oka 2008	Oka K, Tsujino T, Nakao S, et al. Symptomatic ventricular tachyarrhythmia is associated with delayed gadolinium enhancement in cardiac magnetic resonance imaging and with elevated plasma brain natriuretic peptide level in hypertrophic cardiomyopathy. J Cardiol. 2008;52(2):146-53.	Wrong design
Okamoto 2013	Okamoto R, Hirashiki A, Cheng XW, et al. Usefulness of serum cardiac troponins T and I to predict cardiac molecular changes and cardiac damage in patients with hypertrophic cardiomyopathy. Int Heart J. 2013;54(4):202-6.	Wrong design
Orenes-Pinero 2013	Orenes-Pinero E, Hernandez-Romero D, de Torre C, et al. Identification and confirmation of haptoglobin as a potential serum biomarker in hypertrophic cardiomyopathy using proteomic approaches. Ann Med. 2013;45(4):341-7.	Wrong outcome
Panou 2006	Panou FK, Kotseroglou VK, Lakoumentas JA, et al. Significance of brain natriuretic peptide in the evaluation of symptoms and the degree of left ventricular diastolic dysfunction in patients with hypertrophic cardiomyopathy. Hellenic J Cardiol. 2006;47(6):344-51.	Wrong outcome
Park 2012	Park JR, Choi JO, Han HJ, et al. Degree and distribution of left ventricular hypertrophy as a determining factor for elevated natriuretic peptide levels in patients with hypertrophic cardiomyopathy: insights from cardiac magnetic resonance imaging. Int J Cardiovasc Imaging. 2012;28(4):763-72.	Wrong design
Paulus 2001	Paulus WJ. Natriuretic peptides to probe haemodynamic overload in hypertrophic cardiomyopathy. Eur Heart J. 2001;22(15):1249-51.	Wrong publication type
Payá 2008	Payá E, Marín F, González J, et al. Variables associated with contrast-enhanced cardiovascular magnetic resonance in hypertrophic cardiomyopathy: clinical implications. J Card Fail. 2008;14(5):414-9.	Wrong outcome
Penicka 2001	Penicka M, Gregor P, Krupicka J, et al. Tumour necrosis factor-alpha soluble receptors type I are related to symptoms and left ventricular function in hypertrophic cardiomyopathy. Can J Cardiol. 2001;17(7):777-84.	Full text not available
Peukert 1999	Peukert S, Fu ML, Eftekhari P, et al. The frequency of occurrence of anti-cardiac receptor autoantibodies and their correlation with clinical manifestation in patients with hypertrophic cardiomyopathy. Autoimmunity. 1999;29(4):291-7.	Wrong design
Pinamonti 2008	Pinamonti B, Di Lenarda A, Nucifora G, et al. Incremental prognostic value of restrictive filling pattern in hypertrophic cardiomyopathy: a Doppler echocardiographic study. Eur J Echocardiogr. 2008;9(4):466-71.	Wrong determinant
Podzimkova 2019	Podzimkova J, Palecek T, Kuchynka P, et al. Plasma osteopontin levels in patients with dilated and hypertrophic cardiomyopathy. Herz. 2019;44(4):347-53.	Wrong design
Pudil 2015	Pudil R, Horaova L, Vašatova M, et al. N-Terminal pro BNP and echocardiographic parameters in patients with non-obstructive hypertrophic cardiomyopathy. Klinicka Biochemie a Metabolismus. 2015;23(4):175-80.	Full text not available
Pudil 2015	Pudil R, Vasatova M, Fucikova A, et al. Vascular Endothelial Growth Factor Is Associated with the Morphologic and Functional Parameters in Patients with Hypertrophic Cardiomyopathy. Biomed Res Int. 2015;2015:762950.	Wrong design
Qian 2021	Qian D, Zhou X, Liu H, et al. Clinical value of 2D speckle tracking imaging in evaluating the effect of percutaneous intramyocardial septal radiofrequency ablation in patients with hypertrophic obstructive cardiomyopathy. J Clin Ultrasound. 2021;49(6):554-62.	Wrong outcome
Rajtar-Salwa 2017	Rajtar-Salwa R, Hładij R, Dimitrow PP. Elevated Level of Troponin but Not N-Terminal Probrain Natriuretic Peptide Is Associated with	Wrong outcome

First author & year	Reference	Reason
	Increased Risk of Sudden Cardiac Death in Hypertrophic Cardiomyopathy Calculated According to the ESC Guidelines 2014. 2017;2017:9417908.	
Rajtar-Salwa 2017	Rajtar-Salwa R, Tokarek T, Dimitrow PP. Reversed Septal Curvature Is Associated with Elevated Troponin Level in Hypertrophic Cardiomyopathy. Dis Markers. 2020;2020:8821961.	Wrong outcome
Roldan 2008	Roldan V, Marin F, Gimeno JR, et al. Matrix metalloproteinases and tissue remodeling in hypertrophic cardiomyopathy. Am Heart J. 2008;156(1):85-91.	Wrong design
Roncarati 2014	Roncarati R, Viviani Anselmi C, Losi MA, et al. Circulating miR-29a, among other up-regulated microRNAs, is the only biomarker for both hypertrophy and fibrosis in patients with hypertrophic cardiomyopathy. J Am Coll Cardiol. 2014;63(9):920-7.	Wrong outcome
Saeki 2002	Saeki H, Hamada M, Hiwada K. Circulating levels of insulin-like growth factor-1 and its binding proteins in patients with hypertrophic cardiomyopathy. Circ J. 2002;66(7):639-44.	Wrong design
Safrygina Iu 2007	Safrygina Iu V, Gabrusenko SA, Ovchinnikov AG, et al. [Cardiac natriuretic peptides in patients with hypertrophic cardiomyopathy]. Kardiologiia. 2007;47(5):50-7.	Wrong language
Sanchez 2016	Sanchez D, Gregor P, Curila K, et al. Anti-calreticulin antibodies and calreticulin in sera of patients diagnosed with dilated or hypertrophic cardiomyopathy. Autoimmunity. 2016;49(8):554-62.	Wrong outcome
Sari 2021	Sari M, Yazar H, Kocayigit I, et al. Alteration of serum biomarkers in patients with hypertrophic cardiomyopathy with and without atrial fibrillation. Biomark Med. 2021;15(13):1131-42.	Wrong outcome
Sato 2003	Sato Y, Taniguchi R, Nagai K, et al. Measurements of cardiac troponin T in patients with hypertrophic cardiomyopathy. Heart. 2003;89(6):659-60.	Wrong outcome
Saura 2009	Saura D, Marín F, Climent V, et al. Left atrial remodelling in hypertrophic cardiomyopathy: relation with exercise capacity and biochemical markers of tissue strain and remodelling. Int J Clin Pract. 2009;63(10):1465-71.	Wrong outcome
Shi 2020	Shi RY, An DA, Chen BH, et al. Diffusion-weighted imaging in hypertrophic cardiomyopathy: association with high T2-weighted signal intensity in addition to late gadolinium enhancement. Int J Cardiovasc Imaging. 2020;36(11):2229-38.	Wrong outcome
Shigematsu 2011	Shigematsu Y, Hamada M, Nagai T, et al. Risk for atrial fibrillation in patients with hypertrophic cardiomyopathy: association with insulin resistance. J Cardiol. 2011;58(1):18-25.	Wrong outcome
Shimada 2021a	Shimada YJ, Raita Y, Liang LW, et al. Comprehensive Proteomics Profiling Reveals Circulating Biomarkers of Hypertrophic Cardiomyopathy. Circ Heart Fail. 2021;14(7):e007849.	Wrong design
Shimada 2021b	Shimada YJ, Batra J, Kochav SM, et al. Difference in Metabolomic Response to Exercise between Patients with and without Hypertrophic Cardiomyopathy. J Cardiovasc Transl Res. 2021;14(2):246-55.	Wrong design
Silva 2013	Silva D, Madeira H, Almeida A, et al. Tissue Doppler imaging and plasma N-terminal probrain natriuretic peptide for the identification of hypertrophic cardiomyopathy mutation carriers. Am J Cardiol. 2013;112(7):996-1004.	Wrong design
Song 2019	Song C, Wang S, Guo Y, et al. Plasma big endothelin-1 predicts new-onset atrial fibrillation after surgical septal myectomy in patients with hypertrophic cardiomyopathy. BMC Cardiovasc Disord. 2019;19(1):122.	Wrong outcome
Sonnenschein 2019	Sonnenschein K, Wilczek AL, de Gonzalo-Calvo D, et al. Serum circular RNAs act as blood-based biomarkers for hypertrophic obstructive cardiomyopathy. Sci Rep. 2019;9(1):20350.	Wrong design
Sonnenschein 2021	Sonnenschein K, Fiedler J, de Gonzalo-Calvo D, et al. Blood-based protein profiling identifies serum protein c-KIT as a novel biomarker for hypertrophic cardiomyopathy. Sci Rep. 2021;11(1):1755.	Wrong design

First author & year	Reference	Reason
Sonsoz 2021	Sonsoz MR, Yilmaz M, Cevik E, et al. Circulating Levels of MicroRNAs in Hypertrophic Cardiomyopathy: The Relationship With Left Ventricular Hypertrophy, Left Atrial Dilatation and Ventricular Depolarisation-Repolarisation Parameters. Heart Lung Circ. 2021.	Wrong design
Symons 1985	Symons C, Fortune F, Greenbaum RA, et al. Cardiac hypertrophy, hypertrophic cardiomyopathy, and hyperparathyroidism--an association. Br Heart J. 1985;54(5):539-42.	Wrong outcome
Takeuchi 2005	Takeuchi I, Inomata T, Nishii M, et al. Clinical characteristics of heart disease patients with a good prognosis in spite of markedly increased plasma levels of type-B natriuretic peptide (BNP): anomalous behavior of plasma BNP in hypertrophic cardiomyopathy. Circ J. 2005;69(3):277-82.	Wrong outcome
Takeuchi 2015	Takeuchi H. Serum levels of free T3 are associated with ApoA1 and ApoA2, whereas free T4 levels are associated with ApoB and LDL-cholesterol in euthyroid cardiovascular patients. IJC Metabolic and Endocrine. 2015;8:1-6.	Wrong outcome
Tanaka 1997	Tanaka T, Sohmiya K, Kawamura K. Is CD36 deficiency an etiology of hereditary hypertrophic cardiomyopathy? J Mol Cell Cardiol. 1997;29(1):121-7.	Wrong outcome
Taniguchi 2006	Taniguchi R, Sato Y, Nishio Y, et al. Measurements of baseline and follow-up concentrations of cardiac troponin-T and brain natriuretic peptide in patients with heart failure from various etiologies. Heart Vessels. 2006;21(6):344-9.	Wrong population
Tasolar 2014	Taşolar H, Çetin M, Balli M. Relationship between mean platelet volume and hypertrophic cardiomyopathy. Angiology. 2014;65(9):851.	Wrong publication type
Teragaki 1993	Teragaki M, Ohmura T, Iida H, et al. Change of the isoform of creatine phosphokinase MM in hypertrophic cardiomyopathy. Am Heart J. 1993;125(1):251-3.	Wrong outcome
Tesic 2017	Tesic M, Seferovic J, Trifunovic D, et al. N-terminal pro-brain natriuretic peptide is related with coronary flow velocity reserve and diastolic dysfunction in patients with asymmetric hypertrophic cardiomyopathy. J Cardiol. 2017;70(4):323-8.	Wrong design
Thaman 2006	Thaman R, Esteban MT, Barnes S, et al. Usefulness of N-terminal pro-B-type natriuretic peptide levels to predict exercise capacity in hypertrophic cardiomyopathy. Am J Cardiol. 2006;98(4):515-9.	Wrong design
Tomoda 1989	Tomoda H. [Relations of intracardiac dimensions as measured by echocardiography and plasma atrial natriuretic peptide levels in various cardiovascular diseases]. J Cardiol. 1989;19(1):247-55.	Full text not available
Tripp 1984	Tripp ME, Shug AL. Plasma carnitine concentrations in cardiomyopathy patients. Biochem Med. 1984;32(2):199-206.	Wrong population
Trueman 1981	Trueman T, Thompson RA, Cummins P, et al. Heart antibodies in cardiomyopathies. Br Heart J. 1981;46(3):296-301.	Wrong outcome
Tucholski 2020	Tucholski T, Cai W, Gregorich ZR, et al. Distinct hypertrophic cardiomyopathy genotypes result in convergent sarcomeric proteoform profiles revealed by top-down proteomics. Proc Natl Acad Sci U S A. 2020;117(40):24691-700.	Wrong design
Tuluce 2016	Tuluce K, Yakar Tuluce S, Kahya Eren N, et al. Predictors of Future Atrial Fibrillation Development in Patients with Hypertrophic Cardiomyopathy: A Prospective Follow-Up Study. Echocardiography. 2016;33(3):379-85.	Wrong outcome
Varol 2005	Varol E, Ozaydin M, Sahin M, et al. vWf levels as a circulating marker of endothelial dysfunction in patients with hypertrophic cardiomyopathy. Indian Heart J. 2005;57(6):655-7.	Full text not available
Varol 2007	Varol E, Ozaydin M, Altinbas A, et al. Elevated carbohydrate antigen 125 levels in hypertrophic cardiomyopathy patients with heart failure. Heart Vessels. 2007;22(1):30-3.	Wrong design
Vilchez 2011	Vilchez JA, Hernandez-Romero D, Ruiz-Espejo F, et al. Collagen peptides, interstitial remodelling and sudden cardiac death in hypertrophic cardiomyopathy. Clin Chem Lab Med. 2011;49(9):1569-71.	Wrong design

First author & year	Reference	Reason
Wasserstrum 2019	Wasserstrum Y, Barriales-Villa R, Fernández-Fernández X, et al. The impact of diabetes mellitus on the clinical phenotype of hypertrophic cardiomyopathy. Eur Heart J. 2019;40(21):1671-7.	Wrong determinant
Wei 2011	Wei BQ, Zhang J, Yang YJ, et al. [Influencing factors for the plasma concentration of N-terminal brain natriuretic peptide precursor in patients with heart failure due to various heart diseases]. Zhonghua Yi Xue Za Zhi. 2011;91(38):2683-7.	Wrong language
Wei 2021	Wei Z, Zhu E, Ren C, et al. Triglyceride-Glucose Index Independently Predicts New-Onset Atrial Fibrillation After Septal Myectomy for Hypertrophic Obstructive Cardiomyopathy Beyond the Traditional Risk Factors. Front Cardiovasc Med. 2021;8:692511.	Wrong outcome
Wo 2008	Wo JS, Zhao Q, Guo J, et al. Correlation of left ventricular diastolic function to plasma N-terminal pro-brain natriuretic peptide in patients with hypertrophic cardiomyopathy detected by echocardiogram and tissue Doppler imaging. Journal of Clinical Rehabilitative Tissue Engineering Research. 2008;12(13):2459-62.	Full text not available
Yakar Tuluce 2016	Yakar Tuluce S, Tuluce K, Cil Z, et al. Galectin-3 levels in patients with hypertrophic cardiomyopathy and its relationship with left ventricular mass index and function. Anatol J Cardiol. 2016;16(5):344-8.	Wrong outcome
Yamamoto 1995	Yamamoto K, Ikeda U, Furuhashi K, et al. The coagulation system is activated in idiopathic cardiomyopathy. J Am Coll Cardiol. 1995;25(7):1634-40.	Wrong outcome
Yang 2020	Yang C, Zhang C, Yuan J, et al. Prevalence and determinants of elevated D-dimer in patients with hypertrophic cardiomyopathy. Biomark Med. 2020;14(2):131-40.	Wrong design
Yarom 1982	Yarom R, Lewis BS, Lijovetzky G, et al. Platelet studies in patients with hypertrophic cardiomyopathy. Cardiovasc Res. 1982;16(6):324-30.	Wrong population
Yoshibayashi 1993	Yoshibayashi M, Kamiya T, Saito Y, et al. Increased plasma levels of brain natriuretic peptide in hypertrophic cardiomyopathy. N Engl J Med. 1993;329(6):433-4.	Wrong outcome
Zachariah 2012	Zachariah JP, Colan SD, Lang P, et al. Circulating matrix metalloproteinases in adolescents with hypertrophic cardiomyopathy and ventricular arrhythmia. Circ Heart Fail. 2012;5(4):462-6.	Wrong design
Zaitsev 2020	Zaitsev VV, Gurshchenkov AV, Mitrofanova LB, et al. [Clinical significance of different assesment methods of myocardial fibrosis in patients with hypertrophic cardiomyopathy.]. Kardiologiia. 2020;60(3):44-50.	Wrong language
Zhang 2015	Zhang C, Liu R, Yuan J, et al. Significance and Determinants of Cardiac Troponin I in Patients With Obstructive Hypertrophic Cardiomyopathy. Am J Cardiol. 2015;116(11):1744-51.	Wrong design
Zhang 2016	Zhang C, Liu R, Yuan J, et al. Gender-related differences in the association between serum uric acid and left ventricular mass index in patients with obstructive hypertrophic cardiomyopathy. Biol Sex Differ. 2016;7:22.	Wrong design
Zhou 2020	Zhou Y, Yu M, Yuan J, et al. Cardiac troponin I is associated with non-sustained ventricular tachycardia in patients with hypertrophic obstructive cardiomyopathy. Heart Vessels. 2020;35(6):876-85.	Wrong outcome
Zhu 2020	Zhu A, Bews H, Cheung D, et al. Scleraxis as a Prognostic Marker of Myocardial Fibrosis in Hypertrophic Cardiomyopathy (SPARC) Study. Can J Physiol Pharmacol. 2020.	Wrong outcome

Full list of studies excluded during full text screening, with references and reason for exclusion.

Supplemental Table 4: Biomarkers for heart failure

Biomarker	Outcome definition	Study (reference)	Unit	n outcome / n total	Follow-up duration (years)	Result	P-value	Factors adjusted for
Big endothelin-1	Progression to NYHA class III-IV	Wang 2017 (S36)	pg/mL	29 / 207	3 (2-5)	HR 1.65 (1.1-2.47)	**0.017**	
			Tertile 3 vs 1	19 / 129	3 (2-5)	HR 3.86 (1.28-11.68)	**0.017**	
			Tertile 2 vs 1	14 / 145		HR NA	0.154	
						aHR 3.79 (1.25-11.51)	**0.019**	AF and NYHA class III-IV
						aHR NA	0.154	
	HF-mortality	Wang 2017 (S36)	pg/mL	12 / 245	3 (2-5)	HR 1.72 (1.11-2.66)	**0.016**	
			Tertile 3 vs 1	9 / 161	3 (2-5)	HR NA	0.08	
			Tertile 2 vs 1	4 / 162		HR NA	0.311	
BNP	HF-hospitalisation or HF-mortality	Kitaoka 2010 (S16)	per 100 pg/mL	6 / 41	3.2 ± 0.7	RR 1.38	0.18	
	HF-hospitalisation	Yang 2018 (S41)	pg/mL	17 / 98	1.4 ± 0.8	HR 1.039 (1.019-1.060)	**<0.001**	
						aHR 1.028 (1.011-1.045)	**0.001**	LVEF, haemoglobin, red blood cell distribution width
	Acute congestive HF	Kitaoka 2012 (S18)	pg/mL	6 / 36	4.8 ± 1.4	Mean event+ 340 ± 198 Mean event- 243 ± 189	0.29	
	Global systolic dysfunction	Aizawa 2019 (S1)	pg/mL	18 / 434	8.4 ± 6.7	HR 1.001 (1.000-1.002)	0.13	
						Mean event+ 432 ± 332 Mean event- 241 ± 292	**0.005**	
	LVEF <50%, MWT ≥5 mm decrease and LVEDD ≥5 mm increase	Pieroni 2007 (S28)	pg/mL	10 / 40	5.9 ± 0.56	Mean event+ 123.5 ± 58.6 Mean event- 121.9 ± 66.2	NS	
CK-MB	Acute congestive HF	Hamada 2016 (S11)	≥2.5%	22 / 77	17.8 ± 4.0	HR NA	**<0.0001**	
	HF-mortality (at <85 years)	Hamada 2016 (S11)	≥2.5%	16 / 77	17.8 ± 4.0	HR NA	**<0.0001**	
Copeptin	Acute congestive HF	Sahin 2017 (S29)	nmol/mL	16 / 60	range NA-2	aOR 1.22 (1.06-1.42)	**<0.01**	Age, sex, hypertension, interventricular septum thickness, LVOT gradient, copeptin/NT-proBNP
Creatinine	Global systolic dysfunction	Aizawa 2019 (S1)	mg/dL	18 / 434	8.4 ± 6.7	HR 0.381 (0.040-3.589)	0.399	
						Mean event+ 0.9 ± 0.2 Mean event- 1.0 ± 0.5	0689	
eGFR	Acute congestive HF	Kitaoka 2012 (S18)	ml/min/1.73m2	6 / 36	4.8 ± 1.4	Mean event+ 66.7 ± 13.5 Mean event- 76.1 ± 17.7	0.20	
	Global systolic dysfunction	Aizawa 2019 (S1)	ml/min/1.73m2	18 / 434	8.4 ± 6.7	HR 1.022 (1.001-1.044)	**0.04**	
						Mean event+ 71 ± 29 Mean event- 65 ± 20	0.797	
Haemoglobin	HF-hospitalisation	Yang 2018 (S41)	g/L	17 / 98	1.4 ± 0.8	HR 0.962 (0.939-0.986)	**0.002**	
						aHR 1.001 (0.967-1.037)	0.936	LVEF, BNP, red blood cell distribution width
hs-CRP	HF-mortality	Zhu 2017 (S47)	µg/mL	14 / 490	3.7 ± 2.0	HR 1.16 (1.04-1.31)	**0.009**	
						aHR 1.15 (1.02-1.30)	**0.02**	Age, sex, family history of SCD, NYHA class III-IV, unexplained

Biomarker	Outcome definition	Study (reference)	Unit	n outcome / n total	Follow-up duration (years)	Result	P-value	Factors adjusted for
								syncope, MWT, LVOT obstruction at rest and nsVT
						aHR 1.15 (1.02-1.30)	**0.02**	Age, sex and NYHA class III-IV
			>3.0 vs <1.0 µg/mL	11 / 312	3.7 ± 2.0	HR 4.98 (1.31-18.86)	**0.018**	
			1.0-3.0 vs <1.0 µg/mL	6 / 379		HR 1.15 (0.23-5.7)	0.87	
						aHR 4.38 (1.15-16.6)	**0.03**	Age, sex, family history of SCD, NYHA class III/IV, unexplained syncope, MWT, LVOT obstruction at rest and nsVT
						aHR 1.31 (0.26-6.52)	0.75	
						aHR 4.38 (1.15-16.6)	**0.03**	Age, sex and NYHA class III-IV
						aHR 1.31 (0.26-6.52)	0.75	
			Tertile 3 vs 1	NA / 327	3.7 ± 2.0	HR 3.41 (0.93-12.42)	0.063	
			Tertile 2 vs 1	NA / 327		HR 0.32 (0.03-3.11)	0.33	
						aHR 3.19 (0.87-11.63)	0.079	Age, sex, family history of SCD, NYHA class III-IV, unexplained syncope, MWT, LVOT obstruction at rest and nsVT
						aHR 0.37 (0.04-3.57)	0.39	
						aHR 3.19 (0.87-11.63)	0.079	Age, sex and NYHA class III-IV
						aHR 0.37 (0.04-3.57)	0.39	
hs-cTnT	Progression to NYHA class III-IV, HF-hospitalisation or HF-mortality	Kubo 2013 (S20)	>0.014 ng/mL	26 / 183	4.1 ± 2.0	HR 11.16 (2.64-47.21)	**0.001**	
	Progression to NYHA class III-IV, systolic dysfunction or HF-hospitalisation	Hasler 2016 (S12)	>0.014 ng/mL	2 / 91	11.5 ± NA, range 0.5-35	OR 5.11	0.495	
	LVEF <50%	Kubo 2020 (S21)	ng/mL	13 / 157	6.3 ± 2.8	Mean event+ 0.033 ± 0.034 Mean event- 0.018 ± 0.019	**0.012**	
			>0.014 ng/mL	13 / 157	6.3 ± 2.8	aOR 6.467 (1.114-37.554)	**0.038**	LVEF <65%, LVEDD >47mm, LA diameter >47mm, follow-up period >7.2 years
Insulin resistance (Homeostasis model)	Acute congestive HF	Murakami 2004 (S25)	≥2.6	3 / 55	8.8 ± 4.2	OR 2.00	NS	
	HF-mortality	Murakami 2004 (S25)	≥2.6	9 / 55	8.8 ± 4.2	OR 1.25	NS	
Intelectin-1	Acute congestive HF	Yildiz 2018 (S42)	≤291 ng/mL	20 / 87	range NA-2	OR 2.26	**0.02**	
Mean corpuscular volume	HF-hospitalisation	Yang 2018 (S41)	fl	17 / 98	1.4 ± 0.8	HR 0.965 (0.904-1.032)	0.299	
MMP-2	HF-hospitalisation or HF-mortality	Kitaoka 2010 (S16)	>1170 ng/mL	6 / 41	3.2 ± 0.7	HR NA	**0.01**	
			per 100 ng/mL	6 / 41	3.2 ± 0.7	RR 1.87	**0.01**	
						aRR NA	NS	Age, AF, NYHA class, LVEF, eGFR
						aRR 1.91	**0.02**	Age

Biomarker	Outcome definition	Study (reference)	Unit	n outcome / n total	Follow-up duration (years)	Result	P-value	Factors adjusted for
						aRR 1.63	0.06	eGFR
MMP-9	HF-hospitalisation or HF-mortality	Kitaoka 2010 (S16)	NA	6 / 41	3.2 ± 0.7	RR 1.2	0.53	
NT-proBNP	Acute congestive HF	Sahin 2017 (S29)	pg/mL	16 / 60	range NA-2	aOR 1.004 (1.001-1.006)	**<0.01**	Age, sex, hypertension, interventricular septum thickness, LVOT gradient, copeptin/NT-proBNP
	HF-hospitalisation	Maczynska-Mazuruk 2019 (S22)	Tertile 2-3 vs tertile 1	32 / 603	2.3 ± 1.2	aHR 1.76 (1.03-3.0)	**0.037**	Unknown variables (not reported)
	Heart transplant or HF-mortality	Coats 2013 (S5)	log(fmol/mL)	23 / 847	3.5 (2.5-4.5)	HR 3.03 (1.99-4.60)	**<0.001**	
	Progression to NYHA class III-IV, HF-hospitalisation, LVEF <50%, HF-mortality, AF or stroke	D'Amato 2013 (S6)	log(pg/mL)	67 / 183	3.9 ± 2.8	HR 2.73 (1.67-4.4)	**<0.01**	
						aHR 2.6 (1.5-4.3)	**<0.01**	Age, AF, NYHA class III-IV and LA diameter
Red blood cell distribution width	HF-hospitalisation	Yang 2018 (S41)	%	17 / 98	1.4 ± 0.8	HR 1.583 (1.248-2.009)	**<0.001**	
						aHR 1.711 (1.042-2.809)	**0.034**	LVEF, haemoglobin, BNP
Soluble Fas	HF-hospitalisation	Zen 2005 (S44) (restricted to dilated-stage HCM)	≥3.48 ng/mL	7 / 11	1.8 ± NA, range 0.7-2.8	HR NA	**0.0276**	
Tenascin-C	Acute congestive HF	Kitaoka 2012 (S18)	≥39.2 ng/mL	6 / 36	4.8 ± 1.4	HR NA	**0.0009**	
			ng/mL	6 / 36 30 / 36	4.8 ± 1.4	Mean event+ NA Mean event- NA	**0.008**	
TGF-β1	Acute congestive HF	Ayca 2015 (S2)	>4877 pg/mL	9 / 49	range NA-1.5	OR 2.32	0.2	
TIMP1	HF-hospitalisation or HF-mortality	Kitaoka 2010 (S16)	NA	6 / 41	3.2 ± 0.7	RR 24.1	**0.03**	
Uric acid	Progression to NYHA class III-IV or HF-mortality	Zhu 2015 (S46)	mg/dL	NA / 588	5.2 ± 2.4	aHR 1.19 (1.06-1.33)	**0.003**	Age, sex, BMI, controlled hypertension, family history of SCD, NYHA class III-IV, unexplained syncope, MWT, and LVOT obstruction at rest
			Tertile 3 vs 1	NA / 392	5.2 ± 2.4	HR 3.35 (1.87-6.00)	**<0.001**	
			Tertile 3 vs 1	NA / 392	5.2 ± 2.4	aHR 3.46 (1.86-6.45)	**<0.001**	Age, sex, BMI, controlled hypertension, family history of SCD, NYHA class III-IV, unexplained syncope, MWT, and LVOT obstruction at rest
			Tertile 2 vs 1	NA / 392		aHR 1.30 (0.64-2.64)	0.476	
			Per tertile	NA / 588			**<0.001**	

Data presented as means ± SD, medians (interquartile range), odds ratio indicated with OR, adjusted odds ratio (95% confidence interval) indicated with aOR, relative risk indicated with RR, adjusted relative risk (95% CI) indicated with aRR, hazard ratio (95% confidence interval) indicated with HR or adjusted hazard ratio (95% confidence interval) indicated with aHR. Outcomes including surrogate endpoints are italicised. P-values are provided as reported by the original authors or calculated by review authors. P-values <0.05 are indicated in bold.

aHR, adjusted hazard ratio; AF, atrial fibrillation; aOR, adjusted odds ratio; aRR, adjusted relative risk; BMI, body mass index; BNP, ventricular (brain or B-type) natriuretic peptide; CK-MB, creatine (phospho)kinase MB isoform; eGFR, estimated glomerular filtration rate; HF, heart failure; HOCM, hypertrophic obstructive cardiomyopathy; HR, hazard ratio; hs-CRP, high sensitivity C-reactive protein; hs-cTnT, high sensitivity cardiac troponin T; LA, left atrial; LVEF, left ventricular ejection fraction; LVEDD, left ventricular end-diastolic diameter; LVFS, left ventricular fractional shortening; LVOT, left ventricular outflow tract; MMP, Matrix metallopeptidases (metalloproteinases); MWT, maximum wall thickness; NA, not available; NS, not significant; nsVT, non-sustained ventricular tachycardia; NT-proBNP, N-terminal prohormone of brain natriuretic peptide; NYHA, New York Heart Association; OR, odds ratio; RR, relative risk; SCD, sudden cardiac death; TGF, transforming growth factor; TIMP, TIMP (tissue inhibitor of metalloproteinases) metallopeptidase inhibitor.

Supplemental Table 5: Biomarkers for malignant ventricular arrhythmia

Biomarker	Outcome definition	Study (reference)	Unit	n outcome / n total	Follow-up duration (years)	Result	P-value	Factors adjusted for
Big endothelin-1	SCD	Wang 2017 (S36)	pg/mL	7 / 245	3 (2-5)	HR NA	0.707	
			Tertile 3 vs 1	5 / 161	3 (2-5)	HR NA	0.288	
			Tertile 2 vs 1	3 / 162		HR NA	0.556	
						aHR NA	0.332	AF and NYHA class III-IV
						aHR NA	0.421	
BNP	Appropriate ICD therapy, resuscitated cardiac arrest or SCD	Minami 2018 (S23)	>312 pg/mL	37 / 346	8.4 (4.2-12.5)	HR 5.89 (2.99-11.6)	**<0.001**	
						aHR 6.26 (3.21-12.2)	**<0.001**	Age at diagnosis, sex
						aHR 5.71 (2.86-11.4)	**<0.001**	Family history of SCD, unexplained syncope, MWT ≥30 mm, nsVT
						aHR 4.68 (2.30-9.51)	**<0.001**	LA diameter, LVEDD
	Sustained VT, VF, appropriate ICD therapy, resuscitated cardiac arrest or SCD	Sugiura 2019 (S35) (restricted to patients without 2011 ACCF/AHA guideline SCD risk factors)	per 10 pg/mL	7 / 93	4.7 (2.9-7.5)	HR 1.035 (1.005-1.065)	**0.023**	
						aHR 0.962 (0.895-1.034)	0.321	Mitral L-wave > 0.2 m/s, E/e', mean pulmonary artery pressure
CK-MB	SCD (at <85 years)	Hamada 2016 (S11)	≥2.5%	3 / 77	17.8 ± 4.0	HR NA	**<0.05**	
hs-CRP	SCD	Zhu 2017 (S47)	µg/mL	11 / 490	3.7 ± 2.0	HR 1.21 (1.07-1.36)	**0.002**	
						aHR 1.21 (1.06-1.37)	**0.004**	Age, sex, NYHA class III-IV
						aHR 1.20 (1.06-1.36)	**0.003**	Age, sex, NYHA class III-IV, family history SCD, unexplained syncope, rest LVOT obstruction, MWT, nsVT
			>3.0 vs <1.0 µg/mL	8 / 312	3.7 ± 2.0	HR 12.57 (1.54-102.45)	**0.018**	
			1.0-3.0 vs <1.0 µg/mL	4 / 379		HR 3.45 (0.36-33.21)	0.258	
						aHR 11.29 (1.38-92.20)	**0.024**	Age, sex, NYHA class III-IV, family history SCD, unexplained syncope, rest LVOT obstruction, MWT, nsVT
						aHR 3.53 (0.36-34.16)	0.276	

Biomarker	Outcome definition	Study (reference)	Unit	n outcome / n total	Follow-up duration (years)	Result	P-value	Factors adjusted for
						aHR 11.16 (1.37-91.17)	**0.024**	Age, sex, NYHA class III-IV
						aHR 3.70 (0.38-35.73)	0.258	
			Tertile 3 vs 1	NA / 327	3.7 ± 2.0	HR 7.88 (0.98-63.10)	0.052	
			Tertile 2 vs 1	NA / 327		HR 1.94 (0.18-21.39)	0.589	
						aHR 7.24 (0.90-58.08)	0.062	Age, sex, NYHA class III-IV
						aHR 2.15 (0.19-23.82)	0.532	
						aHR 6.75 (0.83-54.90)	0.074	Age, sex, NYHA class III-IV, family history SCD, unexplained syncope, rest LVOT obstruction, MWT, nsVT
						aHR 2.31 (0.21-25.89)	0.499	
hs-cTnT	Appropriate ICD therapy, resuscitated cardiac arrest or SCD	Hasler 2016 (S12)	>0.014 ng/mL	7 / 91	11.5 ± NA, range 0.5-35	OR 1.33	0.71	
	Haemodynamically unstable sustained VT, appropriate ICD therapy or SCD	Kubo 2013 (S20)	>0.014 ng/mL	8 / 183	4.1 ± 2.0	HR 5.81 (0.72-47.3)	0.100	
	Sustained VT, VF, appropriate ICD therapy, resuscitated cardiac arrest, SCD or nsVT	Gastl 2020 (S8) (cohort overlap with Hasler 2016)	NA	47 / 91	3.4 ± 2.6	OR 2.67 (1.1-6.7)	**0.036**	
		Gastl 2020 (S8) (restricted to LVOT gradient <30 mmHg after provocation) (cohort overlap with Hasler 2016)	NA	NA / 46	3.4 ± 2.6	OR 3.5 (0.9-13.0)	0.210	
Insulin resistance	SCD	Murakami 2004 (S25)	Homeostasis model ≥2.6	4 / 55	8.8 ± 4.2	OR 10.1	**0.0414**	
Intelectin-1	Sustained VT	Yildiz 2018 (S42)	≤291 ng/mL	7 / 87	range NA-2	OR 1.0909	0.6	
NT-proBNP	Appropriate ICD therapy or SCD	Coats 2013 (S5)	log(fmol/mL)	11 / 847	3.5 (2.5-4.5)	HR 1.54 (0.91-2.60)	0.111	
		Maczynska-Mazuruk 2019 (S22)	Tertile 2-3 vs tertile 1	[24] / 603	2.3 ± 1.2	HR NA	NS	
TGF-β1	Sustained VT	Ayca 2015 (S2)	>4877 pg/mL	3 / 49	range NA-1.5	OR 3.41	0.4	
Uric acid	VF, appropriate ICD therapy or resuscitated cardiac arrest	Ozyilmaz 2018 (S27)	mg/dL	13 / 115	2.6 ± 1.1	Event+ 7.03 ± 0.96 Event- 6.04 ± 1.42	**0.018**	
	VF, appropriate ICD therapy, resuscitated cardiac arrest or SCD	Zhu 2015 (S46)	mg/dL	16 / 588	5.2 ± 2.4	aHR 1.60 (1.30-1.98)	**<0.001**	Age, sex, BMI, family history of SCD, controlled hypertension, NYHA class III-IV, unexplained syncope, rest LVOT obstruction, MWT
			Tertile 3 vs 1	NA / 392	5.2 ± 2.4	HR 5.34 (1.55-18.33)	**0.008**	

Biomarker	Outcome definition	Study (reference)	Unit	n outcome / n total	Follow-up duration (years)	Result	P-value	Factors adjusted for
			Tertile 3 vs 1	NA / 392	5.2 ± 2.4	aHR 9.19 (2.40-35.25)	**0.001**	Age, sex, BMI, family history of
			Tertile 2 vs 1	NA / 392		aHR 2.98 (0.70-12.73)	0.140	SCD, controlled hypertension,
			Per tertile	16 / 588			**0.002**	NYHA class III-IV, unexplained syncope, rest LVOT obstruction, MWT
	Sustained VT or nsVT	Ozyilmaz 2018 (S27)	mg/dL	24 / 115	2.6 ± 1.1	Mean event+ 6.83 ± 1.19 Mean event- 5.97 ± 1.42	**0.008**	

Data presented as means ± standard deviation, odds ratio indicated with OR, hazard ratio (95% confidence interval) indicated with HR or adjusted hazard ratio (95% confidence interval) indicated with aHR. Outcomes including surrogate endpoints are italicised. Numbers of subject totals in square brackets were not reported specifically but had to be inferred from available data and are therefore subject to inaccuracies. P-values are provided as reported by the original authors or calculated odds ratios. P-values <0.05 are indicated in bold.

aHR, adjusted hazard ratio; aOR, adjusted odds ratio; BMI, body mass index; BNP, ventricular (brain or B-type) natriuretic peptide; CK-MB, creatine (phospho)kinase MB isoform; hs-CRP, high sensitivity C-reactive protein; hs-cTnT, high sensitivity cardiac troponin T; ICD, implantable cardioverter-defibrillator; HR, hazard ratio; LA, left atrial; LVEDD, left ventricular end-diastolic diameter; LVOT, left ventricular outflow tract; MWT, maximum wall thickness; NA, not available; nsVT, non-sustained ventricular tachycardia; NYHA, New York Heart Association; OR, odds ratio; VF, ventricular fibrillation; VT, ventricular tachycardia; SCD, sudden cardiac death, TGF, transforming growth factor.

Supplemental Table 6: Biomarkers for outflow tract obstruction

Biomarker	Outcome definition	Study (reference)	Unit	n outcome / n total	Follow-up duration (years)	Result	P-value
BNP	Septal reduction therapy	Geske 2013 (S9)	pg/mL	69 / 471	1.7 ± 1.9	Median event+ 273 (129-418.5) Median event- 138.5 (50.8-343)	**0.0002**
			per tertile	69 / 471	1.7 ± 1.9	HR NA	**0.001**

Data presented as medians (interquartile range). P-values are provided as reported by the original authors. P-values <0.05 are indicated in bold.

BNP, ventricular (brain or B-type) natriuretic peptide.

Supplemental Table 7: Event rates of composite endpoints (including surrogate endpoints)

Outcome	References	Event rate (%/year)
Congestive HF, MVA	S20, S43, S46	3.3, 3.9, 7.7
End-stage HF, MVA	S29/S42	NA
Congestive HF, MVA, nsVT	S13	NA
Congestive HF, MVA, ICD implantation	S2/S29/S42	NA
Congestive HF, MVA, ICD implantation, myocardial infarction	S2	NA
Cardiovascular events	S10, S12, S15/S17/S19/S20, S26, S31	2.8 (2.7-5.1)

Cardiovascular events, all-cause mortality, ICD implantation, AF	S32	6.1
Congestive HF, all-cause mortality, AF	S4	2.9
Congestive HF, cardiovascular mortality	S14	6.0
Congestive HF, all-cause mortality	S3, S24	4.9, 7.9
Septal reduction therapy, all-cause mortality	S9	11
Cardiovascular mortality	S5-S7, S33/S36/S39/S46/S47, S37/S40	1.3 (1.1-2.1)
All-cause mortality	S2, S5, S9, S11, S22, S30, S34/S38/S39/S45, S37/S40, S43	2.3 (1.5-3.3)

Event rates for reported composite endpoints of heart failure and malignant ventricular arrhythmia, including surrogate endpoints. Cardiovascular events includes congestive HF, MVA and stroke. Centres with potential cohort overlap are indicated with a forward slash (/). In case of potential cohort overlap, only the event rate of the study with the largest sample size was included. Event rates for references S2/S29/S42, S13 and S30 could not be calculated due to missing summary data.

AF, atrial fibrillation; HF, heart failure; ICD, implantable cardioverter defibrillator; MVA, malignant ventricular arrhythmia; nsVT, non-sustained ventricular tachycardia.

Supplemental Table 8: Biomarkers for composite endpoints

Biomarker	Outcome definition	Study (reference)	Unit	n outcome / n total	Follow-up duration (years)	Result	P-value	Factors adjusted for
Copeptin	HF-mortality or SCD	Sahin 2017 (S29)	nmol/mL	7 / 60	range NA-2	aOR 1.11 (0.98-1.25)	0.09	Age, sex, septal thickness, LVOT gradient, hypertension, copeptin
hs-cTnT	Progression to NYHA class III-IV, HF-hospitalisation, HF-mortality, haemodynamically unstable sustained VT, appropriate ICD therapy or SCD	Kubo 2013 (S20)	>0.014 ng/mL	29 / 183	4.1 ± 2.0	HR 12.56 (2.99-52.82)	**<0.001**	
			0.031-0.140 vs ≤0.014 ng/mL	NA / 119	4.1 ± 2.0	aHR 12.14 (2.56-57.49)	**0.001**	NYHA class
			0.019-0.030 vs ≤0.014 ng/mL	NA / 116		aHR 13.74 (2.92-64.59)	**<0.001**	
			0.015-0.018 vs ≤0.014 ng/mL	NA / 116		aHR 7.47 (1.44-38.65)	**0.017**	
Intelectin-1	HF-mortality or SCD	Yildiz 2018 (S42)	≤291 ng/mL	4 / 87	range NA-2	OR 2.53	**0.04**	
NT-proBNP	HF-mortality or SCD	Sahin 2017 (S29)	pg/mL	7 / 60	range NA-2	aOR 1.00 (0.99-1.002)	0.91	Age, sex, septal thickness, LVOT gradient, hypertension, copeptin
Soluble neprilysin	Symptomatic HF, HF-mortality, VF, SCD	Yoshihisa 2019 (S43)	ng/mL	20 / 93	2.8 ± 1.7 range 0.04-7.9	HR 0.931 (0.695-1.248)	0.633	
			≥1.43 ng/mL	20 / 93	2.8 ± 1.7 range 0.04-7.9	HR 1.109 (0.459-2.682)	0.818	
Uric acid	Progression to NYHA class III-IV, HF-mortality, VF, appropriate ICD therapy, resuscitated cardiac arrest or SCD	Zhu 2015 (S46)	mg/dL	100 / 588	5.2 ± 2.4	aHR 1.27 (1.15-1.41)	**<0.001**	Age, sex, BMI, family history of SCD, controlled hypertension, NYHA class III-IV, unexplained syncope, rest LVOT obstruction, MWT

Biomarker	Outcome definition	Study (reference)	Unit	n outcome / n total	Follow-up duration (years)	Result	P-value	Factors adjusted for
			Tertile 3 vs 1	NA / 392	5.2 ± 2.4	HR 3.62 (2.14-6.13)	**<0.001**	
			Tertile 3 vs 1	NA / 392	5.2 ± 2.4	aHR 4.20 (2.38-7.42)	**<0.001**	Age, sex, BMI, family history of SCD, controlled hypertension, NYHA class III-IV, unexplained syncope, rest LVOT obstruction, MWT
			Tertile 2 vs 1	NA / 392		aHR 1.41 (0.74-2.68)	0.291	
			per tertile	100 / 588			**<0.001**	

Data presented as odds ratio indicated with OR, adjusted odds ratio (95% confidence interval) indicated with aOR, hazard ratio (95% confidence interval) indicated with HR or adjusted hazard ratio (95% confidence interval) indicated with aHR. P-values are provided as reported by the original authors or calculated odds ratios. P-values <0.05 are indicated in bold.

aHR, adjusted hazard ratio; aOR, adjusted odds ratio; BMI, body mass index; hs-cTnT, high sensitivity cardiac troponin T; HF, heart failure; HR, hazard ratio; ICD, implantable cardioverter-defibrillator; LVOT, left ventricular outflow tract; MWT, maximum wall thickness; NA, not available; NT-proBNP, N-terminal prohormone of brain natriuretic peptide; NYHA, New York Heart Association; OR, odds ratio; SCD, sudden cardiac death; VF, ventricular fibrillation; VT, ventricular tachycardia.

Supplemental Table 9: Biomarkers for composite endpoints including surrogate endpoints

Biomarker	Outcome definition	Study (reference)	Unit	n outcome / n total	Follow-up duration (years)	Result	P-value	Factors adjusted for
ANP	Heart failure (not otherwise defined), VF, SCD, stroke or peripheral embolism	Kitaoka 2001 (S15)	pg/mL	8 / 46	2.1 ± 0.9	RR 1.017 (1.009-1.025)	**<0.0001**	
						aRR 1.014 (1.004-1.024)	**0.005**	Unknown variables (not reported)
Big endothelin-1	Cardiovascular mortality	Wang 2017 (S36)	pg/mL	21 / 245	3 (2-5)	HR 1.49 (1.03-2.15)	**0.033**	
			Tertile 3 vs 1	15 / 161	3 (2-5)	HR 5.34 (1.20-23.81)	**0.028**	
			Tertile 2 vs 1	8 / 162		HR NA	0.159	
						aHR NA	0.077	AF and NYHA class III-IV
						aHR NA	0.103	
BNP	All-cause mortality	Geske 2013 (S9)	Tertile 3 vs 1	NA / 515	1.7 ± 1.9	HR 6.98 [(2.43-20.0)]	**0.0003**	
			Tertile 2 vs 1	NA / 515		HR 4.88 [(1.58-15.1)]	**0.006**	
			Per tertile	30 / 772			**0.002**	
		Geske 2013 (S9) (restricted to patients without LVOT obstruction)	Per tertile	NA / 497	1.7 ± 1.9	HR NA	**0.001**	
	Progression to NYHA class III-IV, HF-hospitalisation, appropriate ICD therapy, stroke-hospitalisation,	Kitaoka 2011 (S17)	≥188.9 pg/mL	20 / 130	3.7 ± 1.7	OR 7.32	**0.0024**	

Biomarker	Outcome definition	Study (reference)	Unit	n outcome / n total	Follow-up duration (years)	Result	P-value	Factors adjusted for
	cardiovascular mortality							
						aOR 3.1	0.22	NYHA class III-IV, syncope, AF, MWT, LA volume, septal E/e' ≥15, LVOT gradient ≥30 mmHg (rest)
						HR NA	**0.001**	
	HF-hospitalisation, resuscitated cardiac arrest, appropriate CID therapy, stroke, cardiovascular mortality	Shirotani 2020 (S31) (restricted to apical HCM)	pg/ml	34 / 144	9.5 (6.0-12.9)	Median event+ 334.4 (192.7-657.0) Median event- 133.4 (60.7-271.8)	**<0.001**	
			≥226.0 pg/mL	34 / 144	9.5 (6.0-12.9)	HR 4.51 (2.10-9.67)	**<0.001**	
						aHR 3.71 (1.61-8.55)	**0.002**	Age, NYHA class, AF, creatinine
						aHR 4.46 (2.07-9.61)	**<0.001**	Family history of SCD, unexplained syncope, nsVT, MWT
	HF-hospitalisation, LVAD implantation, heart transplantation, sustained VT, VF, appropriate ICD therapy, resuscitated cardiac arrest, nsVT prompting ICD implantation, AF-hospitalisation, stroke or all-cause mortality	Siriwardena 2018 (S32)	log(pg/mL)	42 / 111	6.2 ± 3.4, 6.9 (3.1-9.6)	HR 4.44 (2.19-9.00)	**<0.001**	
						aHR 4.30 (1.97-9.37)	**<0.001**	AF, LVEF, diastolic dysfunction (pseudonormalisation/restrictive), ESC HCM Risk-SCD score
		Siriwardena 2018 (S32) (restricted to patients without event before second measurement)	>100 pg/mL at first measurement	28 / 81	NA	HR 10.9 (2.6-46)	**<0.05**	
			>100 pg/mL at first and subsequent measurement	28 / 81	NA	HR 14.7 (3.5-62)	**<0.05**	
	HF-hospitalisation, sustained VT, VF, appropriate ICD therapy or all-cause mortality	Miyaji 2016 (S24)	Tertile 3 vs 1 Tertile 2 vs 1 (Total)	NA / [77] NA / [77] [14] / 116	1.6 (0.6-2.4)	aHR 4.59E10 (12.495-∞) aHR 1.091E10 (2.849-∞)	**<0.001** **0.005**	Age, sex, indexed LV mass tertile and indexed LV fibrosis mass tertile
	HF-hospitalisation, cardiovascular mortality	Imazu 2020 (S14) (after HF-hospitalisation)	pg/mL	8 / 25	[5.3]	HR 1.00 (0.99-1.00)	0.377	
	Acute congestive HF, HF-mortality,	Ayca 2015 (S2)	NA	14 / 49	range NA-1.5	aOR 1.223 (1.010-1.050)	**0.030**	Age, sex, BMI, hypertension, diabetes mellitus, hyperlipidemia,

Biomarker	Outcome definition	Study (reference)	Unit	n outcome / n total	Follow-up duration (years)	Result	P-value	Factors adjusted for
	haemodynamically significant VT, VF, resuscitated cardiac arrest, SCD, ICD implantation or myocardial infarction							smoking, family history (not defined), septal thickness, LVOT gradient and BNP/TGF-β1
	HF-hospitalisation, HF-mortality, SCD or ICD implantation	Yildiz 2018 (S42)	NA	23 / 87	range NA-2	HR 5.1 (1.81-8.06) [HR 3.8 (1.81-8.06)]	**0.03** **[0.004]**	
	Septal reduction therapy or all-cause mortality	Geske 2013 (S9) (restricted to patients without prior or planned septal reduction therapy)	Tertile 3 vs 1 Tertile 2 vs 1 Per tertile	NA / 329 NA / 322 85 / 471	1.7 ± 1.9	HR 3.89 [(2.02-7.50)] HR 2.56 [(1.41-4.65)]	**<0.0001** **0.002** **<0.0001**	
BNP & cTnI	HF-hospitalisation, haemodynamically unstable sustained VT, appropriate ICD therapy, resuscitated cardiac arrest, stroke-hospitalisation or cardiovascular mortality	Kubo 2011 (S17)	≥200 pg/mL, ≥0.04 vs <200 pg/mL, <0.04 ng/mL	16 / 95	3.2 ± 1.5	HR 11.7 (1.54-88.51)	**[0.0174]**	
			<200 pg/mL, ≥0.04 vs <200 pg/mL, <0.04 ng/mL	4 / 84		HR 2.4 (0.25-23.44)	[0.4453]	
			≥200 pg/mL, <0.04 vs <200 pg/mL, <0.04 ng/mL	2 / 60		HR 1.7 (0.11-26.8)	[0.6998]	
			(Total)	20 / 167				
			≥200 pg/mL & ≥0.04 vs <200 pg/mL & <0.04 ng/mL	16 / 95	3.2 ± 1.5	aHR NA	**0.028**	Age >65 years, sex, AF, MWT ≥30 mm, LVOT gradient ≥30 mmHg
		Kubo 2011 (S17) (restricted to LVEF ≥50%)	≥200 pg/mL, ≥0.04 vs <200 pg/mL, <0.04 ng/mL	NA / NA	3.2 ± 1.5	HR 7.9 (1.00-62.67)	[0.050]	
			<200 pg/mL, ≥0.04 vs <200 pg/mL, <0.04 ng/mL	NA / NA		HR 1.7 (0.16-18.83)	[0.650]	
			≥200 pg/mL, <0.04 vs <200 pg/mL, <0.04 ng/mL	NA / NA		HR 1.7 (0.11-26.87)	[0.699]	
			(Total)	13 / 157				
CK-MB	All-cause mortality	Hamada 2016 (S11)	≥2.5%	27 / 77	17.8 ± 4.0	HR NA	**<0.0001**	
Copeptin	HF-hospitalisation, HF-mortality, arrhythmia-hospitalisation, SCD or ICD implantation	Sahin 2017 (S29)	nmol/mL	20 / 60	range NA-2	aOR 1.19 (1.04-1.31)	**<0.01**	Age, sex, hypertension, interventricular septum thickness, LVOT gradient and copeptin/NT-proBNP
Creatinine	Heart transplantation or all-cause mortality	Zhang 2018 (S45)	pmol/mL	45 / 756	3.7 ± 1.5	HR 1.025 (1.017-1.034)	**<0.001**	
						aHR 1.001 (0.989-1.014)	0.852	Age, AF, LVOT gradient, LVEF and NT-proBNP/creatinine/free triiodothyronine (per tertile)
						aHR 0.993 (0.979-1.007)	0.339	Age, AF, LVOT gradient, LVEF and NT-proBNP/creatinine/free triiodothyronine (continuous)
	Sustained VT, VF, appropriate ICD	Ekizler 2019 (S7)	mg/dL	54 / 411	6.0 (5.0-8.0)	HR 1.185 (0.801-1.753)	0.395	

Biomarker	Outcome definition	Study (reference)	Unit	n outcome / n total	Follow-up duration (years)	Result	P-value	Factors adjusted for
	therapy, cardiovascular mortality							
	HF-hospitalisation, HF-mortality, SCD or ICD implantation	Yildiz 2018 (S42)	NA	23 / 87	range NA-2	HR 1.78 (0.54-5.83)	0.34	
eGFR	All-cause mortality	Wang 2020a (S37)	mL/min/1.73 m^2	80 / 454	3.8 (NA-NA), range 0.1-9.4	HR 0.99 (0.98-0.99)	**0.004**	
		Wang 2020b (S38) (restricted to myectomy patients with diabetes)	≤80 mL/min	9 / 67	2.3 (1.1-4.4)	HR 4.83 (1.42-19.36) [HR 4.83 (1.28-18.2)]	**0.02**	
						aHR 3.23 (1.34-7.24)	**0.04**	Age, sex, NT-proBNP
	Cardiovascular mortality (including intervention-related mortality)	Wang 2020a (S37)	mL/min/1.73 m^2	52 / 454	3.8 (NA-NA), range 0.1-9.4	HR 0.99 (0.98-1.00)	0.128	
	HF-hospitalisation, cardiovascular mortality	Imazu 2020 (S14) (after HF hospitalisation)	mL/min/1.73 m^2	8 / 25	[5.3]	HR 1.00 (0.92-1.06)	0.899	
Galectin-3	All-cause mortality	Song 2020 (S34) (restricted to myectomy patients)	log(ng/mL)	NA / 125	NA	HR 1.17 (1.09-1.32) [HR 1.17 (1.01-1.36)]	**0.039**	
						aHR 1.35 (0.90-1.65)	0.204	Hypertension, MWT, log(soluble ST2)
	NYHA progression ≥1 class requiring medication, HF-mortality, sustained VT, VF, SCD or nsVT	Hu 2016 (S12)	mU/mL	22 / 107	range NA-7	HR 1.02 (0.94-1.10)	0.644	
						aHR 0.92 (0.50-1.72)	0.802	Age, BMI, LVEF, indexed LV end-diastolic volume, indexed LV end-systolic volume, indexed LV mass and presence of late gadolinium enhancement
Glucose	All-cause mortality	Wang 2020a (S37)	µmol/mL	80 / 454	3.8 (NA-NA), range 0.1-9.4	HR 1.12 (1.03-1.22)	**0.008**	
	Cardiovascular mortality (including intervention-related mortality)	Wang 2020a (S37)	µmol/mL	52 / 454	3.8 (NA-NA), range 0.1-9.4	HR 1.11 (0.99-1.24)	0.070	
	Sustained VT, VF, appropriate ICD therapy, cardiovascular mortality	Ekizler 2019 (S7)	mg/dL	54 / 411	6.0 (5.0-8.0)	HR 0.996 (0.989-1.004)	0.325	

Biomarker	Outcome definition	Study (reference)	Unit	n outcome / n total	Follow-up duration (years)	Result	P-value	Factors adjusted for
Haemoglobin	All-cause mortality	Wang 2021a (restricted to myectomy patients)	g/L	26 / 867	2.9 ± 1.4	HR 0.98 (0.96-1.002)	0.079	
	Cardiovascular mortality (including intervention-related mortality)	Wang 2021a (restricted to myectomy patients)	g/L	23 / 867	2.9 ± 1.4	HR 0.98 (0.96-1.000)	**0.048**	
	Sustained VT, VF, appropriate ICD therapy, cardiovascular mortality	Ekizler 2019 (S7)	g/dL	54 / 411	6.0 (5.0-8.0)	HR 0.968 (0.858-1.094)	0.606	
HDL-cholesterol	Sustained VT, VF, appropriate ICD therapy, cardiovascular mortality	Ekizler 2019 (S7)	mg/dL	54 / 411	6.0 (5.0-8.0)	HR 0.993 (0.973-1.013)	0.470	
hs-CRP	Cardiovascular mortality	Zhu 2017 (S47)	µg/mL	30 / 490	3.7 ± 2.0	HR 1.16 (1.08-1.26)	**<0.001**	
						aHR 1.15 (1.06-1.25)	**0.001**	Age, sex, family history of SCD, NYHA class III-IV, unexplained syncope, MWT, LVOT obstruction at rest and nsVT
						aHR 1.15 (1.06-1.25)	**0.001**	Age, sex and NYHA class III-IV
			>3.0 vs <1.0 µg/mL	20 / 312	3.7 ± 2.0	HR 5.88 (2.13-16.2)	**0.001**	
			1.0-3.0 vs <1.0 µg/mL	15 / 379		HR 2.32 (0.79-6.80)	0.124	
						aHR 5.41 (1.96-14.93)	**0.001**	Age, sex, family history of SCD, NYHA class III-IV, unexplained syncope, MWT, LVOT obstruction at rest and nsVT
						aHR 2.57 (0.87-7.55)	0.087	
						aHR 5.41 (1.96-14.93)	**0.001**	Age, sex and NYHA class III-IV
						aHR 2.57 (0.87-7.55)	0.087	
			Tertile 3 vs 1	NA / 327	3.7 ± 2.0	HR 4.25 (1.59-11.33)	**0.004**	
			Tertile 2 vs 1	NA / 327		HR 0.97 (0.28-3.34)	0.955	
						aHR 4.25 (1.59-11.4)	**0.004**	Age, sex, family history of SCD, NYHA class III-IV, unexplained syncope, MWT, LVOT obstruction at rest and nsVT
						aHR 1.14 (0.33-3.95)	0.842	
						aHR 4.08 (1.53-10.88)	**0.005**	Age, sex and NYHA class III-IV
						aHR 1.07 (0.31-3.70)	0.916	
	Sustained VT, VF, appropriate ICD therapy, cardiovascular mortality	Ekizler 2019 (S7)	mg/dL	54 / 411	6.0 (5.0-8.0)	HR 1.043 (1.018-1.068)	**0.001**	
						aHR 1.017 (0.988-1.046)	0.246	AF, nsVT, ESC HCM Risk-SCD score and monocyte:HDL ratio ≥14.57/hs-CRP (per mg/dL)
hs-cTnT	Progression to NYHA class III-IV, HF-hospitalisation,	Gommans 2021 (S10)	ng/mL	18 / 135	5.0 (4.9-5.1)	HR 1.031 (1.004-1.059)	**0.03**	

Biomarker	Outcome definition	Study (reference)	Unit	n outcome / n total	Follow-up duration (years)	Result	P-value	Factors adjusted for
	sustained VT, appropriate ICD therapy, stroke-hospitalisation or cardiovascular mortality							
						aHR NA	0.096	Sex, nsVT
						aHR 1.03 (1.002-1.07)	**0.04**	Family history of SCD, MWT ≥30 mm, unexplained syncope, nsVT, abnormal blood pressure response to exercise
			≥0.014 ng/mL	18 / 135	5.0 (4.9-5.1)	HR 3.4 (1.4-8.7) [HR 3.40 (1.36-8.52)]	**0.009**	
						aHR 4.7 (1.8-12.6)	**0.002**	nsVT
	Progression to NYHA class III-IV, HF-hospitalisation, haemodynamically unstable sustained VT, appropriate ICD therapy, stroke-hospitalisation or cardiovascular mortality	Kubo 2013 (S20)	>0.014 ng/mL	38 / 183	4.1 ± 2.0	HR 5.05 (2.11-12.09)	**<0.001**	
			0.031-0.140 vs ≤0.014 ng/mL	NA / 119	4.1 ± 2.0	aHR 3.67 (1.31-10.25)	**0.013**	NYHA class and LA diameter
				NA / 116		aHR 3.06 (1.02-9.13)	**0.046**	
			0.019-0.030 vs ≤0.014 ng/mL	NA / 116		aHR 2.94 (0.98-8.84)	0.055	
			0.015-0.018 vs ≤0.014 ng/mL					
			log(ng/mL) per 1 SD	38 / 183	4.1 ± 2.0	HR 1.89 (1.13-3.15)	**0.015**	
	Progression to NYHA class III-IV, HF-hospitalisation, appropriate ICD therapy, stroke-hospitalisation, cardiovascular mortality	Hasler 2016 (S12)	>0.014 ng/mL	10 / 91	11.5 ± NA, range 0.5-35	OR 2.51	0.315	
						HR NA	0.16	
Indoxyl sulfate	HF-hospitalisation, cardiovascular mortality	Imazu 2020 (S14) (after HF hospitalisation)	µg/mL	8 / 25	[5.3]	HR 2.52 (1.10-6.04) [HR 2.52 (1.10-5.78)]	**0.029**	
						aHR 2.59 (1.111-6.277)	**0.028**	eGFR
						aHR 2.91 (1.202-7.498)	**0.019**	Age, sex, eGFR
Intelectin-1	HF-hospitalisation, HF-mortality, SCD or ICD implantation	Yildiz 2018 (S42)	NA	23 / 87	range NA-2	HR 14.3 (4.96-35.1) [HR 13.2 (4.96-35.1)]	**<0.001**	

Biomarker	Outcome definition	Study (reference)	Unit	n outcome / n total	Follow-up duration (years)	Result	P-value	Factors adjusted for
						aHR 5.1 (3.5-9.9) [aHR 5.9 (3.5-9.9)]	**<0.01**	Sex and interventricular septum thickness
LDL-cholesterol	All-cause mortality	Wang 2020a (S37)	µmol/mL	80 / 454	3.8 (NA-NA), range 0.1-9.4	HR 0.64 (0.48-0.85)	**0.002**	
		Wang 2021a (restricted to myectomy patients)	µmol/mL	26 / 867	2.9 ± 1.4	HR 0.59 (0.37-0.94)	**0.026**	
	Cardiovascular mortality (including intervention-related mortality)	Wang 2020a (S37)	µmol/mL	52 / 454	3.8 (NA-NA), range 0.1-9.4	HR 0.67 (0.46-0.96)	**0.029**	
Lymphocyte count	Sustained VT, VF, appropriate ICD therapy, cardiovascular mortality	Ekizler 2019 (S7)	$*10^3$ µL	54 / 411	6.0 (5.0-8.0)	HR 1.021 (0.732-1.426)	0.901	
MMP-2:TIMP1 ratio	HF-hospitalisation, heart transplantation, sustained VT, all-cause mortality, atrial fibrillation	Bi 2021 (S4) (restricted to myectomy patients)	≥3.8262	6 / 55	3.7 (3.5-3.9)	HR 7.83 (0.99-67.0) [HR 7.83 (0.95-64.41)]	0.060	
						aHR 19.2 (1.66-222)	**0.018**	Age, sex, PICP/ICTP ratio ≥0.6885
						aHR 17.0 (1.38-210)	**0.027**	Age, sex, AF, septal thickness, PICP/ICTP ratio ≥0.6885
Monocyte count	All-cause mortality	Scott 2019 (S30) (three potentially overlapping cohorts)	≥0.95 K/µL	[65] / 373	NA	3.12 (1.60-6.07)	**0.00082**	
			≥0.95 K/µL	[2287] / 8565	NA	1.55 (1.41-1.71)	**<0.0001**	
			≥0.95 K/µL	[842] / 9573	NA	1.54 (1.20-1.97)	**0.00056**	
	Sustained VT, VF, appropriate ICD therapy, cardiovascular mortality	Ekizler 2019 (S7)	$*10^3$ µL	54 / 411	6.0 (5.0-8.0)	HR 1.002 (1.001-1.003)	**<0.001**	
Monocyte:HDL-cholesterol ratio	Sustained VT, VF, appropriate ICD therapy, cardiovascular mortality	Ekizler 2019 (S7)	$*10^3$ µL/(mg/dL)	54 / 411	6.0 (5.0-8.0)	HR 1.067 (1.037-1.098)	**<0.001**	
			≥14.57 $*10^3$ µL/(mg/dL)	54 / 411	6.0 (5.0-8.0)	HR 3.941 (2.142-7.249)	**<0.001**	
						aHR 3.148 (1.661-5.966)	**<0.001**	AF, nsVT, ESC HCM Risk-SCD score and monocyte:HDL ratio ≥14.57/hs-CRP (per mg/dL)
MR-proANP	HF-hospitalisation, LVAD implantation, heart transplantation or all-cause mortality	Begue 2020 (S3)	log(pmol/mL)	32 / 357	1.9 (1.1-2.5)	HR 4.25 (2.45-7.38)	**<0.0001**	
						aHR 3.77 (2.01-7.08)	**<0.0001**	NYHA class III-IV and LVEF

Biomarker	Outcome definition	Study (reference)	Unit	n outcome / n total	Follow-up duration (years)	Result	P-value	Factors adjusted for
Neutrophil count	Sustained VT, VF, appropriate ICD therapy, cardiovascular mortality	Ekizler 2019 (S7)	$*10^3$ µL	54 / 411	6.0 (5.0-8.0)	HR 1.051 (0.920-1.201)	0.465	
NT-proBNP	All-cause mortality	Wang 2021a (restricted to myectomy patients)	log([pg/mL])	26 / 867	2.9 ± 1.4	HR 4.76 (1.38-16.44) [HR 4.76 (1.37-16.52)]	**0.014**	
	Heart transplantation or all-cause mortality	Zhang 2018 (S45) Restricted to HOCM patients (overlaps with Wang 2021a)	per 100 fmol/mL	45 / 756	3.7 ± 1.5	HR 1.027 (1.016-1.037)	**<0.001**	
						aHR 1.023 (1.006-1.041)	**0.009**	Age, AF, LVOT gradient, LVEF and NT-proBNP/creatinine/free triiodothyronine (continuous)
						aHR 1.023 (1.005-1.041)	**0.012**	Age, AF, LVOT gradient, LVEF and NT-proBNP/creatinine/free triiodothyronine (per tertile)
		Coats 2013 (S5)	log(fmol/mL)	68 / 847	3.5 (2.5-4.5)	HR 2.43 (1.92-3.06)	**<0.001**	
						aHR 2.04 (1.56-2.66)	**<0.001**	Age, sex, NYHA class III-IV, LVEF and LA diameter
		Coats 2013 (S5) (restricted to patients with LVEF ≥50%)	log(fmol/mL)	42 / 673	3.5 (2.5-4.5)	aHR 2.11 (1.54-2.90)	**<0.001**	Age, sex, NYHA class III-IV and LA diameter
		Maczynska-Mazuruk 2019 (S22)	Tertile 2-3 vs 1	14 / 603	2.3 ± 1.2	aHR 5.9 (2.05-17.04)	**0.001**	Unknown variables (not reported)
		Coats 2013 (S5)	>97.5th percentile for age & sex	68 / 847	3.5 (2.5-4.5)	RR 6.7 (1.7-27.2)	**0.0074**	
	Heart transplantation, resuscitated cardiac arrest, cardiovascular mortality (including intervention-related mortality)	D'Amato 2013 (S6)	log(pg/mL)	8 / 183	3.9 ± 2.8	HR 5.56 (1.21-25.5)	**0.02**	
						aHR 5.8 (1.07-31.6)	**0.04**	Diastolic dysfunction (restrictive)
	Cardiovascular mortality (including intervention-related mortality)	Wang 2021a (restricted to myectomy patients)	log([pg/mL])	23 / 867	2.9 ± 1.4	HR 5.25 (1.46-5.93) [HR 5.25 (1.46-18.85)]	**0.011**	
		Song 2019 (S33) (restricted to myectomy patients) (overlaps with Wang 2021a)	ln(pg/mL)	19 / 758	2.6 (1.3-4.8)	aHR 2.788 (1.450-5.362)	**0.002**	Age, sex, LA diameter ≥45 mm, aortic valve procedure, Maze procedure and coronary artery bypass grafting

Biomarker	Outcome definition	Study (reference)	Unit	n outcome / n total	Follow-up duration (years)	Result	P-value	Factors adjusted for
	Heart transplantation, Cardiovascular mortality (including intervention-related, valvular- and myocardial infarction-mortality)	Coats 2013 (S5)	log(fmol/mL)	38 / 847	3.5 (2.5-4.5)	HR 2.48 (1.82-3.39)	**<0.001**	
	HF-hospitalisation, stroke, cardiovascular mortality (including intervention-related mortality)	Mutlu 2006 (S26)	log(pg/mL)	18 / 80	1.6 ± 0.8, range 0.1-2.5	HR 2.9 (1.56-3.24) [HR 2.25 (1.56-3.24)]	**0.0001**	
						aHR 1.6 (1.10-2.55) [aHR 1.68 (1.10-2.55)]	**0.015**	NYHA class III
	HF-hospitalisation, LVAD implantation, heart transplantation or all-cause mortality	Begue 2020 (S3)	log(pg/mL)	32 / 357	1.9 (1.1-2.5)	HR 2.33 (1.66-3.27)	**<0.0001**	
	HF-hospitalisation, HF-mortality, arrhythmia-hospitalisation, SCD or ICD implantation	Sahin 2017 (S29)	pg/mL	20 / 60	range NA-2	aOR 1.003 (1.001-1.004)	**<0.01**	Age, sex, hypertension, interventricular septum thickness, LVOT gradient, copeptin/NT-proBNP
PICP:ICTP ratio	HF-hospitalisation, heart transplantation, sustained VT, all-cause mortality, atrial fibrillation	Bi 2021 (S4) (restricted to myectomy patients)	≥0.6885	6 / 55	3.7 (3.5-3.9)	HR 7.09 (1.29-38.8)	**0.024**	
						aHR 16.8 (2.21-128)	**0.006**	Age, sex, MMP-2/TIMP1 ratio ≥3.8262
						aHR 12.7 (1.46-110)	**0.021**	Age, sex, AF, septal thickness, MMP-2/TIMP1 ratio ≥3.8262
Platelet count	Sustained VT, VF, appropriate ICD therapy, cardiovascular mortality	Ekizler 2019 (S7)	$*10^3$ µL	54 / 411	6.0 (5.0-8.0)	HR 0.999 (0.995-1.002)	0.496	
Prognostic nutritional index	All-cause mortality	Wang 2021b (S40)	per SD 10 × albumin [g/dL] + 5 × lymphocyte count [nL]	66 / 537	4.8 (2.4-6.8)	HR 0.37 (0.29-0.48)	**<0.001**	
						aHR 0.39 (0.30-0.52)	**<0.001**	Age, sex
						aHR 0.46 (0.34-0.62)	**<0.001**	Age, sex, NYHA class, AF, LA diameter, LVEF, eGFR, cholesterol, triglycerides, diuretics
			10 × albumin [g/dL] + 5 × lymphocyte count [nL] ≥48.8	66 / 537	4.8 (2.4-6.8)	HR 0.25 (0.15-0.41)	**<0.001**	
						aHR 0.28 (0.17-0.47)	**<0.001**	Age, sex

Biomarker	Outcome definition	Study (reference)	Unit	n outcome / n total	Follow-up duration (years)	Result	P-value	Factors adjusted for
						aHR 0.39 (0.23-0.68)	**0.001**	Age, sex, NYHA class, AF, LA diameter, LVEF, eGFR, cholesterol, triglycerides, diuretics
	Cardiovascular mortality (including intervention- and myocardial infarct-related mortality)	Wang 2021b (S40)	per SD 10 × albumin [g/dL] + 5 × lymphocyte count [nL]	45 / 537	4.8 (2.4-6.8)	HR 0.39 (0.28-0.53)	**<0.001**	
						aHR 0.39 (0.28-0.54)	**<0.001**	Age, sex
						aHR 0.44 (0.30-0.63)	**<0.001**	Age, sex, NYHA class, AF, LA diameter, LVEF, eGFR, cholesterol, triglycerides, diuretics
			10 × albumin [g/dL] + 5 × lymphocyte count [nL] ≥48.8	45 / 537	4.8 (2.4-6.8)	HR 0.29 (0.16-0.53)	**<0.001**	
						aHR 0.31 (0.17-0.58)	**<0.001**	Age, sex
						aHR 0.47 (0.24-0.89)	**0.020**	Age, sex, NYHA class, AF, LA diameter, LVEF, eGFR, cholesterol, triglycerides, diuretics
Red blood cell distribution width	All-cause mortality	Wang 2021a (S39) (restricted to myectomy patients)	per SD	26 / 867	2.9 ± 1.4	HR 1.84 (1.59-2.11)	**<0.001**	
						aHR 1.76 (1.54-2.05)	**<0.001**	NYHA class, postoperative hospital stays
			Quartile 2 vs 1	6 / 448	2.9 ± 1.4	HR 2.15 (0.39-11.74)	**0.001**	
			Quartile 3 vs 1	8 / 266		HR 2.67 (0.54-13.23)	**(for trend)**	
			Quartile 4 vs 1	16 / 492		HR 7.20 (1.63-31.73)		
						aHR 2.04 (0.37-11.21)	**0.001**	Age, sex, BMI
						aHR 2.39 (0.48-11.97)	**(for trend)**	
						aHR 7.10 (1.60-31.61)		
						aHR 1.19 (0.52-2.72)	**0.001**	NYHA class, LA diameter, haemoglobin, LDL, log(NT-proBNP), postoperative hospital stay, maze procedure
						aHR 0.99 (0.44-2.22)	**(for trend)**	
						aHR 2.54 (1.21-5.34)		
	Cardiovascular mortality (including intervention-related mortality)	Wang 2021a (S39) (restricted to myectomy patients)	per SD	23 / 867	2.9 ± 1.4	HR 1.87 (1.62-2.15)	**<0.001**	
						aHR 1.91 (1.63-2.22)	**<0.001**	NYHA class, septal thickness, postoperative hospital stays
			Quartile 2 vs 1	5 / 448	2.9 ± 1.4	HR 1.63 (0.27-9.75)	**0.002**	
			Quartile 3 vs 1	8 / 266		HR 2.70 (0.55-13.42)	**(for trend)**	
			Quartile 4 vs 1	14 / 492		HR 6.25 (1.39-27.96)		
						aHR 1.52 (0.26-9.16)	**0.003**	Age, sex, BMI
						aHR 2.48 (0.50-12.38)	**(for trend)**	
						aHR 6.06 (1.34-27.27)		

Biomarker	Outcome definition	Study (reference)	Unit	n outcome / n total	Follow-up duration (years)	Result	P-value	Factors adjusted for
						aHR 1.15 (0.51-2.63) aHR 0.98 (0.45-2.17) aHR 2.56 (1.27-5.17)	**0.002** **(for trend)**	LA diameter, septal thickness, haemoglobin, log(NT-proBNP), postoperative hospital stay
Soluble neprilysin	All-cause mortality	Yoshihisa 2019 (S43)	ng/mL	11 / 93	2.8 ± 1.7 range 0.04-7.9	HR 1.194 (0.972-1.468)	0.092	
			≥1.43 ng/mL	11 / 93	2.8 ± 1.7 range 0.04-7.9	HR 1.507 (0.440-5.158)	0.513	
Soluble ST2	All-cause mortality	Song 2020 (S34) (restricted to myectomy patients)	log(ng/mL)	NA / 125	NA	HR 1.02 (1.01-1.05)	**0.015**	
						aHR 1.15 (0.98-1.27)	0.204	Hypertension, MWT, log(Galectin-3)
T3 (free)	Heart transplantation or all-cause mortality	Zhang 2018 (S45)	pg/mL	45 / 756	3.7 ± 1.5	HR 0.111 (0.065-0.189)	**<0.001**	
						aHR 0.216 (0.083-0.559)	**0.002**	Age, AF, LVOT gradient, LVEF and NT-proBNP/creatinine/free triiodothyronine (continuous)
			Tertile 1 vs 3	[36] / 506	3.7 ± 1.5	aHR 4.918 (1.076-22.485)	**0.040**	Age, AF, LVOT gradient, LVEF and NT-proBNP/creatinine/free triiodothyronine (per tertile)
			Tertile 2 vs 3	[12] / 509		aHR 1.78 (0.327-9.700)	0.505	
T4 (free)	Heart transplantation or all-cause mortality	Zhang 2018 (S45)	per 10 pg/mL	45 / 756	3.7 ± 1.5	HR 0.603 (0.146-2.493)	0.485	
TGF-β1	All-cause mortality	Ayca 2015 (S2)	>4877 pg/mL	6 / 49	range NA-1.5	OR 3.73	0.1	
	Acute congestive HF, HF-mortality, haemodynamically significant VT, VF, resuscitated cardiac arrest, SCD, ICD implantation or myocardial infarction	Ayca 2015 (S2)	>4877 pg/mL	14 / 49	range NA-1.5	OR 7.22	**0.003**	
						aOR 1.312 (1.001-1.333)	**0.028**	Age, sex, BMI, hypertension, diabetes mellitus, hyperlipidaemia, smoking, family history (not defined), septal thickness, LVOT gradient and BNP/TGF-β1
						HR NA	**<0.01**	
	HF-hospitalisation, haemodynamically significant VT, VF, resuscitated cardiac arrest or ICD implantation	Ayca 2015 (S2)	>4877 pg/mL	26 / 49	range NA-1.5	OR 4.20	**<0.001**	

Biomarker	Outcome definition	Study (reference)	Unit	n outcome / n total	Follow-up duration (years)	Result	P-value	Factors adjusted for
Triglycerides	All-cause mortality	Wang 2020a (S37)	µmol/mL	80 / 454	3.8 (NA-NA), range 0.1-9.4	HR 0.67 (0.49-0.93)	**0.015**	
	Cardiovascular mortality (including intervention-related mortality)	Wang 2020a (S37)	µmol/mL	52 / 454	3.8 (NA-NA), range 0.1-9.4	HR 0.55 (0.35-0.87)	**0.010**	
TSH	Heart transplantation or all-cause mortality	Zhang 2018 (S45)	µIU/mL	45 / 756	3.7 ± 1.5	HR 1.015 (0.968-1.064)	0.543	
	Sustained VT, VF, appropriate ICD therapy, cardiovascular mortality	Ekizler 2019 (S7)	IU/mL	54 / 411	6.0 (5.0-8.0)	HR 1.025 (1.000-1.051)	0.051	
Uric acid	All-cause mortality	Wang 2020a (S37)	≥450 vs 250-350 nmol/mL	45 / 263	3.8 (NA-NA), range 0.1-9.4	HR 2.56 (1.42-4.64)	**0.002**	
			350-450 vs 250-350 nmol/mL	44 / 309		HR 1.65 (0.91-3.00)	0.098	
			<250 vs 250-350 nmol/mL	29 / 204		HR 2.11 (0.98-4.55)	0.056	
						aHR 2.75 (1.47-5.15)	**0.002**	Age, sex, hypertension, diabetes, chronic obstructive pulmonary disease and AF
						aHR 1.91 (1.04-3.54)	**0.038**	
						aHR 1.58 (0.71-3.50)	0.261	
						aHR 2.73 (1.47-5.07)	**0.001**	Age, sex, familial HCM, familial SCD and NYHA class
						aHR 1.86 (1.01-3.43)	**0.047**	
						aHR 1.64 (0.75-3.58)	0.215	
						aHR 2.73 (1.42-5.23)	**0.003**	Age, sex, chronic obstructive pulmonary disease, AF, NYHA class, LA diameter, triglycerides, LDL-cholesterol
						aHR 1.99 (1.06-3.72)	**0.031**	
						aHR 2.52 (1.13-5.61)	**0.024**	
						aHR 2.59 (1.39-4.84)	**0.003**	Age, sex, warfarin, hydrochlorothiazide, septal reduction therapy and devices
						aHR 1.73 (0.94-3.20)	0.080	
						aHR 2.02 (0.93-4.41)	0.076	
						aHR 2.53 (1.26-5.06)	**0.009**	Age, sex, LA diameter, LVEF, eGFR, glucose, triglycerides and LDL-cholesterol
						aHR 1.72 (0.92-3.24)	0.091	
						aHR 2.72 (1.21-6.10)	**0.016**	
	Cardiovascular mortality	Zhu 2015 (S46)	mg/dL	44 / 588	5.2 ± 2.4	aHR 1.29 (1.11-1.49)	**0.001**	Age, sex, body mass index, controlled hypertension, family history of SCD, NYHA class III-IV, unexplained syncope, MWT, and LVOT obstruction at rest
			Tertile 3 vs 1	NA / 392	5.2 ± 2.4	HR 2.91 (1.36-6.20)	**0.006**	
			Tertile 3 vs 1	NA / 392	5.2 ± 2.4	aHR 3.10 (1.37-7.04)	**0.007**	Age, sex, body mass index, controlled hypertension, family history of SCD, NYHA class III-IV, unexplained syncope, MWT, and LVOT obstruction at rest
			Tertile 2 vs 1	NA / 392		aHR 1.22 (0.47-3.16)	0.684	
			per tertile	44 / 588			**0.007**	
	Cardiovascular mortality (including intervention-related mortality)	Wang 2020a (S37)	≥450 vs 250-350 nmol/mL	27 / 263	3.8 (NA-NA), range 0.1-9.4	HR 4.43 (1.94-10.15)	**<0.001**	
			350-450 vs 250-350 nmol/mL	25 / 309		HR 2.65 (1.14-6.14)	**0.023**	
			<250 vs 250-350 nmol/mL	16 / 204		HR 3.98 (1.49-10.62)	**0.006**	

Biomarker	Outcome definition	Study (reference)	Unit	n outcome / n total	Follow-up duration (years)	Result	P-value	Factors adjusted for
						aHR 5.16 (2.19-12.18)	**<0.001**	Age, sex, familial HCM, familial
						aHR 3.13 (1.33-7.36)	**0.009**	SCD and NYHA class
						aHR 3.10 (1.14-8.40)	**0.026**	
						aHR 4.63 [4.86] (1.94-	**0.001**	Age, sex, hypertension, diabetes,
						12.18)	**0.007**	chronic obstructive pulmonary
						aHR 3.27 (1.38-7.76)	**0.024**	disease and AF
						aHR 3.16 [3.26] (1.17-9.06)		
						aHR 4.38 (1.77-10.85)	**0.001**	Age, sex, LA diameter, LVEF, eGFR,
						aHR 2.97 (1.24-7.09)	**0.014**	glucose, triglycerides and LDL-
						aHR 4.68 (1.71-12.81)	**0.003**	cholesterol
						aHR 4.20 (1.77-9.96)	**0.001**	Age, sex, warfarin,
						aHR 3.03 (1.28-7.18)	**0.012**	hydrochlorothiazide, septal
						aHR 4.13 (1.52-11.25)	**0.005**	reduction therapy and devices
						aHR 4.14 (1.70-10.13)	**0.002**	Age, sex, chronic obstructive
						aHR 3.18 (1.33-7.61)	**0.010**	pulmonary disease, AF, NYHA
						aHR 4.86 (1.74-13.58)	**0.003**	class, LA diameter, triglycerides, LDL-cholesterol
	Sustained VT, VF, appropriate ICD therapy, cardiovascular mortality	Ekizler 2019 (S7)	mg/dL	54 / 411	6.0 (5.0-8.0)	HR 1.041 (0.913-1.187)	0.548	
White blood cell count	Sustained VT, VF, appropriate ICD therapy, cardiovascular mortality	Ekizler 2019 (S7)	$*10^3$ µL	54 / 411	6.0 (5.0-8.0)	HR 1.001 (0.943-1.063)	0.966	

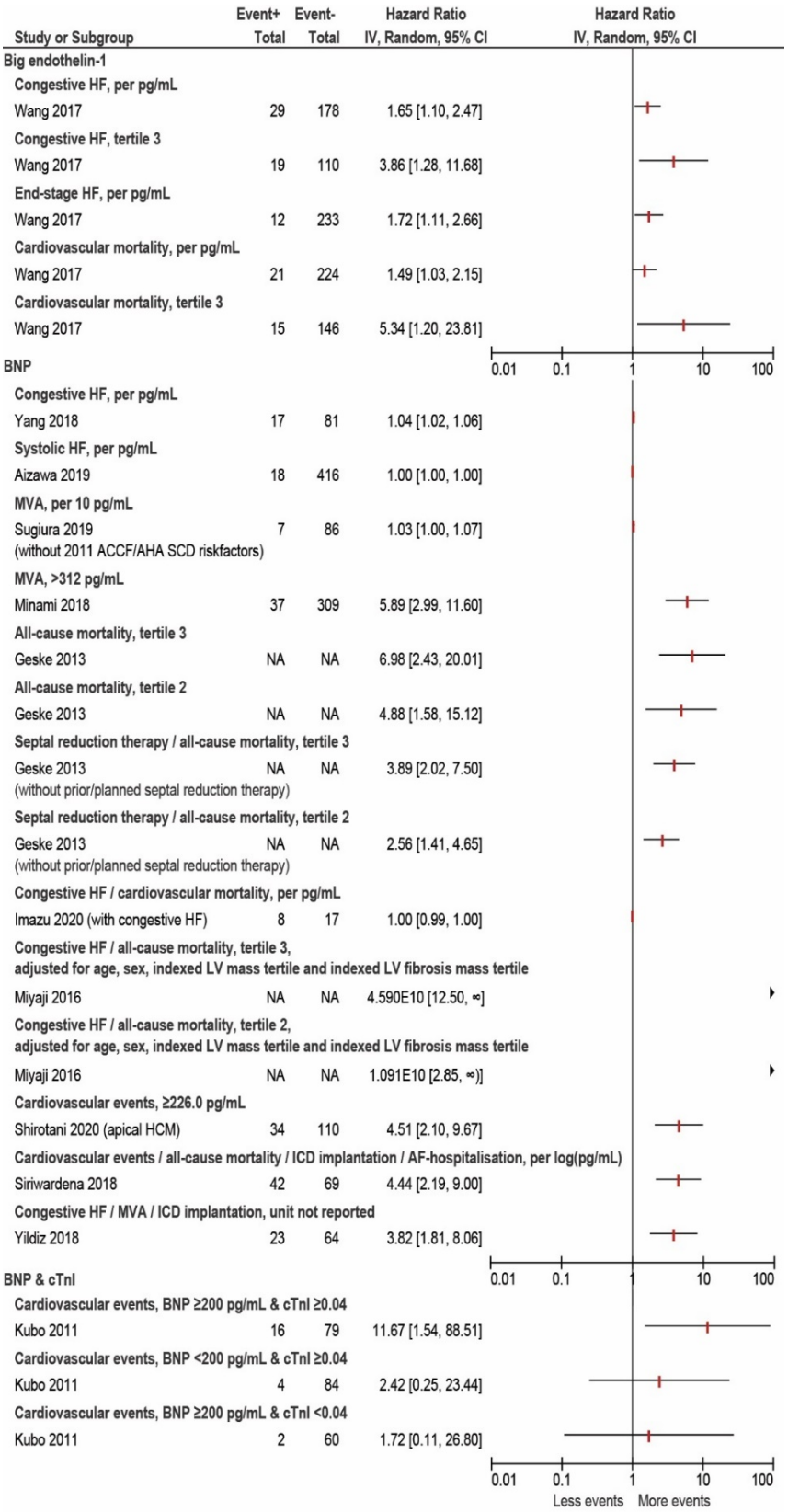
Data presented as hazard ratio (95% confidence interval [CI]) indicated with HR, adjusted hazard ratio (95% CI) indicated with aHR, odds ratio indicated with OR, adjusted odds ratio (95% CI) indicated with aOR, relative risks (95% CI) indicated with RR or adjusted relative risk (95% CI) indicated with aRR. Numbers of outcome-positive subjects, subject totals, follow-up durations, hazard ratios (that could not be reproduced) and 95% CI in square brackets were inferred from available data and are therefore subject to inaccuracies. P-values are provided as reported by the original authors. P-values <0.05 are indicated in bold.

AF, atrial fibrillation; aHR, adjusted hazard ratio; aRR, adjusted relative risk; ANP, atrial natriuretic peptide; BMI, body mass index; BNP, brain natriuretic peptide; CK-MB, creatine (phospho)kinase MB isoform; CRP, C-reactive protein; cTnI/-T, cardiac troponin I/T; eGFR, estimated glomerular filtration rate; ICD, implantable cardioverter-defibrillator; ICTP, C-terminal telopeptide of type I collagen; HDL, high-density lipoprotein; HF, heart failure-related; HR, hazard ratio; hs, high sensitivity; LA, left atrial; LDL, low-density lipoprotein; LV, left ventricular; LVAD, left ventricular assist device; LVEF, left ventricular ejection fraction; LVOT, left ventricular outflow tract; MR-proANP, midregional pro-atrial natriuretic peptide; MWT, maximum wall thickness; NA, not available; nsVT, non-sustained ventricular tachycardia; NT-proBNP, N-terminal prohormone

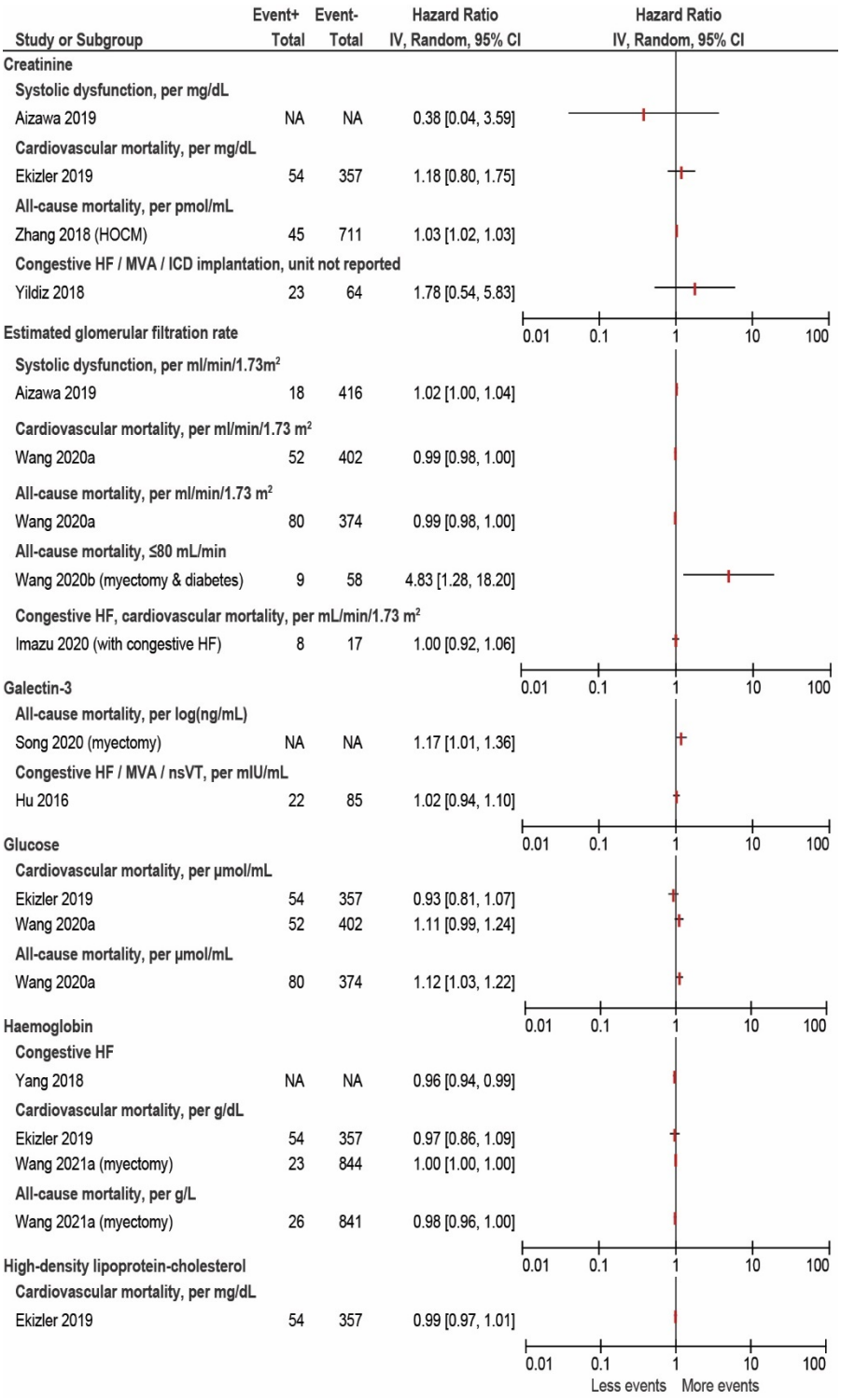
of brain natriuretic peptide; NYHA, New York Heart Association; PICP, Propeptide of procollagen type I; RR, relative risk; SCD, sudden cardiac death; SD, standard deviation; ST2, Suppression of tumorigenicity 2 (also known as interleukin 1 receptor-like 1); T3, triiodothyronine; T4, thyroxine; TGF, transforming growth factor; TSH, Thyroid-stimulating hormone; VF, ventricular fibrillation; VT, ventricular tachycardia.

Supplemental Figures

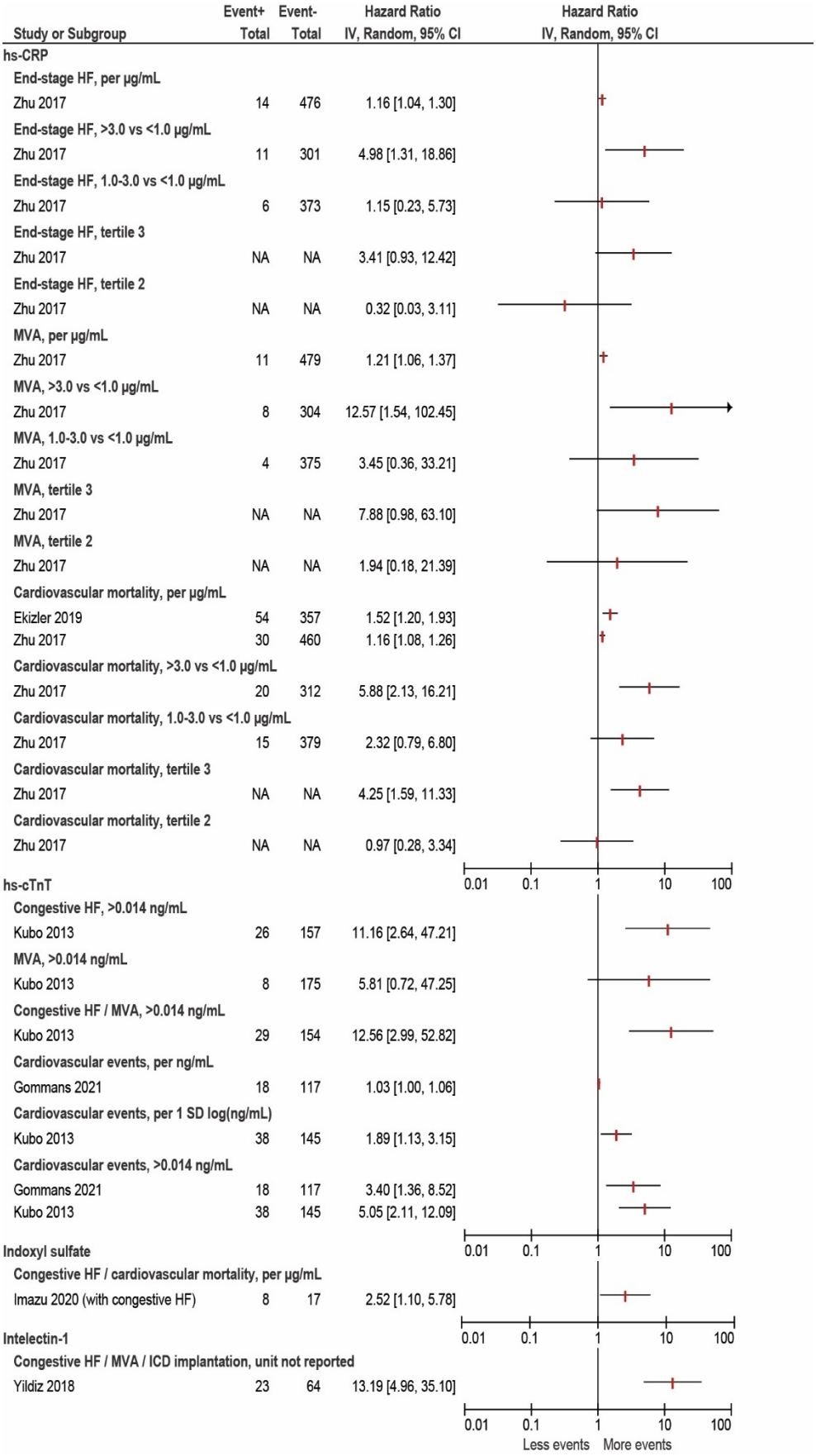
Supplemental Figure 1: Quantitative analysis endpoints



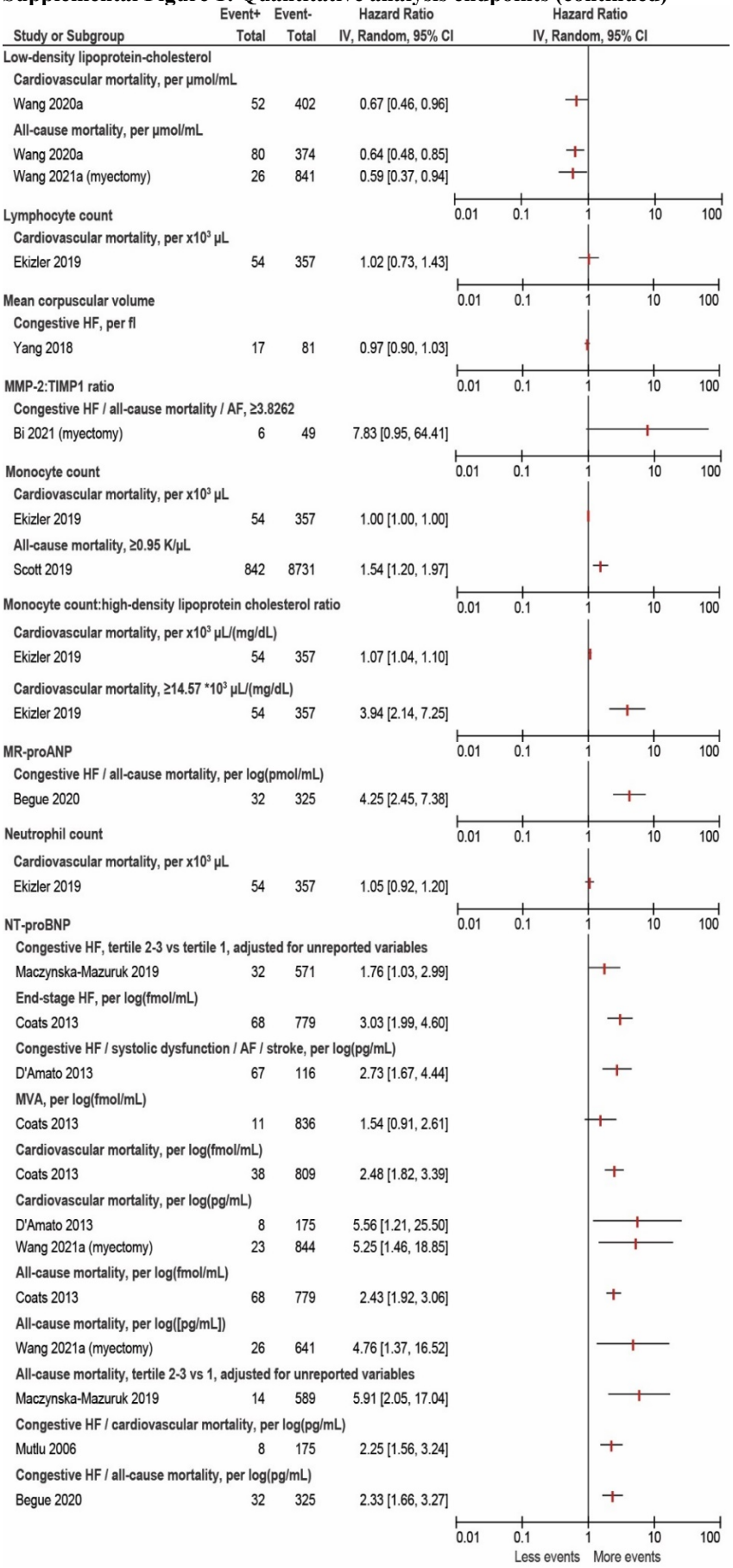
Supplemental Figure 1: Quantitative analysis endpoints (continued)



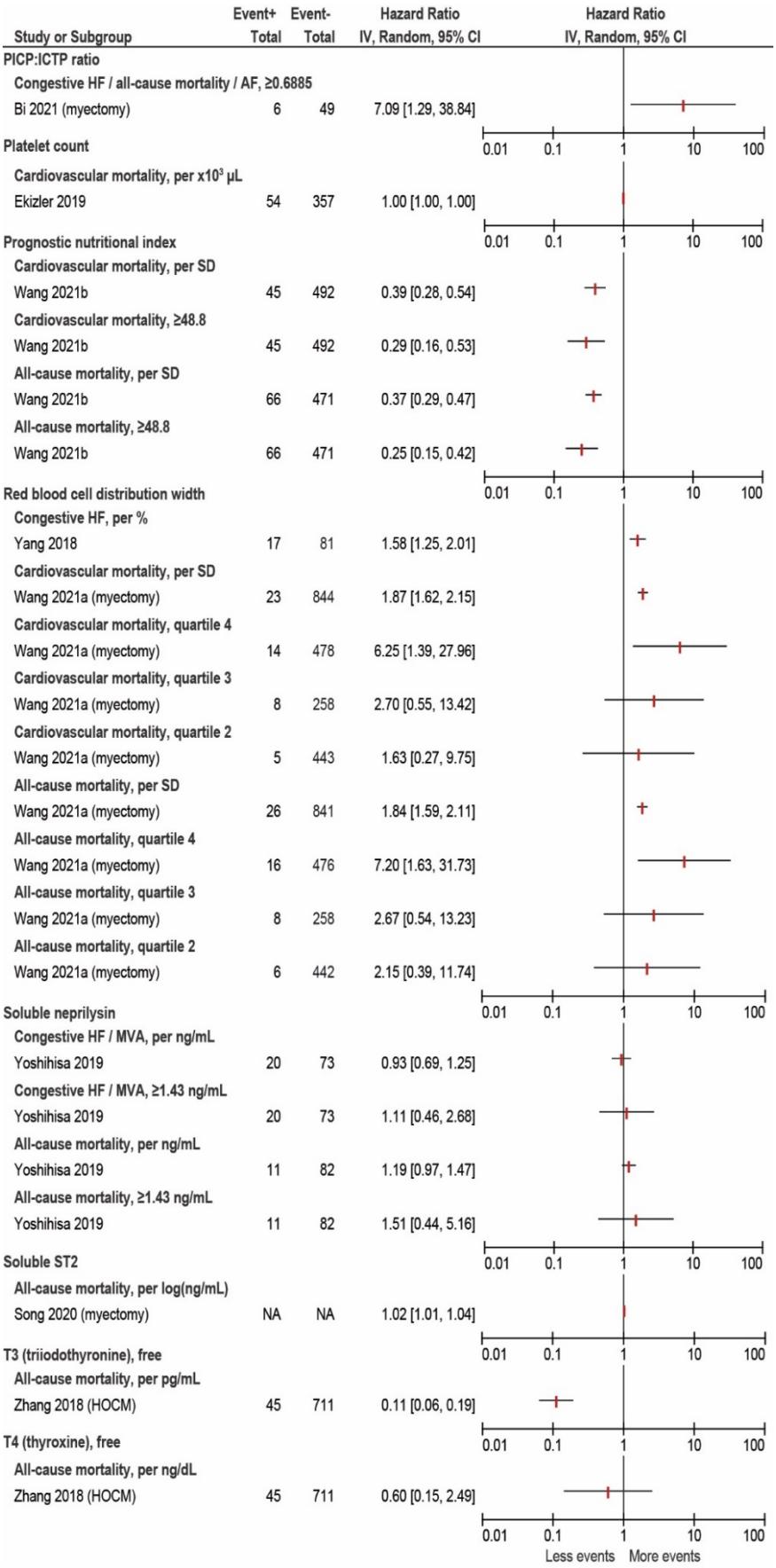
Supplemental Figure 1: Quantitative analysis endpoints (continued)



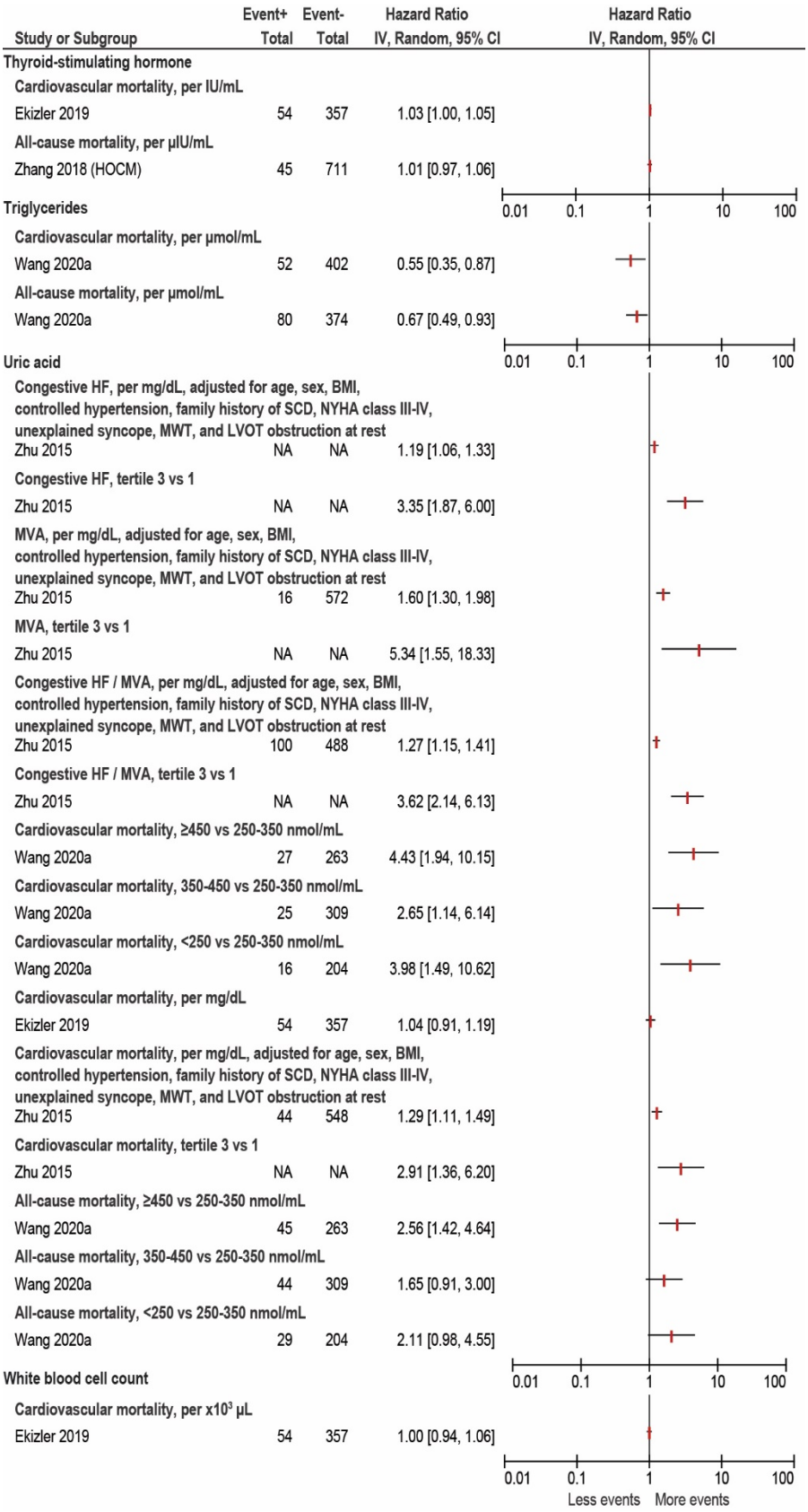
Supplemental Figure 1: Quantitative analysis endpoints (continued)



Supplemental Figure 1: Quantitative analysis endpoints (continued)



Supplemental Figure 1: Quantitative analysis endpoints (continued)



Forest plots of crude hazard ratios (or adjusted when no crude hazard ratios were reported), stratified per biomarker and outcome.

AF, atrial fibrillation; BMI, body mass index; BNP, brain natriuretic peptide; CI, confidence interval; cTnI, cardiac troponin I; ICD, implantable cardioverter defibrillator; ; ICTP, C-terminal telopeptide of type I collagen; IV, inverse variance; HF, heart failure; HOCM, hypertrophic obstructive cardiomyopathy; hs-CRP, high sensitivity C-reactive protein; hs-cTnT, high sensitivity cardiac troponin T; LV, left ventricular; LVOT, left ventricular outflow tract; MMP, matrix metallopeptidase (metalloproteinases); MR-proANP, midregional pro-atrial natriuretic peptide; MVA, malignant ventricular arrhythmia; MWT, maximum wall thickness; NA, not available (not reported); nsVT, non-sustained ventricular tachycardia; NT-proBNP, N-terminal prohormone of brain natriuretic peptide; NYHA, New York Heart Association; PICP, Propeptide of procollagen type I; SCD, sudden cardiac death; ST2, Suppression of tumorigenicity 2 (also known as interleukin 1 receptor-like 1); TIMP1, TIMP (tissue inhibitor of metalloproteinases) metallopeptidase inhibitor 1;

References

1. Elliott PM, Anastasakis A, Borger MA, Borggrefe M, Cecchi F, Charron P, et al. 2014 esc guidelines on diagnosis and management of hypertrophic cardiomyopathy: The task force for the diagnosis and management of hypertrophic cardiomyopathy of the european society of cardiology (esc). *European heart journal*. 2014;35:2733-2779
2. Report of the who/isfc task force on the definition and classification of cardiomyopathies. *British heart journal*. 1980;44:672-673
3. Richardson P, McKenna W, Bristow M, Maisch B, Mautner B, O'Connell J, et al. Report of the 1995 world health organization/international society and federation of cardiology task force on the definition and classification of cardiomyopathies. *Circulation*. 1996;93:841-842
4. JPT H, J T, J C, M C, T L, MJ P, et al. Cochrane handbook for systematic reviews of interventions version 6.0 (updated july 2019). 2019;13-08-2020
5. Luo D, Wan X, Liu J, Tong T. Optimally estimating the sample mean from the sample size, median, mid-range, and/or mid-quartile range. *Statistical methods in medical research*. 2018;27:1785-1805
6. Wan X, Wang W, Liu J, Tong T. Estimating the sample mean and standard deviation from the sample size, median, range and/or interquartile range. *BMC medical research methodology*. 2014;14:135
7. Gersh BJ, Maron BJ, Bonow RO, Dearani JA, Fifer MA, Link MS, et al. 2011 accf/aha guideline for the diagnosis and treatment of hypertrophic cardiomyopathy: Executive summary: A report of the american college of cardiology foundation/american heart association task force on practice guidelines. *Circulation*. 2011;124:2761-2796
8. O'Mahony C, Jichi F, Pavlou M, Monserrat L, Anastasakis A, Rapezzi C, et al. A novel clinical risk prediction model for sudden cardiac death in hypertrophic cardiomyopathy (hcm risk-scd). *European heart journal*. 2014;35:2010-2020
9. Maron MS, Rowin EJ, Wessler BS, Mooney PJ, Fatima A, Patel P, et al. Enhanced american college of cardiology/american heart association strategy for prevention of sudden cardiac death in high-risk patients with hypertrophic cardiomyopathy. *JAMA cardiology*. 2019;4:644-657
10. McKenna WJ, Spirito P, Desnos M, Dubourg O, Komajda M. Experience from clinical genetics in hypertrophic cardiomyopathy: Proposal for new diagnostic criteria in adult members of affected families. *Heart (British Cardiac Society)*. 1997;77:130-132